

REPORT ON PROCEEDINGS BEFORE

PUBLIC ACCOUNTABILITY AND WORKS COMMITTEE

INQUIRY INTO DATA CENTRES

CORRECTED

At Macquarie Room, Parliament House, Sydney, on Friday 22 May 2026

The Committee met at 9:00.

PRESENT

Ms Abigail Boyd (Chair)

The Hon. Mark Buttigieg
The Hon. Scott Farlow (Deputy Chair)
The Hon. Jacqui Munro

PRESENT VIA VIDEOCONFERENCE

The Hon. Peter Primrose
The Hon. Emily Suvaal

CORRECTED

The CHAIR: Good morning. Welcome to the third hearing of the Committee's inquiry into data centres. I acknowledge the Gadigal people of the Eora nation, the traditional custodians of the lands on which we are meeting today. I pay my respects to Elders past and present, and celebrate the diversity of Aboriginal peoples and their ongoing cultures and connections to the lands and waters of New South Wales. I also acknowledge and pay my respects to any Aboriginal and Torres Strait Islander people joining us today or watching online among the masses. My name is Abigail Boyd, and I am the Chair of the Committee.

I ask everyone in the room to please turn their mobile phones to silent. Parliamentary privilege applies to witnesses in relation to the evidence they give today. However, it does not apply to what witnesses say outside of the hearing. I urge witnesses to be careful about making comments to the media or to others after completing their evidence. In addition, the Legislative Council has adopted rules to provide procedural fairness for inquiry participants. I encourage Committee members and witnesses to be mindful of these procedures.

CORRECTED

Mr ALEXANDER HOYSTED, Partner and Strategy Lead, Carbon Zero Initiative, sworn and examined

Ms JAQUELINE MILLS, Senior Climate and Energy Campaigner, Nature Conservation Council of NSW, affirmed and examined

Ms SOLAYE SNIDER, Campaigner, Greenpeace Australia Pacific, affirmed and examined

Dr SIMON BRADSHAW, Senior Climate Adviser, Greenpeace Australia Pacific, affirmed and examined

The CHAIR: I welcome our first panel of witnesses. Thank you so much for making the time to give evidence this morning. I'd like to invite you to make a short opening statement, and I'll start with Greenpeace.

SOLAYE SNIDER: Thank you for the opportunity to attend today. At Greenpeace, we are committed to a safer, more prosperous future for communities across New South Wales. We recognise the vital importance of transitioning away from fossil fuels toward clean energy and industries, and we also recognise the important role emerging technologies can play when they are designed and governed in the public interest. It is in this context that we bring to you some grave and well-founded concerns about the rapid expansion of data centres in this State, and urge the Government of New South Wales to put in place strong guardrails to prevent the derailing of the State's transition to cheap, clean renewable energy, and to ensure future development is appropriately sized, paced, and aligned with the interests of communities.

The claims in our submission that data centre growth could put emissions reductions targets out of reach, prolong the use of fossil fuels and put huge stress on water resources are not speculative. Around the world, we are seeing many cautionary tales of how frenzied data centre growth is driving up electricity prices, draining local water supplies and blowing through carbon budgets. Here in Australia, the warning signs are all too clear. Gas companies like Santos and Woodside are pointing to data centre growth as justification for expanding gas production. Our Federal Climate Change Authority has named rapid growth of data centres as a reason for weakening its advice on Australia's national emissions reduction targets. Australian company Cloud Carrier intends to build a new 700-megawatt, gas-fired power station for its proposed Southern Highlands data campus. I could go on.

We acknowledge that data centres have been a part of Australia's infrastructure for decades and they have enabled digital services that many Australians rely on. But the current data centre frenzy, characterised by a large number of hyperscale facilities, is an entirely different proposition. This new wave of data centres is being built to enable the growth of generative AI, a technology that has been associated with new kinds of harms. Make no mistake, what we are witnessing with this data centre frenzy is a reckless race to expand the power and wealth of a small number of powerful corporations and individuals at immense cost to our climate, communities and the ecosystems on which our lives depend. It's the same extractive mentality that has brought us to the brink of climate catastrophe and we simply cannot afford to continue down such a path.

It's for these reasons that Greenpeace is calling for a series of measures including treating AI data centres as a high-risk category of infrastructure that operators must prove is compatible with the State's climate goals. We must ensure that data centres always use new and truly additional renewable energy, and hit pause on the largest and most energy-intensive data centre projects until appropriate safeguards are put in place. What we need is a major course correction. Put simply, it's time to take a pause until a proper AI strategy and regulatory framework is in place. We thank the Committee for the opportunity to provide evidence to this important inquiry, and look forward to your questions.

JACQUELINE MILLS: Nature Conservation Council of NSW is the peak environmental organisation in our State, with around 200 member groups statewide. The issue of data centres matters right across New South Wales to communities. Following a resolution unanimously passed at our AGM in 2025, Nature Conservation Council wrote to the New South Wales Government calling for a coordinated plan to address the environmental climate and social impacts of data centres in New South Wales. We welcome this inquiry and the consultation informing the development of a New South Wales data centre strategy. We also welcome the announcement from energy Ministers via ECMC of the intent to require data centres to meet their own energy demand and provide demand flexibility.

However, when we speak to government officials here in New South Wales, we hear that energy demand projections are increasingly uncertain due to the growth of data centres. That's calling into question the sufficiency of renewable energy entering the grid and, of course, the timing for closure of our coal-fired power stations. Further potential extensions to the lives of coal-fired power stations is a slap in the face for communities dealing with worsening climate impacts and harms. For communities on the Central Coast, it means the prospect of copping harmful pollutants in the air from coal-fired power for even longer. It means five-million-plus Sydney residents face greater public health risks from toxic wastewater from coalmines at the headwaters of this drinking

CORRECTED

water catchment and it means communities around the State cop hundreds more heavy truck movements on local roads to get coal from mines to power plants.

If we don't know how much data centres will push up energy demand, we can't be sure that our renewable energy target is sufficient to meet our State's energy needs. This could put the whole renewable energy transition into jeopardy. We risk falling even further behind on progress to reach our legislated emission reduction targets. A plan to manage data centres and their growth is essential and Government cannot take a laissez-faire approach in the hope that increased GDP will balance everything else out. We'd suggest that a plan needs to do three things: It needs to align energy demand and supply by requiring data centres to bring genuinely additional sufficient renewable energy into the grid; to speed up the transition of the energy system by leveraging data centre growth to meet energy infrastructure needs; and, of course, to protect nature, water and communities through best practice regulations, coordinated siting of data centres and effective and early community consultation.

Increasing the pace and the scale of the renewable energy transition is essential to meet current and ongoing energy demand across our economy. There are significant economic and emission reduction benefits from lifting the renewable energy target in the roadmap to meet projected energy demand. This will bring down energy prices because, after all, unmanaged data centre growth means higher energy prices for the people of New South Wales. There has been some discussion of siting data centres in renewable energy zones. Should this be practical considering technical needs around latency, it is critical that communities are consulted to ensure confidence that benefits like new jobs and investment will flow to the regions and impacts be addressed and minimised. There is, of course, already concern in some renewable energy zones with impacts from renewable energy infrastructure, as detailed in the parliamentary inquiry into impacts of renewable energy zones on rural and regional communities and industries, so care must be taken to ensure social licence issues associated with renewable energy zones are not compounded by resource, environmental and land price impact from data centre growth in regions.

ALEXANDER HOYSTED: Chair, thank you for the opportunity to appear today. The Carbon Zero Initiative is a not-for-profit organisation focused on decarbonisation. Since 2022 we've existed to decarbonise the energy system, and for the past year we've worked at the intersection of digital and energy infrastructure. In February 2026 we convened a public principles joint statement with 12 signatories from environmental, union and civil groups. Among other things, we called for data centres to act transparently, help our energy transition to a renewable-powered country and invest back into their host communities.

As you're doubtless aware, the scale of data centre investment is significant. There are 90 data centres in New South Wales, while the four big hyperscalers—Microsoft, AWS, Google and Meta—have announced \$600 billion in additional infrastructure investment globally. This is occurring at a sensitive time in Australia's energy transition, and we have to be proactive in ensuring new industry demand bolsters renewable generation, transmission and network affordability. Among others, our submission to the Committee made the following recommendations. First, further integrate data centres into energy system planning. These are system-shaping loads with fast ramp-up and -down times, and they should be incorporated into forecasting transmission planning and renewable energy generation. Since our submission, the AEMC has proposed a ride-through rule change, and it's a really great start to this planning.

Second, location can help solve the energy puzzle. Some 84 per cent of new proposed data centre workloads include AI. A good proportion of these workloads are latency-tolerant workloads that can be located in peri-urban or regional areas. Right now 2.1 gigawatts of renewable energy is being curtailed in regional New South Wales. Creating real incentives for regional location would cut curtailment, reduce transmission costs and anchor new renewable projects. Third, build a reporting framework. Sustainability and efficiency metrics for data centres are today fragmented across industry and State and Federal governments. We need to enable a centralised framework covering energy, water efficiency and emissions to give planners and the public a complete picture of the sector's impact and, crucially, reward operators who lead the charge in helping our energy transition. We appreciate the Committee's interest in this issue and look forward to contributing today.

The CHAIR: Thank you all so much for your excellent submissions. I wanted to start by talking about this thorny issue of renewables and the impact on the renewables rollout. The message from industry seems to be that, yes, we're going to have this additional demand on the grid, but it's chill because we're going to have renewables—if we put to one side the one in the Southern Highlands that's going to have a gas plant attached to it, which shocks me. But if we look purely at the idea that it will all be fine if we can match the amount of data centre demand with renewables projects—I know that both Greenpeace and the Nature Conversation Council talked about this idea of genuinely additional renewables. The question is, is that actually possible—if you're building a renewables project for a data centre, then you're not building it for the existing demand, so is there a case for saying we just need to pause data centres until the total percentage of demand has been met by renewables? What are your thoughts about that?

CORRECTED

SOLAYE SNIDER: It's a great question. It's certainly Greenpeace's view that we need a pause, especially on the largest and most energy-intensive proposals right now because, as you said, we simply don't have enough renewable energy readily available right now for these data centres to meet even their own goals around renewable energy, and that might take some time to come online. That's why a pause is really important—for many reasons. We also want to scrutinise some of the claims that are being made by data centre operators and the data centre lobby around this, because they do put a lot of emphasis on the fact that maybe they even could catalyse the energy transition, and that's just not what we're seeing happen in reality. They seem to be very hypothetical claims. If we actually look at the data, we see the emissions going up from the data centre operators and the big tech companies. We also see the energy demand projections increasing and increasing.

I think it's worth pointing out also that there have been calls from groups like those on this panel for new proposals to meet that goal of 100 per cent renewable energy, and the response from Data Centres Australia, for example, has been resistance to that and saying that that's actually not realistic. I mean, they're saying that themselves—that renewable energy isn't online yet and it's not ready—and so I think we need to listen to that. In that case, that's why we think there needs to be a pause while we sort through some of these issues, because at the same time we have projections that we're not going to meet our emissions reduction targets in New South Wales, and that's without this huge new source of emissions and energy demand. It seems like we can't really rely on these hypothetical claims of renewable energy matching in the future when in the interim we're going to have this huge new source of emissions and we're already not on track to meet our targets.

JACQUELINE MILLS: I think the first thing I would say is that it's totally unacceptable for data centres to be proposing to build fossil fuel generators onsite for year-round or constant energy production, so proposals like the type we've seen from Cloud Carrier in the Southern Highlands should be rejected. We don't want to see data centres derail the renewable energy transition or extend coal-fired power, as mentioned. Data Centres Australia has indicated that around 70 per cent, they claim, of data centre energy consumption is offset currently via PPAs or large-scale generation certificates. But I think this probably includes a mix of projects that are post-financial close, which goes to the point that you made. If agreements are signed with renewables projects post-financial close then it's arguable as to whether the customer, the data centre, is actually pushing forward the development of new renewable energy projects into the grid. I do think those suggestions do need to be interrogated quite closely and that we do need to have a coordinated plan around what this data centre role might be within New South Wales. That needs to be coordinated and involve community consultation.

ALEXANDER HOYSTED: I think if the goal is specifically to have this new demand—and I think we have about 3.5 gigawatts that's expected to materialise on the grid between now and 2035—it's going to be extremely difficult, based on the amount of new renewable energy that data centres are bringing onto the grid, for them to meet that. I think that it's worth noting there are a couple of areas that the industry has pointed out that make it difficult for them to bring new renewable energy online—specifically, contracting PPAs. I think this actually speaks to issues in the energy system where we're trying to get new renewable energy development up. The feedback that I've had from the renewable energy industry is that that's quite difficult.

There's also the issue with generators as well. The fact that most data centres use diesel generators as backup fuel—which they absolutely need—is a problem, especially in this environment, with the Iran war going on. But the issues that the data centre industry themselves outline around their backup fuels don't preclude renewable technology being used as well. A lot of that comes down to the amount of space onsite for using best systems as renewable energy backup. There's a number of different angles that we might approach this issue from, but if you're looking at it straight from are they going to be able to meet their demand with new renewable energy, there are barriers to achieving that.

The CHAIR: In your submission you talk about this claim that the industry meets 70 per cent of its energy needs through renewables and PPAs. But the CZI analysis of public reporting documents puts the actual PPA contribution closer to 20 per cent to 30 per cent, with 40 per cent to 50 per cent renewables. Can you tell the Committee how you arrived at that breakdown, and explain why it matters?

ALEXANDER HOYSTED: I should distinguish between what DCA claims, which is that the industry offsets their renewable energies with 70 per cent PPAs. We looked at the operator's—NEXTDC—sustainability report. They had 20 per cent to 30 per cent of their energy demand covered by PPAs, and 40 per cent to 50 per cent by new renewable energy generation—that's solar panels and the like on their facilities. The problem is that that doesn't add new renewable energy to the actual grid, and we need more PPAs underwritten. We need more renewable energy projects coming to financial close because of agreements like this. I don't know whether that was because there aren't enough viable projects for them to underwrite. I know that it's been difficult for them to have financial confidence in closing some of these. It speaks to a temporal issue that we have in the system, where PPAs are expected to come online a few years into the future, but really, we need that new renewable energy

CORRECTED

generation coming on right now. We think that distinguishing where the actual green electrons are feeding into data centre facilities is really important.

The CHAIR: Ms Mills, you mentioned that Cloud Carrier project. The Lock the Gate submission claims that around 74 per cent of the anticipated annual production from the Narrabri Gas Project would be the amount that that new project would use in the Southern Highlands. What does that say to you about our stated intentions of moving away from fossil fuels in New South Wales?

JACQUELINE MILLS: I think it's quite clear, Chair, that that would seem to be moving in the wrong direction for the energy system in New South Wales. It's very concerning to read that or to hear that in relation to the Narrabri Gas Project, because what we need to be doing in New South Wales is weaning ourselves off gas. We don't need to be opening up new forms of gas supply in the form of Narrabri. As mentioned, that particular proposal—I understand Cloud Carrier put in a first proposal seeking to build this gas plant. They've actually got knocked back in the local council. They're challenging that in the Land and Environment Court. But now they've said—I think just within the last week or so—that they want to beef up that plant to over 600 megawatts, which is huge, as you mentioned. It's very much heading in the wrong direction. I think one of the ironies is that that particular site was knocked back for a proposed coal-fired power plant as well. For the community in that area, it feels like it might be an ongoing battle to try to address these issues. I think what we need to see is data centres that are powered by renewable energy, not fossil fuels.

The CHAIR: The other big issue we've been grappling with here is how we bring the community on board, and adequately consult in relation to each particular data centre. One of the problems has been that we don't really know what each of those data centres are for, so to be able to judge whether or not it is worth the cost in terms of energy and climate and water is very hard for the community to assess. That particular example is of people who have been opposed to the Narrabri Gas Project for so long. The idea that three-quarters of that could be going directly just to power a data centre and we don't know what the data centre is actually doing, how do we crystallise that for people in a way where we can say, "This amount of data centre usage is critical for things like health information and government information"? But we don't want to destroy huge amounts of our country and open up new gas projects just for the generation of cat videos or Trump acting like Jesus. Do you have any reflections on that aspect?

SIMON BRADSHAW: First of all, the example you've highlighted relating to Narrabri does show that we are genuinely at risk of making exactly the same mistakes that we have seen play out in other jurisdictions, notably the US. We know that's the extreme example of a gas-powered AI boom. But also in the UK we see the same problem that has come through rushing and through not taking that pause to intentionally design future infrastructure, to have the appropriate community consultation and to have an appropriately sized and scaled industry. We have the opportunity now to get that right here in Australia.

To the specific question about how we move forward with appropriate community consultation and some transparency on what these centres are used for, first of all, I'd reinforce that communities have every right to ask that question of what is the social benefit from some of these hyperscale AI data centres? We're in no risk of compromising our ability to provide these essential services. We all agree that's important. That has been possible for many years. What we're talking about now is much larger data centres, specifically to power AI, with questionable social benefits—that is, all the well-known potential harms we're aware of these days, including everything from compromising our cognitive abilities through to the much more egregious things around generation of abusive images and assistance in war fighting.

We know there has been some pushback from the industry in saying, "We cannot disaggregate and show what these different uses of computer are for." We've got to be very sceptical about that. We've seen what big tech can do. They can deliver us a targeted ad within seconds of us viewing a particular article. The idea that something as rudimentary as distinguishing what different uses of this computer are for so we can, as you've pointed out, distinguish and have some agency over what we want support and what we don't—I think that's just nonsense. We really have to insist on better transparency there, and put the onus on the industry to be providing that and be designing a good regulatory regime that's going to demand it so we can make informed choices going forwards.

ALEXANDER HOYSTED: It's a good question, Chair. I think part of the issue that the industry sees here is that, if they're not able to take a customer who has a workload that they want to put in one of their facilities, then they're just going to go elsewhere. I think that there is some truth to that. But I think where we're arriving in the national conversation is the recognition that the use case matters. We really do need to start distinguishing between what's a useful workload and what's something that's maybe a little bit more speculative. I think what we're going to see in the next couple of years is that the social licence issue is going to become much more tricky.

I know that industry operators are already looking at the peri-urban regional area to maybe put AI training workloads or inference workloads a little bit further away from city centres. The feedback that I've had from folks

CORRECTED

who work on this specific social licensing issue around renewable energy has assured me that data centres will be seen as basically an extension of the renewable energy industry if they do choose to locate in those areas. You're also seeing in Melbourne some of the issues around being located close to one of these facilities and things like noise that affect people's health and their ability to sleep at night. I think they're going to become much more material in the next couple of years. We need to get better at distinguishing those different workloads, and we need to be aware that these issues are going to arise in the next couple of years.

JACQUELINE MILLS: I agree that we need transparency on what data centres are to be used for and to enable communities to have a say as to weighing up the pros and cons and the benefits, particularly if they're to be located near them. Alex has just mentioned some of those impacts that we see from data centres—heat island effects, noise, water use at vast scale. I think that there needs to be a robust planning system that actually comes up with a plan around what is needed to ensure that cumulative impacts of data centres are addressed and that we're not just looking at a project-by-project basis, but there's essentially a plan around where data centres could be sited in New South Wales. Then there needs to be a process around engaging communities and identifying the benefits that they would want to see. We know renewable energy companies need to work to define and come up with community benefits. There's no reason why data centres should not undertake a similar process.

The Hon. JACQUI MUNRO: Thank you for coming today. My first question is related to this idea of social licence and the lessons that have been learnt through the renewable energy rollout. I was wondering if I could hear from each of you any lessons that can be applied to the data centre conversation in relation to social licence and rollout.

SOLAYE SNIDER: A key lesson is that social licence is really important and that comes from really genuine, careful consultation with communities who are going to be most impacted by where these data centres could be located. I also think social licence can't just be bought; it needs to be earned. If there are serious, concerning impacts—which there are—from all of these data centre proposals, there is going to be a lot of community pushback. That's what we're seeing happen play out in other parts of the world already. In the US, in particular, I think there's over a hundred local groups already organising to try to block data centre development in their area. There have already been many data centres blocked by community opposition in the US, and we're seeing that starting to happen in Australia as well. There was a recent proposal in Perth that was rejected because the community was really opposed to it for very good reason.

The Hon. JACQUI MUNRO: That's also been occurring through the renewable energy rollout—local communities have activated against the new infrastructure, whatever development is around, whether it's farms or solar farms. That sense that local communities are losing their character or their power—have there been good examples that you can point to in the renewable energy rollout where perhaps there's been a particular model of engagement that has allowed community input to shape a project effectively, or developed a community benefit sharing scheme that was useful for that particular area? Or is it better to have a statewide approach that is more top down? Could you comment on that, Dr Bradshaw?

SIMON BRADSHAW: It's absolutely a combination of both. It's an excellent question because we could argue that challenges around social licence have become one of the biggest barriers to achieving our renewable energy goals. I think a really rushed, poorly designed data centre rollout could exacerbate that massively. We do, obviously, need appropriate State and Federal regulations and enforceable guidelines, not just expectations around what data centre development needs to look like. But, crucially, yes, we need as much agency for local communities as possible over their economic future, be that around renewable energy developments and particularly around data centres because we don't necessarily, by default, see much or any benefit flowing directly to the local communities. We do see that, for every dollar invested into data centres, a lot of that straightaway goes offshore, and that doesn't necessarily go into the local community who then, of course, are forced to bear the cost through increased pollution, degradation of their landscapes, stress on the energy and water supply. It's particularly important in this case that we have it rightly paced and as much agency to local communities in making decisions about what they want in their area as possible.

The Hon. JACQUI MUNRO: Just quickly, to push back on that slightly, there is a benefit to grid stabilisation in having these larger power users that are consistent in their use. That is particularly important as our grid becomes "omni-generated", if I can put it that way. I would say there is some benefit, particularly related to the way that energy generation and the grid will operate in the future.

SIMON BRADSHAW: Potentially, yes. But the key here is that we do it with intentional design, at the right pace and in a way that we can realise those benefits. We do see risks to that if we rush into this, and if we go down this road of enormous hyperscale data centres that, in fact, create enormous additional challenges for our energy future. It comes back again to that need to pause, think this through, weigh up the costs and benefits, including the wider social implications of these sort of developments, and take time to get it right.

CORRECTED

JACQUELINE MILLS: I will jump in there, in relation to your question on social licence in regional communities. It's obviously in the data centre industry's interests to prioritise social licence, but I'm not yet convinced, from what I've heard from Data Centres Australia, that they understand that and they see the benefits of that. You asked about examples from the renewable energy transition where there are good examples of communities being engaged and benefits flowing. What springs to mind is the Hay Shire Council in south-west New South Wales. My understanding is that they acted early in the process of renewable energy developers having interest in their region. They brought together the community, key agencies and key stakeholders, and they defined for themselves the benefits that they wanted to see for themselves in the near term and into the longer term.

They were very much empowered when negotiating with individual developers from that point. I think they outlined, for example, that, with the temporary worker accommodation from the renewable energy projects, they would want to see that repurposed into aged-care accommodation so that there's a prevailing benefit at the end there. But I think some of the characteristics of that are that early engagement, the identification of and working with those key stakeholders, and ensuring that different dimensions of the issue were brought to the fore early.

ALEXANDER HOYSTED: In Carbon Zero's other work, we've worked with landholders who host renewable energy and we've seen what good looks like. From our experience, that's meant early engagement, often from the hosts themselves. We do talk about intentional design—and that's important—but it also always comes from the community as well. That's something that we've noted. I think there's an erroneous assumption as well that regional communities are always anti-investment. We've worked with regional economic organisations that are these leaders, and people who work in their economic organisations are really keen to attract investment. They're trying to understand what to do with this infrastructure that's landed on their doorstep.

The sense that we've got is that there is a cautious interest, maybe, in attracting this investment. But, as our colleagues have said, the gap is really in understanding the role that data centres play in the economy and in the digital world as well. I think in that sense there's still a lot of questions left to be answered. What exactly are we hosting here? What's being produced in this warehouse? How is this facility bringing down network costs or reducing burdens on transmission? One other thing on that is that the feedback that we've got from the renewable energy industry and from folks out in regional areas is that endless consultation is not the answer. They really need to be heard and decisive action by industry very early on. I'm certain that the data centre industry will be very keen to learn the lessons of the renewables rollout.

The Hon. JACQUI MUNRO: Are you surprised that there is not already a data centre strategy, given we already had 90 data centres in New South Wales before this boom started? There's been a conversation about AI, data and data centres for at least a couple of years now—let's say 18 months, to be reasonable. Is there some surprise that there hasn't been something formal put out in the world from the New South Wales Government?

SOLAYE SNIDER: Yes, it definitely seems like an oversight and that everyone is scrambling to catch up. It's good that we're having this conversation today and that the issue is finally being dug into properly, but that's exactly why we think we need to be taking a bit more time before considering fast-tracking especially a lot of these enormous proposals before a proper framework is in place. There have been some encouraging signs. We know that the Federal Government has put out some expectations for data centres recently. But for us the key word there is "expectations". We think there really needs to be actually legislated regulation and that it's enforceable before these proposals are being approved, because we simply can't trust the industry to do all these things that they're actively resisting, if not actually forced to do that. We don't think expectations, advice or suggestions will be sufficient, and we really want to take the time to get this right.

SIMON BRADSHAW: I think, yes, we've been surprised. Certainly I think it's not too late to get this right. In our submissions there are many answers to what's needed here. Obviously, as we've all highlighted, that's greater transparency about what this massive amount of new compute is actually being used for, and what we should be supporting and what we shouldn't; a far greater onus to actually be showing real and additional renewable capacity beyond just what's needed to meet the additional demand; and, obviously, a strategy that is properly integrating these into our urban and regional environments in ways that leave communities stronger and are not opening up a whole new world of pain for communities that can really afford it least.

JACQUELINE MILLS: I think it's quite clear that we need a plan to manage these data centres for the public good. I think it's not sufficient just to cater to the prerogative of the data centres themselves. I think that opportunity is there, and those developments that Solaye and Simon refer to are very good, but I think the clock is ticking as well. There is a need to think about the cumulative impacts of data centres alongside other forms of industrial and land use across the State and to ensure that this rollout happens in a much more managed way than what we've seen before.

CORRECTED

ALEXANDER HOYSTED: I want to give the Government a little bit of credit here. It was only in August last year that Scott Farquhar from Atlassian was talking about Australia's potential to be a data centre and renewable powerhouse, so—

The Hon. JACQUI MUNRO: But there were many—I was talking about data centres in the House at least a year ago. Scott Farquhar is important in this field, but let's be real: There were many discussions and prompts about the importance of data centres.

The Hon. MARK BUTTIGIEG: You asked a question and he is answering it.

ALEXANDER HOYSTED: Absolutely. I completely agree. I mean in the sense that the investment that's arrived on the scene has really—it's a little bit like a sneeze. It starts slowly and then, all of a sudden, these issues burst onto the scene very quickly. Would it have been good to get our hands around this a little bit earlier? I absolutely agree with that. I think that what this shows is that, as you put it, as our energy system decentralises and new demands like this come online, we need to start being aware of the different effects that different industry has on the energy system and be proactive about that. If anything, the data centre industry is going to be a case study, because I'm willing to bet that this will happen again in the future.

The Hon. JACQUI MUNRO: For the record, I'm pretty sure that the Treasurer and the Premier were standing up with the founder of an Australian data centre at this time last year to extol the virtues.

ALEXANDER HOYSTED: I think I know the company you're talking about.

The Hon. JACQUI MUNRO: I'm aware that other people might want to ask a question, but I do just want to ask one thing: Is there a set of data that we should be relying on for energy and water use, in your minds? One of the things we've heard is that, basically, it's very difficult to find accurate predictions or data that show usage.

ALEXANDER HOYSTED: I'm happy to take that to begin with. Probably a good place to start would be aggregated PUE—power usage efficiency and water usage efficiency metrics. As you point out, they're quite fragmented at the moment, and there's a whole bunch of different reporting regimes that data centres have to go through. We don't have any broad sustainability metrics that the industry has to report to. That would be a really good start. They point out that there are different technologies, and different facilities have different energy and water requirements. That's true, but I think an aggregated measure would be a good place to start.

SOLAYE SNIDER: I think it's also worth noting that, right now, a lot of it is projections of where this will go and what the data centre frenzy will do to energy demand. AEMO recently released new figures, and they had to revise their projections and increase them again. Even the previous projection was, in the highest scenario, data centres could use up to 13 per cent of the demand by 2040. That's a huge amount. You know, it's becoming its own sector entirely, and what we're seeing in other parts of the world is that those projections are just continuously revised and they're growing, so we just need to be really mindful of the fact that we don't have all that data right now. But if we look at how it's going in other parts of the world, it's just going to keep growing and growing unless we do something about it.

The Hon. MARK BUTTIGIEG: I think the evidence is interesting, and the caution and all the kind of talk that we've got to plan it properly and look forward to the future, but from a practical perspective, let's get real here. The Government is not going to let this stuff pass by. They're very keen to grow this industry and it seems an opportunity—and a competitive race in a lot of ways, to be frank. Your evidence is that there is a gap between our ability to provide this energy-hungry growing industry with renewables and, in the meantime, that could obviously attenuate or retard our progress towards net zero and renewables, and all the rest of it. What would you do, as a government, in terms of legislation, regulation and stimulating investment in renewables? What would you do to fill that gap? That's the crux of the question, isn't it? We want these data centres. You're not going to stop the demand for them. You're not going to stop the progress of them, but ideally what we want to do is supply them with renewables as quickly as possible. How do we do it?

ALEXANDER HOYSTED: I think probably the best place to start is what the data centre industry cares about because we have to work with the incentives that they have, if this is unstoppable. What they care about most is time to power. By "time to power", I mean how quickly they can be up and servicing a customer with one of their facilities. My suggestion, or our suggestion, is that when we're developing guidelines around how much new renewable energy we want them to bring on, what sorts of technologies they use to power their facilities, we're able to use that "time to power" metric as a guide for good behaviour. We're telling the industry from a government perspective what good looks like, what we want as a State, and then rewarding operators that are leading from the front because then they are also incentivised and it can help their bottom line as well.

CORRECTED

JACQUELINE MILLS: I'm reminded of comments that the New South Wales Treasurer made, I think earlier this year, that there was a good level of interest in New South Wales specifically for data centre investment, and that put New South Wales in a position to set the highest possible standards and really ensure that we attract the best players to New South Wales. I think we've covered some of the points that we would see to be really important, but certainly there needs to be regulations to ensure they bring additional energy into the system. That should happen. Requiring them to have PPAs that are post-financial investment decisions to ensure that it's actually genuinely additional, that needs to be a basic regulation. That should be achievable as well and have the effect of bringing new renewable energy into the system.

For example, if a renewable energy generator is signing an agreement with a data centre—so for the renewable energy generator it might be for 70 per cent of their power that they intend to generate when they get going, and they intend to sell 30 per cent into the spot market. Signing that PPA with the data centre as a load could support them getting over the line and getting that project up and running. I think, in that way, there could be a valuable contribution. It has to, of course, constitute 100 per cent of the data centre's needs, but in that way we might see genuine additionality and we might see a speeding up of renewable energy transition.

SIMON BRADSHAW: To make a more general point, what I'd be saying to the Government is take the opportunity to do this right and on our terms, because the world is on a decarbonisation journey. I know there's some bumps, but that's not going to change. There will be demand in future for genuinely zero carbon, responsible compute power, and we have the opportunity at this stage to still set ourselves down that path and to—yes, if the Government is determined to—ride this wave in the right way, but not to be repeating the mistakes we've seen in other parts of the world, not forgoing the opportunity but really doing it on our own terms, taking the same approach as we'd want to take to new green export industries and really using our advantage to do that in a responsible, zero carbon way, confident that there will be appetite and demand for that.

The Hon. MARK BUTTIGIEG: Is there an exemplar jurisdiction? We mentioned the bad example of America. Is there an exemplar jurisdiction where they're doing this?

SIMON BRADSHAW: As we detail in our submission, we do see other jurisdictions taking time to pause at least the largest, most energy intensive data centres in order to get their strategies and regulations right. I don't know if that's quite the answer you're looking for. I suspect the full answer is "No, not yet, because this has happened so quickly." But we do at least see examples of jurisdictions not going headlong down into this and getting themselves into all sorts of trouble, but taking a pause—very often because their local community has demanded it—and giving themselves a bit of time to develop a proper strategy and to ensure that future development is appropriately integrated into their energy transition.

SOLAYE SNIDER: I'd just add that I think one example that people are looking to is the EU, who has been attempting to implement more regulation around transparency and disclosures in particular. I think there's still, unfortunately, some loopholes in how they've actually created that legislation, which means it's not necessarily as effective as it could be, but I think there's some lessons to learn there. They have a lot more, for example, than Australia has, so far.

The Hon. SCOTT FARLOW: This week the Treasurer outlined that it was renewables and data centres that were holding us from a recession and were the only thing that stood between New South Wales and a recession. Following on from Mr Buttigieg's question, if there were to be constraints placed on data centre development, do you think that it would bear out that we would be heading to a recession in New South Wales?

SOLAYE SNIDER: I mean, we're not economists, so I won't pretend to know if I can predict there'll be a recession. It does seem like an extreme statement, but I think I'd just go back to the points we made earlier about this investment and how much it's actually benefiting Australian communities. Like we said, I think it was a *Sydney Morning Herald* article that detailed that for every \$100 coming in, \$70 to \$80 is immediately going overseas again and benefiting other companies. I'm not really sure where it comes from that data centres are driving all of the economic growth in New South Wales. I just don't think that's the reality.

JACQUELINE MILLS: I go back to my earlier mention of the comment by the New South Wales Treasurer that there's a lot of interest. New South Wales is highly competitive for data centres, which would leave one with the impression that it's quite possible to regulate and put some guardrails around that interest in a way that will still allow for solid economic growth.

ALEXANDER HOYSTED: I don't have a strong opinion about whether the economy would be in a recession or not. I'm a little bit out of my depth there. But what I would say about the digital economy more broadly is that this isn't something that's going away. If you have a look at people's data consumption, year over year, it's going up 30 per cent. We do need to build facilities and have an economy that recognises that reality. This is a really big source of consumption in day-to-day life. When we transfer that to AI, a colleague of mine

CORRECTED

had a really interesting perspective on this. If Australia does want to be at the forefront of AI safety and regulation internationally—because this is a global economy, and we're sending signals back to the United States from data centres all the time—then we do want to actually build out a national capacity in AI as well. We want to be a significant exporter of AI tokens if that turns out to be the case. The only thing I'd say to that is that the digital economy will probably play a really important part in our growth in the future, and that wouldn't surprise me.

The Hon. SCOTT FARLOW: One of the other things that has been raised through this inquiry, probably from both ends, is, firstly, who cares if we've got data centres here? You can have them overseas; you can have them somewhere else. They don't have to be here. Secondly, the counterpoint to that is you increase regulation here and then we will send them offshore. From your perspective, particularly considering the nature of this panel and the environmental perspective, is it important for us to have them domiciled here or is it better, in a sense, to have them domiciled overseas?

SOLAYE SNIDER: From a climate and environment perspective, there's a lot of harm that we would avoid by not having some of these massive new data centres in Australia. As we've outlined, they are using a huge amount of energy. They could prolong the use of fossil fuels. They could delay our energy transition. They're sucking up huge amounts of water. There's the heat impacts that people have touched on and the noise. There's an opportunity cost, and we would avoid a lot of those harms by not having the data centres physically located in Australia. I know some people may say that's not realistic but, if you just take it at face value, of course, if they could not exist, we would avoid all of that harm.

The Hon. SCOTT FARLOW: But if they still existed but existed in Malaysia, would that avoid the harm? Is the regulatory environment any better over there or the energy landscape or water environment?

SOLAYE SNIDER: I think we're focused on what's in the remit of the New South Wales Government or the Federal Government here. If we're thinking about our own regulations and how to meet our own emissions reduction targets and how to keep our own communities safe, I think that's what's in the remit of this Government to consider.

The Hon. MARK BUTTIGIEG: It's actually a very pertinent point, isn't it, in the context of your previous evidence. If you put up the shutters, to use a blunt term, and say that we've got to slow things down and get this right, and then these things go offshore and they've actually got a lower regulatory regime for renewables, it sort of defeats the purpose from a global perspective, doesn't it?

SIMON BRADSHAW: I think we've got to be a bit careful with this argument. When we're talking to the fossil fuel industry, we call this the drug dealer's defence: the idea that if we don't do it, somebody else will. It's an overly simplistic argument that probably underestimates two things: firstly, our ability here to still be able to do this right and, secondly, the idea that if we don't rush headlong into this, the opportunity is going to be taken up somewhere else. I think what we're actually seeing from that sort of mentality is other jurisdictions getting themselves into all sorts of difficulties in terms of their energy journey, in terms of community opposition, when this is not going awry. I do understand the question, but I think what we need to focus on, as my colleagues outlined, is getting it right here and being confident that there is a longer term journey here. We don't want to repeat the mistakes of elsewhere. We want to do this on our own terms.

The CHAIR: I'm smiling because, listening to those two arguments—the one that otherwise the industry would go elsewhere and the one that, somehow, if we don't allow it here, it will be done in a less clean way somewhere else—they are the exact arguments the fossil fuel industry have used for decades. What I'm hearing from you is a very strong message of let's learn the lessons from the past and actually be a bit more confident in what we have to offer in Australia by setting these guardrails and saying, "Yes, you can come here but, if you come here, these are the things that you need to do," and setting that high standard. Is that how you would sum it up?

SIMON BRADSHAW: Absolutely, Chair.

The CHAIR: I was going to do commentary on what the Treasurer said, but I won't. I will resist.

The Hon. SCOTT FARLOW: Ask it in a question on notice.

The CHAIR: That's right. I will do a comment disguised as a question. I will call it at that. Thank you very much. I could talk to you all for hours and hours, but that does bring us to the end of the session. To the extent there are questions taken on notice, or I know I already have supplementary questions for you, the Committee secretariat will be in touch about answering those, but that does conclude our time with you. Thank you.

(The witnesses withdrew.)

CORRECTED

Mr JASON KRSTANOSKI, Executive General Manager of Network, Transgrid, sworn and examined

Ms MARYANNE GRAHAM, , Executive General Manager Corporate Affairs, Stakeholder Engagement and Environment, Transgrid, sworn and examined

The CHAIR: I welcome our next witnesses. Would you like to commence with a short opening statement?

JASON KRSTANOSKI: Yes, please. Thank you for the opportunity to appear before the Committee today. I have worked in the electricity industry for more than 20 years, beginning my career as an apprentice cable joiner, later qualifying as an electrical engineer, and now as a senior leader. Over that time my experience has spanned maintenance, operations, planning, commercial and innovation roles across both electricity transmission and distribution networks. Throughout my career, my consistent focus has been ensuring electricity networks deliver consumers safe, secure, reliable and affordable electricity supply across New South Wales and the ACT.

We've brought that same perspective into Transgrid's approach to data centre connections. Data centres are fast becoming critical economic infrastructure underpinning the digital services that drive modern economies. New South Wales is well placed to capture a significant share of this investment provided policy settings are clear, timely and investment ready. At the same time, the scale, pace and geographic concentration of proposed developments represent a material step change in electricity demand. Existing regulatory arrangements were not designed for this level of large, clustered load growth. This creates an opportunity to thoughtfully evolve those frameworks to ensure they continue to support efficient investment while maintaining fairness for existing consumers and the integrity of the power system. Transgrid would encourage policymakers to continue pursuing reforms that strike the right balance between protecting consumers and maintaining the safety, security and reliability of the power system while enabling new investment in a practical and efficient way.

Our perspective is guided by three core principles. First, existing consumers should be no worse off. New large loads should fund the infrastructure required to supply them and contribute appropriately to system reliability. Households and businesses should not bear the cost or risk of that growth. Secondly, policy settings should support investment and economic development. New South Wales should remain an attractive destination for global data centre investment with predictable, transparent and investment ready settings that allow network capacity to be unlocked efficiently.

Thirdly, reforms should be practical and timely. This is a globally competitive opportunity and policy changes should build on existing frameworks and approaches, rather than introduce unnecessary complexity or delay. Consistent with these principles, Transgrid sees merit in targeted reforms to strengthen commitment signals for large loads, ensure investment remains cost reflective and require large users like data centres to contribute to system reliability as our power system transitions from coal to renewables. Taken together, these measures would provide a clearer, more predictable framework for connecting large electricity loads, one that protects consumers, supports investment and advances New South Wales's broader economic and energy transition objectives. Thank you again for the opportunity to appear today. We welcome the Committee's questions and the opportunity to assist you with your deliberations.

The CHAIR: Your submission says that Transgrid has received connection inquiries from data centres totalling more than 10 gigawatts, with approximately six gigawatts progressing to formal applications. Data Centres Australia and the Business Council of Australia have both cited Oxford Economics phantom demand research, which estimates that six in seven megawatts of connection requests won't materialise. What is Transgrid's best estimate of how much actual electricity demand from data centres will materialise on your network over the next five and 10 years?

JASON KRSTANOSKI: What we do know is that the 10 gigawatts I mentioned is the inquiry stage. That's very early on. That's making applications, starting the conversation. The next six gigawatts is when it's progressed to a serious point, where people are actually starting to pay some money, putting their applications through and they're actually progressing through some technical assessments. That's where that's at. We are very close but we haven't had a data centre actually commit yet. We're on the eve of that now. It is hard to predict because you do get these phantom requests across the distribution, transmission and other States as well. What we will do is we will make sure that these new connections don't compromise safety, security and reliability of the network.

The CHAIR: If much of the demand is speculative and you do have people who are putting proposals that then don't eventuate, what is the cost to Transgrid—and, ultimately, to consumers—of conducting those detailed technical assessments for projects that don't end up happening?

JASON KRSTANOSKI: There's a couple of parts to that. The proponents do pay an application fee to go through that connection process. They do pay for that, which is roughly cost reflective, so there's that. That's

CORRECTED

not the 10 gigawatts; this is once they've gone through—that's the six gigawatt figure that we were talking about. They do pay a fee for that, so that funds that part of the process. What we are doing to ensure that we don't have what we call that phantom demand is we are working very closely, more so than ever, with distributors and other parts of government—with DCCEEW, Infrastructure NSW—and with data centre proponents as well to collaborate to make sure that we are sifting out phantom load as best we can. There's been a lot of effort in that space.

The CHAIR: When you undertake that network augmentation to enable a data centre connection, how are the costs of that augmentation allocated?

JASON KRSTANOSKI: Again, there are probably two parts to this. Something you'll hear data centres say is that they'll pay for their connection, and that's true. If someone wants to connect onto the network, they will pay for their connection. Something that we're advocating for at Transgrid is based on our principle that consumers should be no worse off. We think that the data centres should pay for all the augmentations that they trigger. It might not be as evident as what is in front of the front fence where you connect. For example, if there's a data centre that wants to connect in location A, but to enable that capacity to be released it means you've got to put a better plant 200 kilometres away at another location, we think that the data centres should also pay for that augmentation. If they didn't, then that would be part of what we call the shared network and that cost would be spread across all the existing consumers.

The CHAIR: So that then we'll get socialised away from the data centres. It's true, isn't it, that if data centres are directly transmission connected, they're not part of the same recovery pools like your average retail customer? Effectively, they're not even paying their fair share of that augmentation cost that's going to the network more broadly?

JASON KRSTANOSKI: Yes, let me elaborate a little bit. I should probably clarify first that we don't have a data centre connected onto our network as yet—on transmission in New South Wales and Transgrid. At the moment there are a number of small ones connected on distribution networks, but at the moment we don't have one, so I can tell you what we're planning to do and how we're going through it. You're correct: They will pay for their direct connection and that associated fee, which is part of their work. Any other augmentations that happen on the network form part of what we call the user shared network, which gets spread across transmission-connected customers, being the distribution companies. There are some other large-load customers—not data centres, but we do have some other large-load customers connected to the network. They are treated differently to a household on the distribution network.

The Hon. MARK BUTTIGIEG: I want to get a handle, Mr Krstanoski, on where Transgrid fits into this jigsaw puzzle. You've got these data centres—what would be your standard plug-in voltage for a data centre? You guys do down to 132 kV—

JASON KRSTANOSKI: Yes.

The Hon. MARK BUTTIGIEG: —and I just wasn't sure what the adaptability of these data centres was to the voltage, given you're in that transmission space.

JASON KRSTANOSKI: Great question, Mr Buttigieg. It's good to clarify that. We have seen data centres apply for connections at 132 kV, 330 kV and 500 kV—so all voltages. We have 220 kV in the south as well, but we've seen them apply for wherever they can get access. The voltage hasn't been something that swings them. What we see is a—

The Hon. MARK BUTTIGIEG: Just to follow up, Jason, that's really a function of the economics of the geographical location, then they do their own step-down to whatever voltage they need to do, and that's not an issue.

JASON KRSTANOSKI: Yes, it's quite extreme. The steps you can see now are 500 to 33 and all sorts of things. It's quite big steps that we've been seeing. One of the advantages we see with them coming onto the transmission network—we see an advantage. There are technical advantages, but something we're passionate about is the consumer advantages. Transgrid operates under a revenue cap, which has to proportionate costs across customers. When you have an existing network like we already do now and you start to add some of these large loads—these data centre loads—to it, it means that more of the existing cost under the cap can be proportioned to them, and it can provide some relief and benefit to your household consumers. So there can be a good consumer outcome by enabling these connections onto the transmission network.

The Hon. MARK BUTTIGIEG: As opposed to the distribution network.

JASON KRSTANOSKI: There are some nuances to that. It is a little bit different with distribution. Speaking for transmission, the way our revenue model works, it's a fixed cap. This would proportion more costs

CORRECTED

to them as larger users, and it would give relief through the distribution network to consumers that way for the transmission component.

The Hon. MARK BUTTIGIEG: Is it more efficient, from a distribution-loss perspective, to plug in at those higher voltages?

JASON KRSTANOSKI: I know you know this, Mr Buttigieg, but when you are transporting power, the higher the voltage, the more efficient—generally, as a bit of a principle. But the thing to think about is probably having the demand closer to the source is really the trick. Then you've got less distance and less losses no matter what the voltage. But you're right: There is more efficiency at higher voltage. There are economics to that as well. When you're paying for bigger infrastructure, as in high voltages, there are different costs to lower and that sort of thing. But, generally, there are less losses at higher voltages and, as a principle, you'd normally want your demand closer to your source as well.

The Hon. MARK BUTTIGIEG: What's Transgrid's planning on projections given—I think in previous evidence we've heard that there's a bit of a gravitation towards the metropolitan centres, where all of the demand for computing is. Therefore—it's a little bit counterintuitive, I guess—what's Transgrid's view on the evolving nature of the market and how much demand there is going to be for people to plug into the transmission network? I think you said previously that you haven't had anyone connect yet, but there are things in the pipeline, are there?

JASON KRSTANOSKI: Absolutely. All the pipeline is generally Western Sydney. For us, ideally, we would like them to go where there is current capacity because then we can make the most of that, as long it doesn't compromise our principles of consumer no worse off and we do practical things. Again, it is on a foundation of things such as safe, secure and reliable, and making sure that we have good affordability for consumers. For us, the customers, or the data centres, are telling us that they want to be in Western Sydney. There is some capacity in different parts of the network. If they were to go there, that would be great, but they've got their complications with water and fibre and all those other things. But, yes, the strong concentration is in that area, so we're trying to meet that. We will be able to connect some data centres there because we have some—

The Hon. MARK BUTTIGIEG: Is that a function of land price, is it?

JASON KRSTANOSKI: I can't speak for the data centres there. From what I hear, it's a combination of fibre and all those things, but what we're seeing is the requests are for that area. That is their strong push.

The CHAIR: From an economic perspective, or from Transgrid's perspective, what we've heard is that not all data centres need to be metro. A lot of them for certain applications can be very happily out in regional areas, particularly in renewable energy zones. What happens from a practical perspective when you're getting applications? Are you able to say, "Look, if you have another one in Western Sydney, we're going to struggle, but it will be much quicker if you could do it over here." How does that then factor back into the decisions from the data centre company?

JASON KRSTANOSKI: You're absolutely right. That's exactly how it works. When the data centres talk to us, they might say, "Yes, we want to be in the Western Sydney." We'll say, "Okay, we'll process your application", because we're required to do so in a fair and equitable way, which is under the National Electricity Rules so we have to follow that process with the applications. There is some capacity in that western area which is really popular. When I look at how many applications are coming through, I know that we can't meet them all, so there'll be some people disappointed until augmentations happen. Some will get in, and then we would happily encourage them—and we are already—to go to other regional areas to be close to renewables, absolutely. Again, as a principle, if you're close to the source, there are less losses, it's more practical and you don't need all the network in between the supply and the demand. So, practically, that would be good. Not speaking for data centres, but what we hear from them is that, yes, they're looking at land, fibre and water. We're challenging them always to think about the community impacts and the social licence. That's really important to us. So that's where we're pushing them.

The CHAIR: When I asked that question before when we talked about the phantom demand and we talked about how much demand you are anticipating or making assumptions about in order to then make your plans in the next five or 10 years, should the question really have been on a per area basis so that you're looking at 85 per cent of the eventualising in Western Sydney, but not necessarily in other areas. How does that planning work from your perspective?

JASON KRSTANOSKI: Number one for us is to make sure that the network operates safely and then securely. What we mean by securely is you can't have one data centre turn off abruptly and the whole system wobbles and there is a cascading impact. For us, we have those core principles to make sure we're good stewards at operating the network. With that, it means we have to limit how many can be clustered in one area. With that, we know that we will get some in that area, but—I'm sorry, Chair, could you restate the last part of that question?

CORRECTED

The CHAIR: Yes. I'm trying to get to grips with how much the data centre industry is making a lot of claims about, "Oh, yes, it's different, a phantom demand". How does that come into your particular planning?

JASON KRSTANOSKI: We wouldn't invest in actually building anything unless we have full commitment from the data centre, as opposed to other non-regulated where you could invest and build something and wait for them to build it and they will come. We're not like that. We need the full commitment to make sure that it's actually going ahead and they're funding their connection works, and then we can start to proceed. We shouldn't get ahead or put the cart before the horse in that respect. But with planning, what's been really challenging over the last little while with data centres is normally, on a transition network you have small incremental demand grow over a long period of time, and we can have good planning for 10 years-plus. We can have plans for 30 years ahead that are a bit more looser, but we can have strong plans for 10 years ahead, which are with AEMO and the DNSPs and that sort of thing. The step change in demand we've seen with data centre requests has been a challenge. That's what we're collaborating with government, AEMO and other distributors to manage. But we're just going to stick to our principles where we make sure that we don't compromise the network and so we only will connect what's safe, secure and reliable.

The Hon. PETER PRIMROSE: I heard you talking earlier about potential benefits of the connections, and I missed a bit of that. Can you just elucidate on that? What actually are the potential benefits?

JASON KRSTANOSKI: Yes, I can. If done well, connecting the large data centres to the transmission network can deliver a lot of benefits to the electricity system itself and then also ultimately to consumers—if we do this well. They are a large, somewhat stable load, which is good for the power system, and it's good for the power system that we're evolving into, that we're transitioning into. If you think about it, in 2015—just 10 years ago or so—more than 90 per cent of our generation was coal. It was a nice, strong, stable, synchronous generation that we had. In 10 years time, around 2035, we see the generation mix in New South Wales being more than 90 per cent renewable. With that renewable, you have the intermittency of the renewables; you have a different type of system. What helps that system with sticking, having less variability and less volatility, is nice, stable loads, which is what we see come out of data centres. Data centres can help support us in a newly transitioned world that's renewables dominant. Also, if the data centres can have some flexibility, which is what we're starting to have conversations about now—

The Hon. MARK BUTTIGIEG: So people understand, Jason, when you've got a supply regime which is variable—wind, solar, whatever—and you've got a variable load, like air conditioning not being used at certain times of the day or cooking or whatever, then that's instability, right? But when you've got data centres churning 24/7, it settles down that variability of supply. That's what you're saying, isn't it?

JASON KRSTANOSKI: That's right. A number of years ago, the demand was very predictable. It was very straightforward. You could forecast it really well. What we see today is that the way people use energy is very different. On the supply side, the variability is extreme now. We see a high volatility compared to previous years. We need to have a strong, stable power system. We need to have a strong voltage waveform to make sure that the grid is safe, secure and reliable. On the load side, having a stable, predictable, forecastable load like data centres is very appealing to help stabilise and help you have a strong power system.

The Hon. SCOTT FARLOW: Thank you very much, and thank you for your submission and your education today. I think it's been very helpful to the Committee. I'm interested in your table of international comparisons and different jurisdictions. I note there's not one that ticks all the boxes. Is there a deliberate reason for that? Are there things that are more complementary than others in terms of the arrangements, regulations and requirements around systems?

JASON KRSTANOSKI: There's a number of things we've seen. We just wanted to show that we're not making up this story on our own. Data centres have been coming around to different countries over a number of years, so we've been looking over the fence to try and learn what others are doing, and to make sure that we take the learnings from other jurisdictions and apply them to New South Wales. That was the intent of having a bit of a look around the world. There are some things we would recommend that we think would apply to New South Wales and be beneficial for New South Wales consumers and then also the power system. If I just talk through a few of those, for example, one is, if we go back to phantom demand, we think that having a bit of a higher up-front application fee might weed out some of those, and we could focus our efforts and our resources on the more serious ones that are in front of us, that have more certainty on their land and other things.

Another thing that we see a benefit of is we don't want data centres to go and ask for a gigawatt of connection and then not use it. That creates issues as well. We think that, if they say they're going to use a gigawatt of power, they pay for gigawatt of power and if they don't meet it—it stops them overbidding their capacity and overstating what they want, because that can cause a challenge as well. A third thing is what we talked about a little bit earlier. We strongly feel that the data centres should pay for all the augmentations that they initiate to

CORRECTED

protect consumers so that consumers are, again, no worse off. Also, I heard this earlier in the inquiry as well. We think it's important that these new large loads bring new generation with them as well.

When I say new generation, I don't mean the renewables that are already being built as part of the energy transition, because we've got to remember that in New South Wales coal is closing. No-one is really building new coal in New South Wales. Coal is closing. It's still the dominant supply at the moment, but we're moving to renewables. There's already a good plan to transition to renewables with the REZs and other connections that are outside the REZs. We have a good plan. If we're going to have a step change in demand now, we need to match that with a step change in generation. It's not just signing PPAs; it's new PPAs so that we can keep the supply-demand balance right. Again, we need to be mindful of the impact that can have on wholesale prices and then, eventually, household consumers who we want to protect.

The last bit is we should also take advantage of the opportunity that a data centre could provide. What I've heard from the data centres, when questioning their applications and their connections, is they're saying that there is some opportunity. They might not need the air conditioners on all the time, 24/7. Maybe they can turn them on in the middle of the day to soak up some of the solar, and then they can turn them off in the evening peak where domestic demand is high. There could be some flexibility there to use with the data centres, which would be beneficial to the grid as well.

The Hon. SCOTT FARLOW: In terms of the phantom demand—and this is one of the arguments we've heard throughout the inquiry. Some will say there are so many applications. I think we heard from Data Centres Australia at the beginning that not all of these proceed. From your experience—and I note the table in terms of the inquiries and then the applications—how many end up going from inquiry to application to then actually a connection at the end of the day?

JASON KRSTANOSKI: What we're seeing right now on our network is those figures at the moment. It's a bit more than 10 now since submitting that. It's probably closer to 12 or 13 gigawatts in the inquiry stage, but it's still around six, or six and a bit, for the application stage. We haven't actually fully connected one yet.

The Hon. SCOTT FARLOW: I noted your point before.

JASON KRSTANOSKI: But we are close.

The Hon. SCOTT FARLOW: Are there any other observations you have from other jurisdictions as to what that figure actually ends up coming to?

JASON KRSTANOSKI: Nothing material, but what I do want you to know is that we don't react—we don't invest in new equipment, new assets, long lead items or anything—until we have that full commitment. Even though it's phantom demand, we wouldn't waste resources or funds on phantom demand until it's actually certain and it's committed. What I will say as well is the whole assessment of the phantom demand is maturing really well, because we've had strong collaboration between ourselves, Infrastructure NSW, DCCEEW, other parts of government, the distributors, AEMO and the data centres as well—Data Centres Australia. We are working really closely to try and sift that out. It's a work in progress, but I feel like it's maturing well.

The CHAIR: On the six gigawatts of data centre applications, they're the ones that are currently progressing. They're not phantom. We think they're going to actually happen?

JASON KRSTANOSKI: Where they're at is that means they've reached a serious stage where they're telling us their technical requirements and that this is what they need, and we're assessing how that will impact the network. It doesn't mean we've guaranteed them access to that six gigawatts. What it means is they've got to a serious point where they're saying, "Here's what our load will look like. Does this work for you?" We look at the network, because we don't want to go and connect a gigawatt if it means there's not enough capacity there for it. We also don't want to connect a gigawatt if it means that, when it glitches or if it trips off, it's going to wobble the rest of the network and cause a cascading effect. We need to model and study that. None of these applications are a quick tick box: "Yes, you've got"—

The CHAIR: No, but they look likely to happen.

JASON KRSTANOSKI: They're pushing it to the next level. They're pushing hard to test it.

The CHAIR: Six gigawatts is—and I don't have the knowledge of Mr Buttigieg. My understanding is that's the same as, or similar to, the Central-West Orana REZ capacity. That's an entire renewable energy zone for these ones that are currently in application.

JASON KRSTANOSKI: Yes.

The CHAIR: That's huge.

CORRECTED

JASON KRSTANOSKI: It is huge. You're absolutely right, and it's all in one area. It's in that city west area, really. What will happen is that six gigawatts will not be given out the way the system currently is. A small portion of that will be given to the first one or two data centres, or the first little bit will be given out. Then, after that, the door will be closed until there are augmentations, until we have new lines, new substations and new assets. The first part will go out because there is some capacity, but it won't be the full six.

The Hon. JACQUI MUNRO: Can I ask on that as well? Are people paying at the moment if they generate their own rooftop solar? For example, people are paying to sell that back into the grid at the moment. How much of that excess capacity is going to be soaked up by data centres? What is that calculation?

JASON KRSTANOSKI: It depends on how the data centres operate. Some will be a flat load that never changes. That's still good because it means more demand in the middle of the day. But some will vary their demand, from what we've heard from the data centres. The ones that can turn their load on, increase their load in the middle of the day, will help the system because, if you don't have that and you have that excess solar, what it can do is it can cause higher voltages on the network—which is okay, we can manage it—but if it was closer to where it should be, it would be a better power system and more efficient. So, there is some benefit. I haven't got the exact calculation for how much they would soak up, but I think the principles are good where it's just trying to reduce volatility and variability in the network. If the demand on the power system was just constant all the time, it would be very easy to plan for, but it's not. But that's okay.

The Hon. JACQUI MUNRO: Data centres can act like a bit of a sponge, basically. Could the result of that be that people are actually able to sell their energy back into the grid rather than having to pay for it to go in the grid?

JASON KRSTANOSKI: There's a tremendous opportunity here. It would be a market-driven opportunity. I think it's important that there are market signals that show in the middle of the day there's a higher amount of supply, whether it's from rooftop solar or other renewables like—generally solar at that time of day. I think it's good if that's a market signal that the data centres respond to and say, "I'll put my load on now." In the evening, as the demand goes up and the solar drops off, they respond to that market signal with higher pricing and then they use their minimum. It's a market solution which will help the network. Good market signals will mean a better power system and also better use of the energy, so it's better for consumers.

The CHAIR: Did you have any others?

The Hon. JACQUI MUNRO: No, I'm okay. My other ones were also about phantom demand.

The CHAIR: Can we just tease out that? From what you were saying, obviously, we need to make sure there's demand before we can approve these projects, but until we know—because we don't want to be building out for things where there's a phantom demand—we're not building things until there is the demand. There's a bit of chicken and egg. How are you managing that? Particularly, how do you ensure then that the costs of building out the things that aren't coming online are not borne by consumers but end up being borne by the data centre companies?

JASON KRSTANOSKI: There's a few parts to that. Again, we won't go and build network and then wait for them to come. That's good. We are responding to them. The challenge has been just the step change. Normally it's okay if demand trickles up over years. If you forget data centres for now, normally the demand would trickle up incrementally over a number of years and we would watch that happen. The market would respond because, as the demand gets closer to supply, it means the wholesale price of electricity would go up. Then that would get a renewable generator to say, "Oh, great. Prices have gone up. My project will commit, and then we'll get a new generator on." Normally, the market can self-solve slowly over time, so it works when you don't have these big step changes in demand. This is something that we're working through now.

The CHAIR: Again, the question then is how do we effectively ensure that the consumers are not the ones bearing the brunt of this. How do you ensure that?

JASON KRSTANOSKI: There is a bit of work to do there. We recommend that we make sure that the data centres pay for all the other augmentations, which at the moment wouldn't occur, so we recommend that that does happen. At the moment they would pay for their direct connections, but there could be upstream augmentations that they've triggered, or it could even be a project that was always going to happen but was going to happen much later. There could be a project that was going to happen in 2037 but, because the data centres have come on and increased the demand, now that project has got to be built in 2033. So there's a new asset that's going to be on the network for four years more than it needed to be. The cost of operating and maintaining that will be borne by consumers. Even the acceleration costs, we think, should be borne by the data centres. At the moment that's something we need to solve for, I think.

CORRECTED

The CHAIR: Can you tell us about Lumea—what that does, when it was established and what it's for.

JASON KRSTANOSKI: Lumea are part of the Transgrid group but they are at arm's length from myself. Where I work is in the Transgrid network regulatory asset base. Lumea are at arm's length, so they have their own unregulated—

MARYANNE GRAHAM: They provide connection services to large users, and data centres would be one of those large users. But currently, as we said, there are no connections to our transmission network.

The CHAIR: But it's fair to say that Lumea's finances would be better the more data centres are coming on, because its business is in connections for those data centres.

MARYANNE GRAHAM: It operates in an open market, so it has to compete against all of the other connection businesses. It runs in a very different model, given the transmission network is a regulated network.

The CHAIR: That's only for transmission connected, not for—I don't know. Apologies, I'm sure Mr Buttigieg could give me a lesson in how all of this works. If you've got a new data centre in Ryde or somewhere like that, are those data centres connecting to transmission? Is that connection something Lumea would do, or is that a different type of connection?

JASON KRSTANOSKI: If you're a data centre in that area—Ryde, or wherever—you would apply for a connection to the local network. You could apply for the distributor or the transmission company, or both. That's where we get phantom demand as well. We'll think we've got half a gig of inquiry, and so does Endeavour or Ausgrid. That's why, when I was saying we work closely with distributors, we're trying to sift out the people who have gone dual track. We're very aware of that, and that's something we're trying to mitigate there. As a data centre, when you're trying to make your connection, you need somebody to do a contestable substation in front of your connection and that, so Lumea could win some of that work in that competitive market, or it could also be someone else. Lumea, in that competitive environment, are winning that work, but that's at arm's length and separate. They are under the Transgrid group, but I'm on the network-regulated side.

The CHAIR: I know I keep coming back to this question about phantom demand, but when you are, as you say, sifting it, do you do that based on "Well, we assume that X percentage of these projects won't go ahead"? Or is it a case of actually looking at each project and then saying, "Is there a possibility that you're actually going to go to a distributor instead of us?", so you're actually dealing with the facts rather than—

JASON KRSTANOSKI: Yes, absolutely. We're definitely having those conversations with them. We look at how serious they are. We talk to them and ask, "Have you made this application with the distributor as well?" We also talk to the distributor. I think we're always going to get this sort of noise, because you'll get data centres that are prospecting in multiple areas and trying to do multiple connections. I think we'll always have some of that noise, but I think the answer will be continuing the collaboration we're doing with all of the different parties and then actually not investing until we have that firm commitment, unless it's the application process which is funded by the data centre. It will be a challenge, but we'll just keep collaborating as we are.

The CHAIR: Just a final question, and I suspect what the answer will be. I note that in the *2025 Transmission Annual Planning Report* you do mention that many of those large projects are targeting Western Sydney because of the international submarine internet cables, which make Sydney a preferred location. That implies to me that a lot of those centres are designed for offshore capability rather than for needing to be near urban areas. What is Transgrid's role in trying to shepherd some of those centres elsewhere and actually making the most of that suburban network for houses?

JASON KRSTANOSKI: We are definitely encouraging them to spread out. Ideally, if we could choose where to put them, it would be where there's existing capacity, and they would be spread out right across the State. If we could choose, we would do that. But what the data centres are telling us is that they want to be in that area, so we can only respond to their applications and follow the National Electricity Rules to process those requests. But, ideally, what I think will happen is we'll get the first couple in the Western Sydney area and then, once that area is relatively full, until augmentations come, I think that will naturally push them out to the regional areas. It's probably just a bit of staging. Everyone wants to grab the first Western Sydney bit and then, after that, I expect they'll move out and we'll get some regional ones as well.

The CHAIR: Which is why we need a broader plan where we can actually say these ones should go here and these ones should go there.

The Hon. MARK BUTTIGIEG: A quick follow up on that. You spoke about the multiple connections between the distribution and the transmission network, and you've obviously got a different outlook than, for example, Ausgrid or Endeavour Energy because they'll be looking at this from the perspective of, "We've got all these renewables coming onto the market, which means that we've got our community batteries going, our solar

CORRECTED

and our locally distributed generation." Analysts would look at this and say that the transmission network is going to be increasingly irrelevant because you don't have to pump all of this electricity for hundreds of kilometres anymore because of all the renewables coming on at the load centres. So Transgrid's business model, I assume, would be to try to encourage as many of those data centres to connect directly to your network to attenuate some of that drop-off. How do you see all of this being coordinated? It seems to me as though there are competing market interests here, depending on which perspective you're coming from.

JASON KRSTANOSKI: For us, if you mention our incentives or what we're on about, it's having a safe, secure, reliable power system that's affordable to consumers. We're looking after the regulated network. I'm not out there trying to grab all the data centre business type of thing. I'm simply responding to the data centre enquiries that are coming through. If they want to connect onto our network, then that's fantastic. We'll facilitate that, and we do see the benefits. I'm less qualified to talk about the economic benefits for New South Wales—I do believe in those—but from the network perspective, I can see the technical benefit of having that large load in a future where we've got high renewables as part of our generation mix. I can also see the tremendous benefit to household consumers, because I do believe that adding this new load can proportion some of the cost of transmission towards them and relieve it from household consumers, which I think is a tremendous benefit for the community. That's our driver.

The Hon. MARK BUTTIGIEG: What role is the renewables supply playing in your network? With all these REZs and that sort of stuff, are they pumping in at your level, or is that the Essentials of the world?

JASON KRSTANOSKI: There's a combination. CWO REZ will connect into our network and then you've got things like Hunter-Central Coast REZ, which will go more or less directly into Ausgrid's network. So it's a combination. But if you step back at the high level of New South Wales as a high level, you will see—remember, just only 10 years ago 90 per cent coal. That comes from, currently, 12 coal units spread across four power stations very close to Sydney. They're all closing soon—in the blink of an eye, in just a few years—so coal is closing. We are right now in the deep transition. We're moving to the renewables, and they are much further away, because they're in the REZs.

The Hon. MARK BUTTIGIEG: I guess that's a roundabout way of me saying, Jason, you could envisage the transmission network still being of use because you're pumping renewables in at that level. Then when the coal-fired power stations drop off, you're still going to need that transmission network because of the existing infrastructure and your renewables pumping in at that level.

JASON KRSTANOSKI: You're absolutely right. The transmission is the backbone of the network. It gives you a strong power system. If you just rely on lots of small little things going on, it will be weaker. Having a strong transmission backbone which connects the REZs and other renewables to the demand centre—because remember, in Sydney, Newcastle and Wollongong, just that small area is more than 75 per cent of the entire demand for all of New South Wales.

The Hon. MARK BUTTIGIEG: Yes.

JASON KRSTANOSKI: But all those renewables are much further away, so you're going to need a network to connect the two to bring it in. Having a strong transmission backbone means the system can stay safe, secure and reliable.

The Hon. MARK BUTTIGIEG: Okay. Thanks.

The CHAIR: That does bring us to the end of our session. Thank you so much for making yourselves available. I have learnt a lot during that session, thank you. I am also keenly aware of how much I don't understand about electricity. That concludes our time with you. To the extent there were questions taken on notice or supplementary questions, the Committee secretariat will be in touch. Thank you very much.

(The witnesses withdrew.)

(Short adjournment)

CORRECTED

Ms DANIELLE FRANCIS, Manager, Customer and Policy, Water Services Association of Australia, affirmed and examined

Mr DARREN CLEARY, Managing Director, Sydney Water, affirmed and examined

Mr PAUL HIGHAM, Head of Business Development, Sydney Water, affirmed and examined

The CHAIR: I now welcome our next witnesses. I'd like to give you the opportunity to give a short opening statement.

DARREN CLEARY: Thank you, Chair and members of the Committee, for giving us the opportunity to appear here today. Sydney Water welcomes this inquiry and the opportunity to contribute to a shared understanding of how the rapid growth of data centres can be supported in a way that delivers long-term benefits for New South Wales. Data centres are key infrastructure underpinning services across health, defence, finance and the digital economy. Sydney Water is responsible for delivering water and wastewater for 5.6 million customers across Greater Sydney and the Illawarra. We are already managing competing demands in a climate-constrained environment and are planning to sustainably meet future needs through significant and carefully staged investment. This includes new trunk infrastructure and a greater use of rainfall-independent sources such as recycled water.

While the data centre sector presents a different demand profile from the broader community because of clustered volumes of large demand and high reliability requirements, Sydney Water is well equipped to manage this. We have considered the demand expectations and risk profile of the emerging sector and have adopted a set of principles to cater for the emerging sector while sustainably servicing the community. These principles are that residential users and existing customers should not bear any additional cost or risk to their supply; charges for very large water users should reflect the economically efficient cost of supply; we will prioritise long-term system resilience through greater use of rainfall-independent sources; cost recovery risks should be borne by the parties best able to manage them; and pricing mechanisms should reflect long-term demand or cost uncertainty.

Sydney Water is responding to the scale and pace of the emerging demand and what this means for water use, cooling, pricing and cost recovery and, importantly, long-term water security. Forecast and actual demand remains Sydney Water's greatest risk factor in planning for data centre growth. We are reliant on proponents' forecasts, as evidenced by applications. Existing data centres do not provide a precedent for water use planning, as the applications that we are now receiving are for significantly larger scale developments and, therefore, water use. As we plan for this emerging demand, Sydney Water is working with government, industry and the community to enable this growth in a way that is sustainable, equitable and in the public interest. We are committed to ensuring that residential customers are no worse off.

This is underpinned by innovative and resilient supply planning, efficient and appropriate allocation of risk, and pricing mechanisms designed to manage the uncertainty while protecting affordability and service outcomes. Supplying water to new data centres provides opportunities. Better utilisation of recycled water has the potential to reduce pressure on constrained wastewater management systems and our waterways. We are ensuring that these sources represent new supply in the long term, not supplanting the potential sources of rainfall-independent drinking water that will be necessary to cater for population growth. Sydney Water looks forward to supporting the Committee's work and contributing to a balanced, future-focused approach that supports economic growth while safeguarding water security for the people of Greater Sydney.

DANIELLE FRANCIS: The Water Services Association of Australia represents urban water utilities across Australia and New Zealand. We recognise that data centres bring economic and other benefits. Our focus is on how to enable their growth, and the step change in cooling demand from AI, in a sustainable way. The scale and pace of the data centre boom here in Australia has been striking. Australia received the second highest global investment in data centres in 2024, second only to America. That's total investment, not per capita. WSAA aims to support policymakers in rapidly building knowledge in this dynamic space and to help establish clear, practical settings that support the speed-to-market needs of data centres.

One advantage for Australia is that we are not starting from scratch. The boom has emerged later here than in some markets, and we've been able to see how those markets responded. There are trends worth noting in Europe, Asia and America, places that can have a similar water context to us. Many places that initially took a relatively permissive approach are now shifting towards clearer expectations, particularly around water efficiency and reporting. We think that's a positive. Australia has made a strong start with the Federal and New South Wales expectations. They build in some important principles and provide early clarity in a fast-moving space, which can give confidence to investors and communities.

CORRECTED

The first principle is early engagement so that water feasibility can be factored into site selection. Water is often a relatively small component of data centre costs, but it can become a significant strategic and logistical constraint. Early engagement allows better outcomes—for example, seeking opportunities to co-locate at or near water recycling plants for reliable recycled water supply, and also avoiding site selections that limit options. For example, if a proponent comes to us having already purchased an inland site, that tends to rule out desalination as an option. The second principle is water efficiency. In locations where multiple large facilities are seeking to establish, often close together, clear efficiency expectations can help ensure that the demand on system capacity is sustainable.

The third principle is recycled water, where feasible. Recycled water can provide a rainfall-independent supply, but it requires significant infrastructure and lead times. These investments must be planned carefully, with costs appropriately allocated rather than passed on to existing customers, as captured in the New South Wales principles. The fourth principle is transparency. This is a rapidly evolving space with strong links between water, energy and technology. Clear, consistent and easy-to-access reporting on water use will help build a shared evidence base, support better decision-making and maintain community confidence. We see a strong opportunity to integrate water efficiency into the existing NABERS energy framework to provide a practical and nationally consistent approach.

Finally, on behalf of Brendan Guiney of the Water Directorate, who couldn't be here today, there is a regional perspective. Most of the interest so far has been in big cities, but most water and wastewater systems in regional New South Wales are not currently configured to support very large industrial demands, especially in drought. There has also been limited visibility of local infrastructure constraints in broader planning processes. If regional data centres are planned, regional planning, infrastructure capacity and local government inputs need to be closely integrated into decision-making. Finally, this is not a simple policy challenge, but it is a manageable one. With clear expectations, early coordination and thoughtful planning, we think Australia is well placed to enable data centre growth while safeguarding water security for communities.

The CHAIR: Thanks very much to both organisations for coming along, and for Water Services Association of Australia's submission. Just firstly for Sydney Water, your data and forecasts in relation to water usage have been cited in at least eight submissions to the inquiry by environmental groups, industry bodies, water utilities, infrastructure providers and local councils, and your projections about data centre water demand have featured in Infrastructure Australia's 2026 priority list and in lots of media reporting. Why did you choose not to make a submission to this inquiry?

DARREN CLEARY: We chose to appear to talk about the issues. We are working with the New South Wales Government on the principles and the consultation paper that they have put out for public comment and using that as the basis of our policy response back to the industry.

The CHAIR: The water Minister, in a budget estimates session earlier in the year, said that they were putting together submissions to this inquiry. It seems that lots of people were and then suddenly didn't. Were you planning on making a submission and then decided not to?

DARREN CLEARY: No, we were not. The issue about demand, we feel—as you started with, the preface to your question—has been publicly well debated and we're very happy to continue to talk about that. We are working with the Government on their policy response, and we're here to talk about—obviously, this is rapidly emerging—how we are aligning to that policy response.

The CHAIR: Sure, but you're only here because you were invited to appear. Some Committees take the view that if you don't get a submission, you don't then invite a witness, so it was a little bit perplexing that we didn't end up with a submission, but we will move past it. DCA say that they estimate 1.9 per cent of water will be used for data centres, which is very different to your 25 per cent. I asked them about this when they appeared. What's your take on why the forecast is so different?

DARREN CLEARY: I'll provide an opening comment, and I will pass to Mr Higham to provide some more detail. We are basing our forecasts on the information that we are receiving from the applications that are coming in. The issue that is raised often, and the challenge, is around the uncertainty of particularly the longer term forecast when we get towards 10 years, and the issue often of phantom demand is raised. We do take out obvious applications that are duplicative or clearly entirely speculative, but our forecasts do reflect applications that are coming in that are planning to go ahead. There's clearly uncertainty when we get towards peak of that 10-year time frame. When we're talking about what's happening in the next five years, we have much higher degrees of certainty because the proponents we are dealing with are further advanced in their planning cycles.

What I would say about the uncertainty in the long-term forecast, that's true of demand forecasts of all types. There is uncertainty in these longer term forecasts, but that is the best information we have and that's not a

CORRECTED

reason not to plan. Our view is we need to acknowledge that's a potential outcome and then work with the industry about how we deal with that uncertainty. We will be applying the principles we've stated, which means much of that uncertainty around what infrastructure should be provided and who pays for that should actually be borne and managed by the proponents. I think that also provides a strong commercial incentive then for us to be working with them to say, "How do we get as accurate as we can, and how do we design infrastructure that is adaptive as possible?" Because, as in any commercial activity, there's always going to be uncertainty when you talk out 10 years. Paul, do you want to talk in a bit more detail about the projections themselves?

PAUL HIGHAM: Yes. Sydney Water has received approximately 90 service requests from industry. In part of that service request process, we ask proponents to provide the staged water demand forecasts and their ultimate capacity, so what they envisage to be required for their ultimate campus development. That is used in our planning process. As Darren's alluded to, clearly things in the near term are far more certain than things in the longer term. But when a data centre is talking about being a 500 megawatt or a 300 megawatt or, potentially, even a gigawatt facility, it's really important that we understand what the long-term goals are in order to plan long-term water infrastructure to meet those needs.

The CHAIR: Can I just ask a technical question? Obviously you're responsible for the resilience and the integrity of the water system as a whole. If we're looking at somewhere like Ryde where they've got a bunch of data centres already and they've got a bunch more that are in the planning process, again, I'm not a water engineer, but is there a point where there is so much additional demand that you need to upgrade the actual pipes and things and that infrastructure? How do you prepare for that and work that out?

DARREN CLEARY: The short answer is yes, and that's quite common, particularly with the large-scale data centre proposals that we are now seeing. They are very large water demands on our water network, but also it does have a demand on our wastewater network, because there's generally a discharge that comes from their water-cooling systems, or there can be. They're not designed with large amounts of spare capacity; that would be inefficient. What we are doing is we are working with the proponents and saying, "What infrastructure is required to ensure that we can sustainably service this development?" I would point out that data centres are different just because of the scale of the demand and the speed with which this is emerging as a new part of our economy. But this issue of new growth coming into our system and having to service it and provide new infrastructure is common with urban development, particularly greenfield development.

We have already processes and policies in place for adding to that infrastructure and for proponents to be funding a proportion of that infrastructure and delivering that infrastructure. We will do the same with data centres. In many cases, there will need to be new infrastructure provided in the local network. I would say there are two broad issues. There's the capacity of the local network to get water, whatever the source is, to the data centre, and then there's the question of where does that water come from. We are strongly encouraging data centres to take new sources. There needs to be a new source of water. A recycled water product is what we are strongly encouraging. They will need to be sourcing it from that new water source or potentially, if recycled water isn't feasible, another water source.

The CHAIR: I just want to get these basics in. For example, in Lane Cove, I think it was, a couple of months back there was a burst water main, and I understand that that's quite an old part of the water network. Was any of that caused by the extra demand coming on from nearby data centres?

DARREN CLEARY: No.

The CHAIR: When Minister Jackson made an announcement, I think last year, in relation to an extra investment in the waste or the sewerage network in response to—it's such a bad name for them—the poo balls incident, she also mentioned that this would help to bolster the system for data centres. It was a little throwaway line in the release. I guess the question, then, is this: If you were having to plan for a certain forecast demand, putting in place this sort of infrastructure, how are you getting back any contribution from data centres for that work and that cost to the network?

DARREN CLEARY: At a high level, we're doing—as we have been doing—strategic planning across the whole region and over long time frames for how we deal with a whole range of challenges. That has identified as part of that work, for example, where strategic locations would make sense to have recycled water facilities. That's work that we should be doing regardless, because there are lots of different challenges. What we are now seeing is this emerging new demand, and it's coming quite rapidly, and we're building off that broader planning work then to say, how do we specifically service data centres? What we are now doing is entering into specific agreements with the data centre proponents to then fund the planning and design work to specifically service them. It's based off a broader strategic framework, but there'll be specific solutions that will be required—to your point, planning and design work, and then infrastructure. Our pricing principles are, "The data centres will fund that." Paul, do you want to talk briefly about planning agreements?

CORRECTED

PAUL HIGHAM: We currently have seven planning agreements in place with data centre proponents to undertake the work that Darren has just described about what infrastructure is required to meet the services that they are asking for. We have a further five currently in negotiation in order to allow us to do that detailed planning: Where does the next source of water come from, how would you move the water from where it's produced to where it's used, and what additional infrastructure is required in order to achieve that?

The CHAIR: The \$3 billion over 10 years is for the Malabar System Investment Program, which is what I was referring to. How much of that \$3 billion would be recovered from data centres?

DARREN CLEARY: That particular investment was something that we were undertaking to reduce the load on the wastewater system, so the short answer is none, because that investment is delivering outcomes for our existing customers. What the Minister was referring to is what that investment was facilitating a new recycled water facility at Liverpool, and an investment at Glenfield to enable that. What we're seeing with data centres is that that's an investment that's justified based upon our current needs. The data centres then could use that as an opportunity to take that recycled water product. The infrastructure to get the recycled water from that facility to them would be funded by them.

The CHAIR: So they're not paying any of that \$3 billion.

DARREN CLEARY: They're not paying for that, but that's because that's required anyway. There are other potential schemes we are dealing with where the recycled water infrastructure would be entirely new. In that case, the data centres would be paying for it. Or there are other schemes where we do have plans for recycled water infrastructure, similar to the Malabar investment you were talking about, but it's in the future and the data centre demand would require that investment to be substantially brought forward and delivered much earlier. Under our principles, the cost of bringing that investment forward would be borne by the data centres.

The CHAIR: I want to bring you in, Ms Francis. Looking at this issue of—again, I wish I was a water engineer, but I'm not. My understanding is there's the high-quality drinking water that ideally we'd want to go to households, and we want to make sure that data centres aren't using that and there's a variety of ways that we could try and achieve that. How realistic is it for all data centres to use recycled water?

DANIELLE FRANCIS: Probably not all of them, but certainly a good amount of them could. When I say not all of them, there are cases where recycled water isn't as feasible. For example, if you're a long way from a recycled water pipeline, then the cost of building a main to get to you is really expensive. In water, it's the transporting that can be really expensive, more than the water itself. Also, some data centres have onsite water tanks, for example, where they capture rainwater, so some of them will use that as their first source of water. Then they might need some tap water—drinking water—to top that up, for example. Others might have stormwater detention systems in their roof and things like that where they can use the recycled water.

There are others that might use desalinated water. There's a place in Finland that uses raw seawater. It doesn't even desalinate it; it actually just uses the raw seawater, for example. But as a general rule to your question, it is pretty feasible that most places could use recycled water. There's also another option, which is called offsetting, which is quite strongly used around the world. What that means is—it's a bit like the carbon offset framework. It basically says if it's not feasible to actually get recycled water physically from a plant or a nearby main to you for whatever kind of operational reason, maybe you could sponsor or fund a project elsewhere in the catchment that restores that benefit to the overall water levels without it having to physically attach to you.

The CHAIR: This is the recycled water certificate idea?

DANIELLE FRANCIS: Yes, the offsetting framework.

The Hon. SCOTT FARLOW: On that point, what sort of requirements are there in terms of data centres and the quality of water? Can you use recycled? Can you use greywater? What are the requirements that you're seeing from data centres?

DANIELLE FRANCIS: Basically you can use any quality of water in a data centre, but the quality of the water will influence how much of it you use, for example. It also depends on the type of cooling system that the data centre is using. To give you an example, a really high-purity water like drinking water—or you can even get higher purity water than drinking water—a data centre won't use as much of that. Let's say they have an evaporative cooling system. About 85 per cent of that water just gets evaporated, but the remainder, because it's very pure, circulates through the system. And because it's very pure, you can use it for quite a long time before you have issues like scaling or biofouling, and things like that, and you have to replace it. You can use recycled water instead. If it's high-purity recycled water, again, you can get a lot of what's called cycles of concentration where you can re-use that water for a long time.

CORRECTED

But if it's low-purity water—a very low-grade recycled water, for example—you're not going to be able to use it as often because it already has more contaminants in it when it's been used, so it's going to more quickly create those system issues. You'll use it for maybe four cycles of concentration, and then you'll flush it through with fresh water. Whereas with that higher level water, you might be able to use it for longer without needing to re-flush. So you can technically use all levels, but the higher the purity of the water, the better it is for the system integrity. That's why, generally, there's a preference for using higher qualities of recycled water.

The Hon. SCOTT FARLOW: We've also heard throughout the inquiry that there can sometimes be trade-offs in terms of energy use and water use. Effectively, you've got higher energy use and you can have less water use. How is that trade-off made, effectively? Is it done on a cost basis? Is it done on an availability basis?

DANIELLE FRANCIS: Yes, it is generally done on a cost basis, but it's also on a technical need. For example, water basically has a better property for thermal discharge. Air cooling works to be able to reduce the heat from a hot premises, but water tends to work better. It's just got better thermal discharge properties. For example, that's why in cooler climates, you can often see that data centres can last longer with air cooling. But with the newer generation of AI that's coming through, it's leading to smarter chips, and smarter chips are hotter chips. So the servers and all the facilities are—

The Hon. SCOTT FARLOW: Not very smart of them, is it?

DANIELLE FRANCIS: Yes, so they're getting hotter. What we're finding is that this new generation of AI technology needs basically a more sophisticated system of cooling. What we're finding is that, particularly as the ambient temperature in the climate in whatever geographic region you are in is higher, air cooling is less able to cope with it. So that's part of the drive towards liquid cooling systems or water cooling and so on. What you often hear about is this trade-off where people say, "Okay, I can have a really water efficient cooling system—like a closed-loop system, for example—but it can lead to slightly higher energy use." It's unclear how much higher the energy usage. Some people say it's a lot higher; others say it's actually not that much higher. But there is certainly a trade-off.

What we generally say in the water sector is that rather than trying to apply blanket rules of thumb, water is really location specific and we have to assess an application that comes to us based on the local situation. While we obviously have a general inclination towards a water-efficient system, there might be spots, for example, where you're in a really difficult spot to provide energy supply to a particular data centre. That might be a particular location, a particular corner of a particular suburb, where it actually makes sense to have a more water intensive cooling system, because it might save the need to build a really, really high cost node of energy provision or something like that there. So it is a little bit location specific as to what's best to optimise the resource use.

DARREN CLEARY: If I could add, Mr Farlow, our approach is to apply these principles and get cost information, pricing information, to the industry so that they can make that determination. We don't have the information—neither should we—to specify, once we get to a minimum efficiency standard, what is that right trade-off. But let's make that information clearly available, with clear expectations about the costs and the obligations on the providers, and they can make that decision.

The Hon. SCOTT FARLOW: I know we've heard from Lane Cove residents so far about some of their concerns with the data centre proposals in their area and the number of data centres in the surrounding area. One of their criticisms or concerns was in terms of water pressure and water outages. Is that something you're seeing which has been linked to data centre development in certain areas where there's a high concentration of data centres?

DARREN CLEARY: The developments that are being proposed—as I've already outlined, we will be requiring new proponents to fund the infrastructure that's required to ensure that their service need can be met, and that that new connection does not reduce the service to the existing customers. That's currently what we do with any development. Any concerns that residents have about current service of water outages or pressure, that's not due to data centre development proposals, and that's something Sydney Water—we should be working with those residents, to be clear.

The Hon. SCOTT FARLOW: I think part of their concern is about existing data centres. There's one that's proposed but there's also one that's already within the precinct. Then there are others in surrounding precincts, which I take it would be serviced by a similar water network. They were some of the concerns that were expressed to us last time.

DARREN CLEARY: New development of any sort that connects, we require them to provide the infrastructure that's necessary to meet their needs without reducing the level of service we provide to customers, particularly in terms of the obligations we have with respect to our standard levels of service and our operating licence.

CORRECTED

The Hon. SCOTT FARLOW: We've got the UDIA coming next, and I thought it probably worthwhile putting some of their concerns to you in advance because I'm sure they'll make those known here. Some of their concerns are around the trade-off between housing and data centres. I'm just interested in terms of how Sydney Water assesses that trade-off, so to speak, and is there any prioritisation you do in terms of an application as to whether it's servicing a data centre or another industrial user or if it's for housing?

DARREN CLEARY: We do not do that prioritisation, and that's not a decision that Sydney Water can or should make. That's really up to planning authorities to decide what's an appropriate development. Our job is to enable growth of the region and to provide those services, so we do not prioritise. I would say, from our infrastructure perspective, we are confident with the application of the principles that we are talking about and the ability of industry to fund that infrastructure that's required. We can service this new data centre development without adversely affecting the ability to provide infrastructure for housing.

The Hon. SCOTT FARLOW: I know that some data centre developers have looked at alternative water provision services to Sydney Water or to supplement Sydney Water's service provision. What is Sydney Water's approach to such applications or such proposals?

DARREN CLEARY: We'll work collaboratively with third parties who are going to provide that service. That's absolutely up to the proponents to decide how they want to work with us or work with a third party.

The Hon. JACQUI MUNRO: I was curious about whether the Investment Delivery Authority has been engaged with Sydney Water around the approvals of data centre projects.

DARREN CLEARY: We have provided input to the Investment Delivery Authority.

The Hon. JACQUI MUNRO: Did they reach out to you asking for advice on the proponents' plans or proposals?

DARREN CLEARY: We've provided advice about our overall general approach and how we are managing these. Paul, did you want to comment?

PAUL HIGHAM: Yes. We've been part of the consultation process with the Investment Delivery Authority. We're involved in that process as a consultation activity. In some cases, we've been able to articulate whether there's water available or not. Equally, where there isn't—I mentioned earlier, the planning agreements—we've encouraged proponents to come and enter into planning agreements in order to understand the ability to service in future. So we're part of that consultation process.

The Hon. JACQUI MUNRO: But there obviously isn't a data centre strategy at this point. So, in terms of how you're considering a strategic approach to the approvals of data centres and their resource requirements, how does that operate in this IDA process?

DARREN CLEARY: I'd say, generally, like with any significant development, the specifics of whether infrastructure is provided is largely considered once the development is coming through into the later stages of approval because that infrastructure can be provided as part of the development. At the highest level, when we're talking about data centres and their needs, we can always build more infrastructure to service that development. We are providing advice about what those obligations may look like, so it's provided as part of that approach. From a strategic perspective, in terms of Sydney Water and our particular service provision, what we are doing now—the guidelines the Government is putting out are exceptionally helpful because they set a clear expectation with industry around what they may be required to do. Then, given the scale of this industry, we are working with them through these planning agreements to specifically say, "Let's work together to determine what infrastructure is required specifically for your needs." We know that these proponents do tend to cluster together for a whole range of reasons, particularly around infrastructure, so that provides the opportunity to work with them about that service provision.

The Hon. JACQUI MUNRO: In terms of the difference between the infrastructure—as in pipes, for example—and the actual resource of water, how are you managing approvals based on the future forecasts? When the IDA is coming to you and it's saying—perhaps this is the wrong process; correct me if I'm wrong—"Hi, we have all these proposals. Which ones do you think are going to be suitable?", are they asking you for guidance on how much water each of these proposals will be required to—

DARREN CLEARY: No, because we're working towards the principles. I think what we are seeing, and what government has clearly identified with Sydney Water and working with government has identified, is that what we are seeing—and it has happened relatively quickly—is the emergence of a large new water-intensive industry that's coming into the city. The question of the new water source is now starting to be fleshed out in these policy principles, which is that there's a very strong preference for recycled water. That's certainly what we are working with many proponents on right now. If recycled water wasn't economically feasible—and we haven't had

CORRECTED

to grapple with this in any specific proposals right now—then there would still need to be identified a new additional water source to meet that demand.

The Hon. JACQUI MUNRO: I'm basically trying to understand at what point the Government is looking at water in relation to these proposals. Are they going, "Okay, yes, we have the land here, we have someone with the money, we go through all these things," and then down below, "Okay, now we have to think about the water"?

DARREN CLEARY: Up until recently with the scale of the proponents that we are now seeing, it is very different to what we've seen historically, even with data centres—the scale of the data centres and therefore their water demand. What has happened previously is they've been dealt with in the same way that any other commercial industrial development has, and that then relates back to the Greater Sydney Water Strategy and our long-term planning of how we deal with water security. What we have seen, and the Government has recognised, is that this is significantly outside of that planning, and now we have this new approach, which we are now applying to the proposals that are coming through. It has happened in parallel, but we've been having this conversation.

The Hon. JACQUI MUNRO: Is water considered in one of the first considerations of whether a data centre proposal is going to be feasible?

PAUL HIGHAM: It is a similar planning process to what currently already exists. In considering whether a data centre is feasible, water is one of the components, but it also considers energy supply and land use. It does continue along the typical pathway.

The Hon. JACQUI MUNRO: I'm trying to understand where the IDA makes its decisions separate to the existing planning process.

PAUL HIGHAM: My understanding is that the IDA is a facilitation process rather than an approvals process.

The Hon. JACQUI MUNRO: But the facilitation is obviously not accepting every proposal.

DARREN CLEARY: For some of these matters, we can't speak on behalf of the IDA.

The Hon. JACQUI MUNRO: I understand. At what point in this IDA process is Sydney Water engaged?

PAUL HIGHAM: We're engaged in the consultation process with the proponents, but the responsibility is upon the proponents to provide the information to the planning authorities about their ability to provide water services and their ability to supply energy services. This is where the planning agreements come into effect. The proponents are approaching us at that point and saying, "Help us understand how we can provide water services for our development." We're not an approvals pathway per se. We're facilitating the proponents' applications through the planning processes.

The Hon. JACQUI MUNRO: Is it different to the existing planning process really at all?

PAUL HIGHAM: I think it facilitates an opportunity to go through that process, but that would be a matter for the IDA.

The Hon. JACQUI MUNRO: On non-potable water and existing systems, one of the submissions spoke about the millennium drought and infrastructure that was developed at that time for the distribution of non-potable water. Is it accurate that that system exists but it's no longer under the same pressure or it's not being used to the same degree that it was at that time?

PAUL HIGHAM: There is a third-party operated scheme in south-west Sydney that is providing recycled waters for industrial uses at the moment. Obviously, the owner of that facility will be exploring options and opportunities to maximise that value.

The Hon. JACQUI MUNRO: Do they work with operators or proponents directly, or what's the relationship between Sydney Water and a private operator?

PAUL HIGHAM: In this case, they would work with a customer—a proponent—but equally they would work with Sydney Water as the ultimate source of the water, being the wastewater from Sydney Water's system.

The Hon. JACQUI MUNRO: Is there a general principle that it is good to work with private operators to achieve the outcomes that we're trying to get to now?

PAUL HIGHAM: Yes.

DANIELLE FRANCIS: I was just going to add that there's something called the Water Industry Competition Act, which came in in about 2006 during that millennium drought. That actually has provided a quite expansive regulatory framework by which public utilities obviously already exist, but private utilities can

CORRECTED

participate and they can get licences to operate as well. You have quite a competitive market and a rich market in which lots of players can act. Obviously, they all have to interact in the sense that you can have a private utility but they still need to tap into the resources of the public water utility, who has the actual wastewater, and then they can work together.

The Hon. JACQUI MUNRO: It sounds like AI models being foundational and everyone gets to use the data and systems. Are those private networks going to be important for the delivery of water resources to data centres?

DARREN CLEARY: There's one at the moment that absolutely has the potential to be so. With respect to much of the other proposals, that's a possibility.

The Hon. JACQUI MUNRO: I mean, does it take off pressure—

DARREN CLEARY: But it's certainly not reliant on those, so I suppose fundamentally I think the response is Sydney Water working with the data centre proponent certainly has the capacity and the capability to deliver the infrastructure that's required. Then it's really up to the data centre proponents to decide whether they want to work with Sydney Water or they want to go and work with a third party. It's not necessary to have a third party involved.

The Hon. JACQUI MUNRO: I guess I'm getting to: Is it useful to have these private operators involved in this to take pressure off Sydney Water?

DARREN CLEARY: Like, it's not necessary. I mean, like I said, we certainly—

The Hon. JACQUI MUNRO: But is it helpful?

DARREN CLEARY: Not necessarily, no. It's really a matter for the proponents.

The Hon. JACQUI MUNRO: It's not necessarily cheaper for the taxpayer.

DARREN CLEARY: No, not if we apply these principles. Then it won't be cheaper for the taxpayer.

The CHAIR: Can I just go back to that question—then I'll come to Mr Buttigieg—about the phantom demand. I note that in your submission, Ms Francis, there's the comment there that although when we're talking about energy connections, you can have duplicative proposals and speculative proposals that might not actually eventuate, but when it comes to water proposals you said they're typically past that stage and fairly mature, so water demands are more likely to be real. Can you explain that? How should we view these competing water demand forecasts?

DANIELLE FRANCIS: Everybody's trying to grapple with this question of how much demand is phantom demand. It's also complicated because of the fact that applications, and proponents when they make applications to utilities, it's commercial in confidence. Obviously, utilities are seeing everything, but they would never—and can't ever—talk about what they're seeing to other proponents. So you can't try and gauge that. But there has been some work that's been happening to try and assess how much of the demand that is being proposed might actually be phantom. There is some reporting around that quite a lot of the energy demand—like, a higher proportion; I think about 85 per cent or something like that—might actually be phantom and it might actually not eventuate. We wouldn't comment on whether that is true or not for energy because that's not our space.

What we know is that there is a perception that to get energy, you have to apply really, really early and so there can be quite a lot of duplicative application processes. But by the time places get to applying for water, because it's a smaller element of their resource need base, they've typically secured that energy connection. We think there is less of a case where the demand that's coming to a water utility is phantom. We think that probably more of the applications that come to water utilities are likely to materialise because they're a little bit more mature in their stage than maybe some of the energy.

The CHAIR: Mr Cleary, do you stand by the 25 per cent forecast demand?

DARREN CLEARY: We do, and we acknowledge the uncertainty. But one point I do want to make is that what will happen in 10 years, there is uncertainty on that. Right now, we have applications for proponents that are well advanced, that are asking for substantial new water demand. The water demand is the equivalent of a new suburb connecting to our system. There is no doubt—and we can demonstrate that—that we have a large new industry developing in Greater Sydney that will add significant new demand to our system. Will it be 20 per cent, or will it be 5 per cent or 10 per cent? That is uncertain. But there is no doubt that it is significant, and we need to plan for it, and we need to work with the industry about how we deal with the uncertainty of those longer term projections. What will happen is there will definitely be a new large demand that will be put on our system that we have to cater for—and we think we can, applying these principles.

CORRECTED

The Hon. MARK BUTTIGIEG: In terms of that demand projection, what implications does that have on the supply of water relative to the price for the consumer to cross-subsidise?

DARREN CLEARY: We will be applying these principles, which means that there won't be a cross-subsidy to the consumer. If a recycled water facility is being developed, that cost will be borne by the proponents. The uncertainty on the longer term demand—so therefore the life over which you recover those costs, for example, will cater for the fact that there is uncertainty on demand, and therefore we'll be looking to recover most of those costs early in the proponent's life cycle. That commercial framework is quite important to manage that risk that the broader consumer base doesn't pay.

The Hon. MARK BUTTIGIEG: Does that account for the fact that if we go through another dry spell, as we did 20 years ago, when we built the desal plant—the sky was going to fall in and then it started raining again. If we go through another of those periods and supply is restricted, so demand goes up, so the cost of the commodity—water, in this case—has got to go up, does it account for all that?

DARREN CLEARY: It will, with respect to if it's recycled water, then it's a new source of water, so it's not affecting the water security for our existing community. If it isn't recycled water, then we'll have a mechanism for that contribution towards a new water source. There's a question of restrictions policy. How will water restrictions in times of drought apply to data centres? It's a matter for government. That's a Government policy perspective. But that's also the other lever that's available: How do restrictions apply?

The Hon. MARK BUTTIGIEG: We heard evidence from previous witnesses on the closed-loop systems. Have you got any commentary around the potential contribution they'll make to efficiently using water?

DARREN CLEARY: What I can say is where there are closed-loop systems—we've certainly seen some of them, and they have very small water requirements from us. Certainly, it's not our position to mandate any particular technology, as Ms Francis has touched on.

DANIELLE FRANCIS: Yes. One of the things that we think is really important in looking at efficiency standards is that they be outcome based. The idea is that you achieve a level of efficiency that is deemed reasonable, but we want that to be technology agnostic. We don't want to say, "You must use this particular—closed loop, immersion, adiabatic." We don't want that because that can stifle innovation. We just want to say that you need to achieve an outcome, and there's lots of good work happening about that. Just one thing with closed loop, it's important to understand that closed loop is half of the cooling system. Closed loop means you put some water in a pipe and that pipe is sealed, so you only ever use that water basically once. That is really water efficient.

That piping system touches the servers and the chips and it absorbs the heat from them. That's one half of the cooling system. But then you still have to find a way to reject the heat that has been absorbed, that is in that cooling system. There's a sort of secondary cooling system in facilities—in data centres—and that can go different ways. Some of them will have a closed-loop system, which is water efficient. And then, for example, to reject that heat that is absorbed, they might have an air-cooling system which uses no water, and so that's very efficient. Others might then have an outer loop to actually use evaporative cooling to discharge that heat, which is quite water intensive. Closed loop can be very water efficient or it can not. It depends on the secondary cooling system that's in place. Closed loop is only half of the story.

The CHAIR: Unfortunately, we have come to time. It has been very informative. To the extent that there were questions taken on notice—and I know I've already got supplementary questions—the Committee secretariat will be in touch. That concludes our time with you. Thank you.

(The witnesses withdrew.)

CORRECTED

The Hon. STUART AYRES, Chief Executive Officer, Urban Development Institute of Australia, sworn and examined

Mr GAVIN MELVIN, Executive Director Policy and Strategy, Urban Development Institute of Australia, sworn and examined

Mr DAN HUNTER, Chief Executive Officer, Business NSW, affirmed and examined

Mr DAVID BORGER, Executive Director, Business Western Sydney, affirmed and examined

The CHAIR: I welcome our next set of witnesses. I would like to give the opportunity to start with a short opening statement.

DAN HUNTER: Chair and members of the Committee, thank you for the opportunity to appear today on behalf of Business NSW. Business NSW represents more than 50,000 members and businesses across metropolitan and regional New South Wales, operating across sectors including manufacturing, logistics, construction, tourism, professional services, technology, energy and retail. All of those businesses have one thing in common: They rely on data centres as essential infrastructure, and that reliance will only grow into the future. For Business NSW, this inquiry is fundamentally about economic productivity, competitiveness and investment attraction. It will require balance, though.

Data centres are now critical economic and enabling infrastructure for the modern economy. They underpin artificial intelligence, cloud computing, telecommunications, financial systems, logistics, cybersecurity and the digital operations of businesses across virtually every sector of the economy. Access to scalable, secure and reliable digital capability is now also essential to attracting investment, supporting high-productivity industries and enabling businesses to grow, innovate and compete. New South Wales is well positioned to lead in this space. Sydney is already Australia's largest data centre market and one of the leading digital infrastructure hubs in the Asia-Pacific region. This presents a major opportunity for New South Wales to attract investment, strengthen digital capability and support long-term economic growth.

However, the scale and pace of projected growth also creates significant infrastructure and planning challenges. The challenge for government is no longer whether the sector grows; the challenge is ensuring growth occurs in a coordinated, infrastructure-ready and economically sustainable way. Ultimately, Business NSW believes New South Wales can continue to strengthen its position as Australia's leading digital infrastructure hub while also improving productivity, attracting investment, supporting regional development and maintaining reliable and affordable infrastructure systems. Achieving that outcome will require clear long-term strategic direction, better infrastructure coordination and a stronger focus on ensuring the benefits of growth are shared across the broader community. Thank you, and we welcome the Committee's questions.

STUART AYRES: Thank you, Chair and members of the Committee, for the opportunity to appear today. UDIA NSW represents over 450 members across the development industry, spanning developers, financiers, builders, consultants, contractors, engineers, architects and government bodies. We recognise the importance of data centres to the New South Wales economy. They are critical digital infrastructure and will play a role supporting business, technology, cloud services and the broader industrial economy. The issue for this inquiry is not whether New South Wales should accommodate data centres—it should. The question is whether they are planned, serviced and approved within a transparent and disciplined planning framework that does not compromise housing delivery or other strategically important land uses. The key issue for our members is the interaction between data centre growth, water servicing capacity, infrastructure cost recovery and housing delivery.

Sydney Water has estimated that, based on service requests received to date, data centre demand could increase total water demand by up to 25 per cent over the next 10 years if all proposals proceed. A medium-sized data centre using around five megalitres per day can have the same water demand broadly equivalent to around 9,000 homes. This matters because housing delivery in Sydney is already constrained by unresolved water and wastewater servicing. In south-west Sydney alone, the department of planning's Urban Development Program identifies around 121,000 housing lots still to be created across three growth areas. Survey work we undertook representing more than 36,000 planned dwellings found that 22,400 of these dwellings, or 62 per cent, are unable to proceed because of a lack of committed water or wastewater infrastructure.

If data centre demand is not managed properly, it will compete with or delay the same infrastructure capacity needed to unlock housing. That is why we are recommending a clear policy framework built around five core principles. First, data centre growth should not displace housing servicing capacity. Where water, wastewater or recycled water capacity is constrained, housing and broader urban development needs to be prioritised. Second is full cost recovery. The cost of servicing data centres, including dedicated infrastructure, accelerated upgrades

CORRECTED

and enabling works, should be borne by the proponents—not households, existing water customers or housing delivery.

Third, recycled water and other rainfall-independent supply should be the default servicing model. Potable water should not be a routine operating supply for data centres. Fourth, Government should establish an up-front servicing cap by water volume, either system-wide or by region. Without a capped servicing envelope, there is no incentive for proponents to reduce water use, and it becomes harder for governments to manage the cumulative demand and investment risk. Fifth, these principles need to be embedded through consistent and enforceable approval conditions across all relevant pathways, including State significant development and projects facilitated through the Investment Delivery Authority.

UDIA also sees an opportunity for data centres to become an early test case for partnership-based water infrastructure delivery. Where dedicated recycled water or rainfall-independent infrastructure is needed, government should be open to structured partnerships involving Sydney Water, private capital and specialist providers, while preserving public ownership, regulatory oversight, customer protections and cost recovery discipline. Handled poorly, data centre growth could add pressure to constrained water systems and crowd out housing capacity. Handled well, it could help bring forward new water infrastructure, strengthen recycled water networks, protect households from cross-subsidy, and create a model that could then be applied to the larger task of servicing housing growth across Greater Sydney. That is the constructive policy opportunity that UDIA believes this inquiry should seize.

The CHAIR: Mr Hunter, your submission opens by stating that data centres are critical economic and enabling infrastructure, which grounds much of the argument for having streamlined approvals in the Security of Critical Infrastructure Act 2018. The CEFC Baringa report—which you cite in your submission as well—estimates that 65 per cent of new data centre capacity in Australia will be hyperscale facilities and that 84 per cent of that hyperscale capacity will specifically host AI workloads, which I understand to be predominantly generative AI such as image, video and text generation—or cat videos, as we are sort of using as a shorthand. Can you explain why you believe that a facility which has the primary purposes of generating those AI-produced images should attract the same critical infrastructure protections as a facility that hosts emergency government communications or financial system backbone services? Should we be looking at different types of designation for different types of data centres?

DAN HUNTER: I think we can look at different types of designation and that the planning and approval process can certainly look at that. I don't know that we should be making the link that these hyperscale data centres are being built for the purposes of entertainment—the cat video. I think that they are being built for slightly more serious purposes, which are around supporting businesspeople, jobs and communities that flow on from that. I think that the entertainment side of things is one part of it. I think there's a sweet spot here. Having those data centres here in Australia provides an opportunity. It's critical infrastructure. There's data sovereignty issues, which I don't know have been spoken about yet either, whereby a lot of government needs these data centres onshore for data sovereignty purposes. A lot of private enterprise needs that as well.

If we simply leave it to offshore providers, we're providing our data to other jurisdictions where it can get accessed by governments. We are seeing, as a very stable government, a very low risk from sovereign intervention. I think that's a positive for us. There's an opportunity here for us not only to service Australia's data needs and maintain that sovereignty of data. There's also an opportunity for us to become a hub, if you like, for this part of the world. There needs to be balance in that, Chair, and the balance needs to be around energy and water use, and the planning and locations and where those go. That's where we need good planning and approval processes. We need a good digital and data centre plan from the New South Wales Government. This is coming, we are going to need these things and we shouldn't shy away from having these conversations. I welcome these sorts of inquiries.

The CHAIR: What I'm seeing when I look over at the backlash to data centres in the US, but also when you look at some of the issues that we're having in the regions with communities that are seeing renewable energy projects arise, is that there are certain things that people are willing to accept an encroachment on their lives for. We often speak to people who are having land compulsorily acquired near critical transport infrastructure. They say, "We don't disagree with the fact that our land needs to be taken away." Similarly, if you go and speak to people who are faced with a data centre next door and you say to them that this is for sovereign purposes—it's for government data or it's for health data or it's for banking infrastructure—I think that's a different thing for people to understand versus this is so that we can have more generative AI. Do you think that by making that distinction between what is actually critical and what isn't, we might end up bringing the community more on board?

DAN HUNTER: I think that's exactly the conversation we should be having. If we use the example of the internet two to three decades ago, we probably wouldn't have called that critical infrastructure. We had electricity and water. Now internet has become critical infrastructure. It's about having a conversation around how data

CORRECTED

centres are going to become critical infrastructure and what they're actually going to be used for. When it comes to the community conversation as well—and I think this is the role that government should be playing—there's data centres and there's data centres. What I mean by that is there's good quality ones and there's poor quality ones. There's good quality ones that are advanced. They refresh their technology. They're getting better and better at energy and water efficiency. We just heard from some of my former colleagues from Sydney Water about the improvements that are happening at the moment. The community conversation should be around what is critical infrastructure and why it's critical infrastructure, but also that the Government's going to play a role in making sure that we are getting quality data centres built and maintained on an ongoing basis in our communities.

The CHAIR: I'll bring you in as well, Mr Ayres. We have heard a lot from people around the Ryde and Lane Cove area, where they've got a large number of data centres either built or in the assessment process. We've also heard evidence that a lot of those data centres are being built in Sydney because of the submarine cables and being able to access that. Clearly, they're not being built for sovereign reasons. That data is for transmitting offshore. Coming back to that question of critical infrastructure, should we be making that distinction in terms of which ones we're accepting where?

DAN HUNTER: I think that has to be part of the development application process or the planning approval process. We have not necessarily seen evidence from people we've spoken to—and, Dave, you might want to come in here—that having those data centres located near the submarine cables effectively means that we're using them for export purposes. I think that's too simplistic an argument. Data centres need to be located near good fibre and good fibre-optic networks. That includes the submarine cables. They go back and forth. I think there are parallels being drawn here between gas and exporting gas offshore. I don't think we're building data centres and just exporting the data offshore. It goes two ways and it's quite a complex environment. We should be having the conversation, as part of the approval process, about what these data centres are for.

DAVID BORGER: I'd just say that some of the data centres in Macquarie Park and Ryde are sovereign data centres. They only store data within Australia. It's not fair to say that none of them are sovereign.

The CHAIR: How do you know that, though? How do you know what they're using it for?

DAVID BORGER: I know one of them is a sovereign data centre. They do a lot with Federal government data, highly classified information. They have a requirement to store data within Australia and not to let it leave. The other interesting thing about data centres is they definitely use electricity. They are large users of electricity. This is a challenge. We know it's a challenge. It's one that this Committee is dealing with at the moment to balance these things out. But in terms of renewable energy zones and large battery energy storage systems, they're often looking for long-term contracts with very large electricity users, so there's a relationship between data centres and the feasibility of making some of these projects viable, because they need these very large contracts. Even though data centres are using electricity and it's a challenge for us, they're also part of the solution of getting renewable energy zones up and running.

The CHAIR: Can I ask a question on that, and then I'll come to Mr Ayres, because he has been left out. In your submission, you make this point that many data centre operators are supporting new renewable energy generation through PPAs. It's something that the Data Centres Australia submission also said. Others have said to us that they can't actually name any examples of that taking place. Do you have an example of where a data centre has actually supported new renewable energy generation through a PPA?

DAVID BORGER: I don't have an example on me, no.

DAN HUNTER: We can take that on notice, Chair. I believe there are some in the planning process. There is a sweet spot here where there are big data centres and battery storage around renewable energy zones. That's obviously the future. We can take that on notice for you, have a look at that and try to provide some examples, if they're available.

The Hon. MARK BUTTIGIEG: Can I just ask a quick follow-up on the data sovereignty stuff? I just want to get my head around this. I struggle a bit trying to visualise it. When you talk about sovereign data centres, you've got this lump of infrastructure with all these digital zeros and ones sitting there on the chips. Then you've got comms pipes, fibre optics, to government instrumentalities, departments or whatever. Once that connects to the web, it's a function of security, isn't it? Just like internet banking, for example. I don't get the whole data sovereignty thing, because if it's a function of cybersecurity, then why is it so important to have it domiciled here?

DAN HUNTER: I'm not an IT person, but my simplistic explanation of that is it's where the data is housed. Data centres house our data. Our sovereign data centres that only have Australian data, they house it here in Australia. There's a pretty big question as to whether we want our government data and some private sector data housed offshore in other jurisdictions. That really needs to be looked at carefully. Having those Australian-only data centres does solve that problem. Just one more point on that, Chair, if I may—it's not just

CORRECTED

government that are required to keep their data onshore here in Australia. It's often anyone in a private sector high-secure environment—banks, anyone that deals with government in a contracting environment, you have to have your data housed here in Australia. It is much broader than government, and it's also a private enterprise issue as well.

The CHAIR: One of the things in your submission that stuck out was how the Investment Development Authority could throw things out of whack. My understanding is that the IDA is based on the Housing Delivery Authority, which I understand you have experience with. What is your objection with the IDA, and how do you understand it to be working?

STUART AYRES: We don't have an objection with the IDA. I think any vehicle that allows New South Wales to be more competitive at attracting investment into the State is a positive thing. What I do think the IDA should do is operate within a framework that is transparent and everyone knows what the rules of the game are. Our starting point and a core premise of our submission and the evidence that we'll be providing today is that the principles that have been set out by Sydney Water form a foundational framework which should be part of Government policy, and that policy should apply to the IDA as well. If you're trying to work out how you prioritise whatever type of infrastructure it is—and the topic of this conversation has been primarily about water—if there are requirements on the proponent investing in New South Wales to be able to support that through some sort of contribution or the delivery of their own infrastructure, that should be dealt with transparently by the IDA.

The Hon. JACQUI MUNRO: Just off the back of that, then, the IDA is attracting projects that are related to data centres, sure, but also energy and housing. How does that regime work in relation to the existing planning system and potential tensions between housing delivery and data centres?

STUART AYRES: I think the IDA can work complementarily with other forms of policy, provided it's playing by the same set of rules. There's a great opportunity here to be able to facilitate investment into the State. That investment can be concierged. Often there's complexity when you're dealing with multiple agencies, but the rules for any form of development should be clear and transparent for everyone and, for investors, that's what they want as well. They want clarity for the rules they'll be playing by. The IDA can provide that, provided it sits within the planning framework that exists across the rest of New South Wales.

The Hon. JACQUI MUNRO: Is there a risk that some types of development, like data centres, end up becoming more important than housing delivery?

STUART AYRES: The absolute premise of what we're talking about here is how you ensure that policy settings in New South Wales do not create an either/or proposition. We've got a capacity to have houses and data centres too, and to have houses and renewable infrastructure. These are not either/or propositions, and it would be a failure for legislators if they took us down a pathway that made it an either/or proposition. This needs to be about us setting a framework that allows for both of those critical assets that are critical to the economy and critical to how humans function in daily life. You've got to be able to develop policy that allows both of them to coexist.

The Hon. JACQUI MUNRO: But you're suggesting in your submission that that clarity isn't apparent at the moment, as in a pathway to achieve that is not clear at this point.

STUART AYRES: Yes, and the framework that I think you should be looking to follow is the framework that's been set out by Sydney Water. If you set that as a framework, it's embedded into development application decisions. Then you've got a set of rules. The people who are assessing those applications can assess against that clear criteria, that clear set of rules, and they also get the opportunity to balance against the contestability that exists in the community.

The Hon. SCOTT FARLOW: Just to that point, prior to your attendance I put the premise to Sydney Water that you'd put in your submission about the potential balancing of housing opportunities when it came to data centres. Their position on that was that it was effectively a position of the planning system, not one for Sydney Water. One of the things we've heard in this inquiry in the past has been a suggestion that there actually be special zoning for data centres, so that data centres can't get in off the back of either commercial or industrial zoning. I'm interested in what you think about having specific designations for data centres?

STUART AYRES: I don't know that you need to go to a special zoning designation. I think that sounds too onerous, is my immediate response to that. What you should be able to do is prioritise how land gets utilised by setting a clear framework and a set of planning policies that are delivered when any development application is assessed. If you undertake a rezoning for the purposes of commercial land, for instance, and there's a priority around the strategic use for that land—perhaps it's more closely associated with intermodals or freight and logistics assets. There's only so many places you can build an intermodal, and only so many places that you can build on land that's close to freight and distribution locations. You probably should build within your framework to prioritise commercial activity that supports the other infrastructure that's surrounding that, rather than allowing

CORRECTED

data centres to utilise that land when it could be used as other commercial land. We don't have a shortage of land in Sydney, or Australia. It's a big country. We don't often have a shortage of zoned land, particularly from a housing perspective. Our shortage is in serviced land, where the infrastructure can support the development.

The Hon. JACQUI MUNRO: I was interested in the partnership-based infrastructure delivery element of your submission. Could you expand on that further, please?

STUART AYRES: If we're operating in a constrained infrastructure environment, we think that there's always opportunities to allow innovation to exist in the market. Often these will be supported by private enterprises. We're very comfortable with the Government's position around who owns those assets, but that doesn't mean that we can't utilise private capital and private innovation around how you might be able to deliver water networks to support data centres. If that reduces the demand on the public, or if it allows for the prioritisation of, particularly, potable water assets to housing, then we think that would be a sensible approach to be able to do that. We also think there's really good opportunities around utilising infrastructure investments that come with data centres, to continue to expand the recycled water network across Sydney.

The Hon. SCOTT FARLOW: In discussions with Sydney Water earlier, they said that was effectively captured under the Water Industry Competition Act and that they would work cooperatively. Do you see that happening at the moment or do you think there needs to be either changes to that Act or changes to the model of operation?

STUART AYRES: I think there's always an opportunity, if you're operating in an infrastructure-constrained environment, to allow other sources of capital to be able to come into that market. Particularly, if the ownership of the asset and the regulatory framework and the standards around the construction of that asset are maintained, then that allows for an expansion of the asset. In an environment where Sydney Water doesn't have as much capital allocation as it would like, I think there's a real opportunity to bring innovation to the market.

The Hon. JACQUI MUNRO: Do Mr Borger or Mr Hunter have a comment on that element of private capital invested into the infrastructure required?

DAN HUNTER: I do, if that's okay. In a previous life I was a public servant and worked for Sydney Water, amongst other departments. Sydney Water has a long history of partnerships around wastewater. Desalination is probably the big one, and there are some great models around partnerships with the private sector to provide both potable and recycled water supplies. Again, I go back to some of my earlier comments. There's a sweet spot here where a lot of new housing developments now have compulsory water recycling attached to them. Recycled water is perfect for data centres. They are getting more water efficient all the time, data centres, but if we get the planning frameworks right, there's a sweet spot here for private sector to partner with government on these things. We have an example—who pays and how that happens and the pass-through needs to be worked out, but we have a desalination plant sitting there at the moment that is designed to double in capacity. It currently provides 15 per cent of the water supply; it could get doubled to provide 30 per cent. That may be part of the answer. I'm not suggesting for one minute that households should have to pay for that if it's intended for data centres, but there are solutions out there, I think.

DAVID BORGER: I think Sydney Water very much would like to look at doing more purified recycled water. California had this challenge. There was a meeting of the minds between the business community and the environmental community to actually convince the public that, in many areas, they had to actually use purified recycled water as potable water, but certainly for industrial uses. The other thing that's interesting is we mentioned industrial lands and some of the challenges that we've got there. We have a constrained environment in terms of the amount of industrial land that we have. We have amongst the highest prices in the world. We have very low vacancy rates, at around 2 per cent, so there's a broader challenge around how, as a State, we can actually generate more serviced industrial land. Queensland seems to be leading the way. It's doing a much better job than we are at the moment. They have targets. The Government actively invests and gets involved in coordinating the provision of infrastructure. I think, whilst data centres certainly have demand for industrial lands, the bigger challenge is actually having a better system that produces more industrial land and doesn't see businesses go, frankly, to Victoria or to Queensland.

STUART AYRES: Demand is a positive thing. It's a reflection of the fact that there are new customers coming into the market. Whether you're electricity or water, the real issue here is the capacity to expand supply. Our principle around utilising private investment to be able to complement or supplement the existing public investment, we think, provides a degree of protection for existing water customers that allows for new water assets to be embedded and supplemented into the system. This is just a reflection of being able to expand supply to meet the demand. The same thing would exist in electricity as well.

CORRECTED

The Hon. JACQUI MUNRO: Given our conversations about social licence—and we've made the comparison to renewable energy zones—do you think the business community has a role to play in engaging with local communities about the benefits of data centres? Are you considering that? What does that look like?

DAVID BORGER: One of our recommendations is actually that data centres establish a community benefit framework. We know some data centres are actually doing this already. As part of their development application process, they're working out how they can have a relationship with local universities to provide spaces for training and to upgrade skills. But we've seen other industries where this community benefit framework is required, so we think there's probably a role for government to try to work with data centres to establish that—so it's not just the good data centres that are doing this; it's all data centres that are contributing to local communities.

The Hon. JACQUI MUNRO: Can I just go a little deeper on that? Yes, community benefit sharing schemes, but you've already outlined that data centres have a benefit for the local business community. Separate to those types of community schemes, is there a role for the business community to be having, or to be a part of this public conversation on how data centres service small business and local business in the things we do every day, and how something that has been really talked about as an ephemeral, cloud-based thing actually requires land to help us operate, day to day, more efficiently?

DAVID BORGER: I think there's a challenge in that people often think data centres are there for data centres, but it's actually your data and my data and every company's data that's being stored there securely. We're all relying on these things, even though we see them as being separate to us.

DAN HUNTER: I agree that there is absolutely a role for business to play in the community engagement space. Again, I think it comes back to the critical infrastructure and the benefit sharing.

The CHAIR: Would you support mandating local content requirements?

DAN HUNTER: I would need to see the detail on that, but I would be happy to look at it.

The CHAIR: In terms of painting that picture of a more direct benefit to local community.

DAN HUNTER: Yes. If I can take that on a slight tangent, Chair, I think there are enormous opportunities here for the regions. In New South Wales, we've got some fantastic industrial hubs that are in our regions. Tomago, north of Newcastle, is a good example of that. Some of these regional industrial zones are needing to reinvent themselves and are needing to look at new industries and new places to go, and I think data centres is one that we should be entertaining. It's a good regional opportunity.

The CHAIR: I guess it's also the job creation side of this given that we know that past the construction phase, maintaining and running the data centre doesn't require a lot of jobs. Some people are saying it would be great if we had local content requirements because at least then you could see that it would have a direct impact on local jobs. Is that something that you—

DAN HUNTER: We would have a look at it. We haven't seen how that would look but we'd definitely have a look at it and form a view, yes.

The Hon. SCOTT FARLOW: Mr Hunter, in your submission you talk about suboptimal locations and the danger, effectively, if we don't have proper planning to work out where data centres could be that we might get suboptimal locations. Throughout this inquiry we've heard the City of Sydney talk about data centres not being a great use in the middle of the city, particularly pointing to the one in Pyrmont which you'd see if you're going on the Western Distributor. We're hearing from the City of Ryde this afternoon, who'll be making submissions on the data centre concentration along the metro line and how that's not optimal in terms of land use. Segueing on from your point about regional New South Wales, where do you see these locations being best for data centres? Where will they provide the best economic benefit and where will they have the lowest impact on other economic uses?

DAN HUNTER: I think that depends on the enabling infrastructure. They need good water supply, electricity and good fibre. At the moment, anyway, there is some demand for them to be near population bases. There's no point in building them in the middle of nowhere if they're not going to be utilised. Again, I think this comes back down to balance: balancing the needs of the local community, housing, infrastructure needs and finding that sweet spot. Dave, you might have a comment on that, given the Western Sydney impacts.

DAVID BORGER: The other challenge is that Oxford Economics produced a good paper which looked at the speculative nature of many applications for data centres, and that there were a number of developers who were applying for data centres that had no experience in data centre operation and probably there was a low likelihood they would ever actually build a data centre. I think if government could find a way of getting some assurance whether some proposals were actually fair dinkum or not, that would actually help us from a public

CORRECTED

policy point of view. My sense is that a lot of the activity is people applying to inflate the value of their land, which is fair enough—they can do that in a private market—but it's not very useful for government then knowing where the actual data centres are going to be located.

The Hon. SCOTT FARLOW: I don't expect you to have done any of this modelling, but you may have seen it or come across it. What sort of multiplier effect is there to hosting data centres in a geographic location? Is there an economic advantage dividend for New South Wales in being able to host—I think we've heard through this inquiry—the second highest number of data centres in the world?

DAN HUNTER: I can't comment on the actual numbers or the multiplier. I haven't looked into that as yet. What I can comment on is the opportunity. Just with the way the world is going, and at the moment, there's an enormous opportunity here for Australia to be some sort of centre of excellence. I'll preface it always by saying it needs to be balanced with the energy, water needs and housing but if we don't take advantage of this, someone else will. I won't go off onto the Federal budget tangent at the moment and how hard it is for business and to attract investment but, for various reasons, it is getting difficult in this State to attract new businesses and investments. We have an opportunity here with data centres. It needs to be done right, but we have an opportunity here for onshore investment, for some local jobs and for an industry that will be competitive on a global scale. If we pass up that opportunity, one of two things will happen: one, they'll go offshore; or, two, they'll go to another State. Now, we've already got pretty good facts that New South Wales is leading the way, and I think we need to take advantage of this opportunity.

The Hon. SCOTT FARLOW: In a sense, when we look at all industries, particularly new and emerging industries, we look at the ecosystem around them that supports them and those other opportunities. They might be opportunities that perhaps are more aligned with some of your membership. Can you tell us some of those opportunities, potentially, that you're seeing with data centre development?

DAN HUNTER: Yes, I'll go high level, and then I'll pass over to David on that. It's all the startups and those new businesses that are going to be heavily reliant on AI and data centres that will spring up around. It will act as a base and as an enabling infrastructure to attract those new ventures and attract international ventures who are looking for somewhere to start up, even though they're going to be paying more capital gains tax—sorry, I couldn't resist.

The Hon. SCOTT FARLOW: That's all right. Don't worry, all the hyperscale data centres are being used for Albo memes at the moment, so it's all right.

DAN HUNTER: That's right. It's those flow-on effects on all the businesses that, if we do this right, it will attract, and we will see new industries mushroom if we do this right.

DAVID BORGER: I think every business uses data centres, so we all can benefit from them. Certainly there are some data centres in the application process that are trying to find real connections with local communities. I have one where there's an MOU with a local university that tries to provide a pathway for students to learn onsite and to use some of the facilities to understand data centres. They're also working with the local TAFE as well to try and design programs that allow students at the local TAFE to get some real-world experience in a data centre and maybe come on board at some stage and work there.

The Hon. JACQUI MUNRO: This is a little bit of a cheeky question. It seems like there has been furious agreement amongst the vast majority of witnesses that there is a pathway forward, and a lot of the suggestions that have been made in different submissions are pretty similar. Is it just about getting on with it? The solutions are reasonably clear. We just need a process and a framework put in place as quickly as possible so there is clarity.

STUART AYRES: Yes, I think that's a pretty fair observation. I think the solutions to this challenge are available, and a lot of them have been put in submissions that have been provided. What doesn't exist is the framework for how you deliver those. To Mr Farlow's question earlier around contestability on land use, permissibility probably shouldn't be the only policy setting for how a data centre is determined within a commercial or industrial zoning. I mentioned earlier if there are commercial or industrial zonings that are alongside other strategic infrastructure, like transport, rail, intermodals and ports et cetera, you'd probably want to create a policy framework that prioritises industries that benefit from that other significant investment made by the public—probably complemented by the private sector—to ensure the multiplier out of that land is utilised effectively, and you create a hierarchy. That can be done by a policy setting. If you make it clear, data centres know where they are a higher priority and where they are a lower priority.

Once again, I come back to the point around having an established framework. I think Sydney Water has provided a strong starting point for the Committee to make a recommendation, particularly around how water should be prioritised. I think it's really important—the challenge exists here because we're operating in a constrained infrastructure environment. It's not just about saying, "Let's get on with implementing." It's about

CORRECTED

recognising that this challenge is going to exist because of other constraints that are already in the system. Sydney Water's IPART determination means that they're funded for 100,000 fewer homes in their servicing area across Sydney over the next five years than what the National Housing Accord target requires Sydney to deliver.

How do you meet the National Housing Accord target when your primary water servicing provider in the largest metropolitan location in the country doesn't have the water servicing capacity provided to it, from a capital allocation perspective? That's why people get a bit nervous around where assets are going to be deployed and what type of utilities are going to be allocated to them. Without that framework, you're going to see more and more of that contestability. The same thing exists for housing. We mentioned earlier that we already have lots of available land to build houses on, but 121,000 lots are constrained through a lack of water and sewer infrastructure.

The CHAIR: One of your recommendations that stuck out was this one in relation to an up-front, total servicing cap on water volume for data centres, either system wide or by region. How do you envisage that happening? Would it be a hard cap beyond which you can't approve any more data centres? Or would it be more like a soft cap and trigger additional requirements beyond that?

STUART AYRES: That you put a cap on the asset that's utilising the water?

The CHAIR: Yes.

STUART AYRES: If data centres are suggesting that there's a presentation of more water being required than what they actually believe they need, then a cap probably shouldn't be a problem, should it, if the amount of water that they think they're going to utilise is going to be significantly less. We think that provides an assurity and an incentive to ensure that data centre proponents are looking at ways to minimise their water use. This has been a fairly common practice across all forms of development. We try to minimise our water use. We live on the driest continent on the planet. You were talking about sovereign infrastructure. There's probably no more important form of sovereign infrastructure than our water utilities and water assets. We should always be asking people to try to minimise their water usage and be more efficient at it.

The CHAIR: And drive innovation as a result of—

GAVIN MELVIN: I think the other observation on that is that that actually helps Sydney Water's capital planning. Whether they deliver these assets themselves or whether they partner, they need to plan. They're probably going to plan more to a network level or a subregional level, so those caps will help. They can be set in consultation with the data centre sector to set a reasonable usage. But that helps Sydney Water to know what assets they need to build. If that's not in place and they're asked to put in more than they'd expected, then we have this tension. They've got limited resources, so where are they going to deploy their capital in time to deliver infrastructure? It benefits everybody to have that cap.

DAN HUNTER: Chair, may I go back to the previous question? I think there is a pathway through here, and there's a lot of agreement in a lot of the submissions that the role of government now is to get the planning settings right. I do think we need a digital infrastructure roadmap in New South Wales to bring this to fruition in the right way. As an eternal optimist, I think that the Government, the Opposition and the Independents will go away and bring this to fruition in a really sensible way. I want to make the point because I feel like we're—

The Hon. JACQUI MUNRO: That's right. You are an eternal optimist.

The Hon. SCOTT FARLOW: You are an optimist.

DAN HUNTER: I am an optimist. I'm also optimistic about the public service, as you know, as well. Given we're coming close to a close, Chair, these sorts of opportunities aren't necessarily about corporatising and big business; they're about people, community and jobs. A lot of us with children don't know what jobs they're going to be doing in the future, and these sorts of things are critical infrastructure for the future jobs of the next generation.

STUART AYRES: Perhaps I used to be more of an optimist when I was employed in this building than my time outside of it. If I took something like the Housing Accord targets, in the last 12 months we're at around 44,900 homes delivered, against an average housing target over the course of the five years of the Housing Accord of 75,000. If you're significantly underweight on that, and there's contestability on base-level infrastructure, I think it's absolutely critical that you are providing a policy framework that ensures potable water is prioritised for the thing that is most requiring potable water, that being a house, I think there are really strong opportunities to leverage further investment into recycled water, which benefits everyone, and we've spoken about the benefits of the cap around the servicing plan for Sydney Water. We are continuing to operate in this constrained environment where we don't build enough homes, both houses and apartments. If you're not doing that—and one of the main reasons for that not happening is a lack of serviced land, and the number one service that causes a constraint on

CORRECTED

housing delivery in New South Wales is water servicing—you've got to have a functional framework to ensure competition for the scarcity of that asset is managed properly.

The CHAIR: Has this Government got their priorities backwards, then, if they've got 15 data centres being fast-tracked and facilitated but we've got tens of thousands of homes that can't be built because of a lack of water infrastructure?

STUART AYRES: What are we? We're halfway there on meeting our housing accord targets, but we're probably living on a prayer to get to the other end. I don't think it's about priorities; I think it's the capacity for them to be able to manage the complexity of what exists there. We're not asking the Government to fund the water infrastructure to support data centres; we're asking the Government to create a framework so that that water infrastructure can be most appropriately utilised. Like any other developer who's making a contribution to infrastructure, we're prioritising their contribution into the water assets that best suit New South Wales so that you don't have to take the scarcity of potable water and try to spread it too thinly.

The CHAIR: We often talk about resource constraint driving innovation. In most cases, that appears to still hold true in today's economy. We've been talking about driving innovation when it comes to water use and energy use and other things by capping or being responsible with planning. At what point do we also look to limit the amount of data centres in order to drive AI model efficiency, so they don't need as much space?

STUART AYRES: My immediate response to that, and I have to say it's probably more instinctive than something that's immediate to the UDIA for that instance, is if we thought retrospectively around other infrastructure and said, "Should we have constrained technology advancements because that technology tested and challenged the way we thought about the world"—what if we had constrained telecommunication towers when mobile phones came on board? Would we have ever got to the iPhone? My natural instinct here is, when technology drives innovation and drives productivity, it's always going to stretch us and test us and challenge us. But that's what we elect legislators for: to create a framework that allows businesses and citizens to coexist in an environment that's beneficial to all of them. We're just here to help you.

DAN HUNTER: I think it's interesting to look at the example of Singapore. Singapore had a data centre boom. It's still got one now. It paused things for a while and recreated a system to try and get the best data centres built in Singapore, with the highest environmental standards and the most economic efficiency. I think there are probably some lessons there from other jurisdictions where they've said, "This is a critical enabling infrastructure. It's a business we want to be in. It's good for our economy, but we know it's also competing with other areas of the economy. So how do we actually ensure that there's competition within the market to get the best data providers located?"

The Hon. SCOTT FARLOW: Are you advocating for a pause?

DAN HUNTER: No, I'm not. I'm definitely not advocating for a pause. I'm just saying that not all data centres are equal. There are different types of data centres. Not all data centres have the same environmental standards. There's probably an opportunity for government to require the sector to improve over time. It would be useful to look at the example of Singapore and what they've done there. They've said, "We need this industry. It's got to be here, but we don't want every single data centre to survive. We want the data centres that are going to have the most impact on jobs and the economy and the best outcome for environment and resource use."

STUART AYRES: For the avoidance of doubt, what Singapore presents as an example is the benefits of a strong strategic planning framework—the benefit of clear policy settings that allow for the prioritisation and use of land and strong investments in alternative water resourcing. I think we would benefit immensely in New South Wales and across the country if we recycled as much water in New South Wales and Sydney as what they did in Singapore. Then the concept of the constraint on water usage around data centres would probably not be as significant an issue. The example out of another market like Singapore is that strong policy frameworks and clear and transparent settings are in everyone's interests.

The CHAIR: I think we'll end on that note, on that nice, collective common ground.

The Hon. SCOTT FARLOW: Mr Hunter's optimism has come to the fore.

DAN HUNTER: Look, I'm infectious.

The CHAIR: In relation to the questions taken on notice and supplementary questions, the secretariat will be in touch. That concludes our time with you. Thank you very much.

(The witnesses withdrew.)

(Luncheon adjournment)

CORRECTED

Mr PETER EPHRAUMS, Member, Lane Cove Responsible Planning Group, affirmed and examined

Mr PAUL TRAINOR, Member, Lane Cove Responsible Planning Group, sworn and examined

The CHAIR: We are now back with our next witnesses. Would you like to commence with an opening statement?

PAUL TRAINOR: Chair and members of the Committee, thank you for the opportunity to appear today on behalf of the Lane Cove Responsible Planning Group. Peter and I represent a community which will live permanently with the consequences of government planning decisions that allow clustering of hyperscale data centres in Lane Cove. Whilst we recognise the importance of digital infrastructure, we do not support reactive tick-the-box planning based on outdated regulations. Instead, we're advocates for strategic and holistic planning that delivers fit-for-purpose infrastructure for all stakeholders.

In 2025 Lane Cove was named the most liveable suburb in Australia, valued for its safety, our beautiful national park, wildlife and strong sense of community. Yet it is now on track to become one of the largest concentrations of hyperscale data centres in suburban Australia. A recent proposal is from Goodman Group at 12 Mars Road. Located at the residential end of Lane Cove West business area, the site is zoned to bring a diversity of business and employment to the area and has building regulations reflecting that adjacency to conservation areas, residential communities and one of the busiest sporting spaces on the North Shore. Operating 24/7, the proposed facility is 22,000 square metres, contains 49 diesel generators and massive cooling infrastructure, and will be substantially higher than LEP regulations despite homes being less than 20 metres away, a public school being within 160 metres and it overshadowing the Lane Cove Community Nursery. In short, it will permanently dominate the surrounding area.

Goodman's proposal is only one of several being considered within this small precinct. In the planning process, however, the cumulative impacts of multiple projects are not being assessed. If all proceed, data centres will occupy around 40 per cent of the Lane Cove West Business Park. Such unfettered development is not an economic miracle; rather, it reflects a failure of planning. Community concerns have driven a strong public response to the Goodman proposal. However, these concerns are not just theoretical. Recent issues surrounding the nearby AirTrunk facility have already exposed serious gaps in the current regulatory system, yet expansion continues. If the system is struggling now, how will it cope with much larger clusters, more industrial-scale diesel generators, further degradation of national park land, greater pressure on electricity and water utilities, and cumulative impacts from excessive noise? Communities are increasingly concerned they will carry permanent impacts while private corporations benefit.

Lane Cove is becoming a case study in what happens when infrastructure growth moves faster than strategic planning. Governments around the world are already recognising impacts of hyperscale data centre clustering. New South Wales must not be left playing regulatory catch-up. Ultimately, if data centres are truly critical infrastructure, they must be planned to a far higher standard and not cluster beside homes, beside schools and beside recreation areas without enforceable statewide planning rules and cumulative impact assessments. Therefore, we call for an immediate moratorium on further hyperscale data centre approvals in established residential areas until appropriate safeguards are in place.

The CHAIR: Did you want to add anything, Mr Ephraums?

PETER EPHRAUMS: No, Paul has covered the essence of our submission very well.

The CHAIR: We heard a little bit about Lane Cove when we had the council in the last hearing that we had. I think there's so much in what you've said in your opening statement and in the submissions that really goes towards the very real impacts on community, many of which the community still isn't even aware of. You already have a number of these data centres in your local area, and there are now more planned. What's your impression of how involved the community has been before this date when it came to approving those existing data centres? Do you think people know in the community?

PETER EPHRAUMS: I think it would be fair to say that the level of community awareness around the first data centre that went into the business park was probably quite low. That's the AirTrunk facility that's currently in operation and currently planning an expansion. The level of interest has ramped up since the Goodman proposal was lodged because it's so close to schools, residential areas and Blackman Park. I think you can gauge some sense of the community interest in the fact that there's a petition about the Goodman data centre that has currently received in excess of 1,200 signatures and that, with a fairly mangled notification process in terms of advising people about the data centre and the timescale in which they had to respond, we still managed to get 358 submissions from people in the community to the planning authorities in opposition to it. I think community interest has ramped up quickly and is still growing.

CORRECTED

The CHAIR: I've been to the site of the new proposed Goodman one, and I stood there and I had the site across the road from me and then down the road was the school. Can you explain to me why if I had a child at that school, I should be concerned? What impact could it have on my child who's 60 metres, or whatever it is, away from the facility?

PETER EPHRAUMS: In terms of the school, the concern from the school community expressed by the P&C was consistent noise, and remember the kids spend a fair bit of time outside every day. They'll be somewhat sheltered in the classrooms, but they won't be sheltered outside. The school operates from quite early in the morning till at least six o'clock in the evening, so the kids are going to be exposed to that in their after-school care and in their time outside of the classroom. Frankly, I'm pretty sure the classrooms aren't air-conditioned, so I expect they want to open the windows at certain times of the year. If this goes ahead, a lot of extra investment is going to be required to protect the amenity of the kids in those rooms.

The other issue that the parents have—and I haven't read their proposal in detail—is around the noise from the diesel generators if they come into operation. This facility is classified as mission critical, which means that the service provider has a limited time frame to restore service in the event of an outage, and essentially the surrounding community is coopted into whatever nuisance, whatever noise, whatever disruption is required to bring that about. Those generators will come on at any time of the day or night if needed, and they will be very loud. How loud? We don't know because we haven't been told. It has been sort of put to one side as, "This will hardly ever happen; therefore, we don't think it's reasonable that we meet community noise standards, so we're not going to."

The CHAIR: But also the pollution, presumably.

PETER EPHRAUMS: The pollution is a significant issue if—if—you get to a situation where those generators run for a reasonable amount of time. Notwithstanding that power outages are assumed to hardly ever happen, based by the way on the experience of Eastern Creek—I don't know why they didn't use data for North Shore because we have plenty of power outages in Lane Cove—the generators will all come on and they will use their bunker of fuel in 1.2 days. There's 1,014,000 litres of diesel being stored onsite to run those generators. On the published consumption figures for those generators, that's only enough to run them for 1.2 days before they need tanker replenishment. In the environmental impact statement, because this is assumed to never happen, they explicitly state that they won't be fitting any mitigation for noxious gases to the equipment, nor will they be fitting any monitoring equipment to measure how much is produced. The residents of the school are quite concerned that, in a significant outage, particularly with the wind blowing in the direction of the school, the kids will be bathed in exhaust gases.

The CHAIR: At that one, you've got 49 diesel generators, on my understanding.

PETER EPHRAUMS: Yes.

The CHAIR: But perhaps we could talk, Mr Trainor, of the idea of the cumulative impact because it's not just them, is it? Every one of those data centres has it.

PAUL TRAINOR: That's correct. As it stands, we've got AirTrunk, which is a 120- to 140-megawatt centre that's expanding. It's been approved for another 45. They've got another facility that is with the department of planning at the moment, so I think that's another 140; and then you've got Goodman, of course, which is around 90, but of course it varies in the documentation. That's another sticking point for all our community. Then you've got the one that's going through the IDA process under DCA Alliance, and there are no details around that at all. Then there's a sixth facility that was sold very, very recently on the basis that it is suitable for a data centre. That, on the same size and scale if you build it out, is probably another 140 as well. Extrapolate the number of diesel generators using, say, 49 for Goodman at 90, you're getting up to 400, or something like that, potentially in that little, small precinct. The furthest away is about 500 metres. As you say, Chair, if you've got all those running at the same time, what is the impact in the direction of the wind? There's absolutely nowhere cumulative impact statements within—everything's run on a singular basis, so there's nothing that says what the true damage could potentially be from noise, noxious fumes et cetera.

The CHAIR: We've heard on a previous day of hearings that we have a tendency for data centres to cluster. Once you have one, you do tend to have a few around the same area for a variety of reasons, but from what you're saying in your submission, the approvals are done as though that's the only one in the area.

PAUL TRAINOR: Correct.

The CHAIR: What you seem to be saying is that the cumulative impact of things like noise and the potential of using generators and things mean that these should really be considered on a cumulative basis?

CORRECTED

PETER EPHRAUMS: We certainly think they should be treated on a cumulative basis. Dealing with them as they arrive, one on one, seems to be the process that needs to be followed, according to the rules. But everybody knows that there are four more coming down the track. To me, it seems fairly clear that they must be assessed altogether. In the current sound standards that apply, which date from 2017, the cumulative noise impacts are generally a small adjustment based on the noise of the rest of the industrial area. The reality for the Goodman proposal, if it goes ahead, is the facility next door to it won't be a normal industrial facility for the Lane Cove industrial park; it'll be an even bigger data centre. How big? We don't know, because the details have not yet been revealed. But we do know, from what has been mooted, that we could end up with data centres from the eastern boundary of the industrial park all the way down to the boundary on the Lane Cove River.

The CHAIR: We've been talking a lot about the idea of rationing out the licences for particular amounts of data centre capacity. We've had comments that maybe we need to cap the amount of water usage for data centres in a particular area. There have been ideas around that. It sounds like we also need to have a cap or some other mechanism of saying, "No, in this particular area we already have the capacity for this amount of diesel pollution or this amount of noise," and we really ought to be capping it on a cumulative basis. Is that how you'd see that cumulative assessment working?

PAUL TRAINOR: Yes, I absolutely would say that. It appears at this stage that the only constraining factors will be around electricity and probably more water and the amount of water that's required. It's through basically a narrow path up through the suburban housing of Lane Cove West or Lane Cove itself. By that being the constraining factor, it sort of has held back, I suppose, any consideration around, again, noise, noxious fumes or whatever. If you go through all the planning processes, as Peter just mentioned, there are no hard and fast rules. The constraining factors seem to be more the practical things rather than actually really thinking more broadly, more deeply about the whole planning process in such a small, constrained area versus the other precincts around the Sydney Basin and beyond the Sydney Basin that might be more appropriate. We might be able to get more efficient use of land and more efficient use of utilities being delivered to those data centres, rather than just the cut and paste, one by one.

The Hon. SCOTT FARLOW: The AirTrunk facility is operating in the area at the moment. What has been your experience with that facility so far? I note that it is situated a fair bit further away from the residential properties than the Mars Road proposal.

PETER EPHRAUMS: One of the key impacts of the AirTrunk facility, thanks to the reporting of the local newspaper, has been to alert the local community to the deficiencies in the regulation of data centres in Australia at the moment. That facility is currently exceeding its forecast sound levels during the day and substantially exceeding them at night. There are other deficiencies in their documentation which have not been rectified. The AirTrunk facility, while it is impacting people living on Epping Road, is possibly not impacting—from a sound point of view—too many people in Lane Cove West, at this point. But it has certainly alerted us to the risk of hypothetical forecasts of noise converting into actual disasters when they get built.

PAUL TRAINOR: Can I just add, an extension of that is one of the pieces of work that should have been supplied to Fire and Rescue NSW has not been done. Fire and Rescue have actually stated that in an open forum—that it hasn't been supplied to them. Again, to Peter's point around the risk that potentially could occur further into the industrial park with the Goodman proposal, it's nearly 200,000 kilograms of lithium ion batteries as well as the fuel load that sits there. That's across all these different data centres. There doesn't seem to be any oversight by any of the regulatory authorities that puts penalties in place for AirTrunk, who have missed all their obligations. The community just sits back and says, "Well, what can we do?"

The Hon. SCOTT FARLOW: Mr Trainor, you mentioned the challenges when it comes to water infrastructure, and particularly the limited water capacity. I know that we heard commentary from the council last week and I know that your local MP, Anthony Roberts, has raised concerns about the water supply levels in the area as well. Has there been an impact by the data centres already in location on the local community when it comes to water supply issues, or is this something which is more of a concern going forward?

PAUL TRAINOR: It probably goes to highlight the secrecy behind some of the work that's being done in our neighbourhood. To get water to AirTrunk, it has to come through our streets, into the industrial park, under the roadway, all the way to AirTrunk. They've been working since September 2024 on trying to get that new piping in for AirTrunk. They use our piping—as in, the community's piping—in the interim until they get it sorted. But month after month, they're still having problems, still having leaks. Even as we came here this morning, we had to stop for the go-stop person, because they're still digging deep to try to find out where the problem is. That's 19 or 20 months worth, and that is one facility. I'd say it's probably a four- or five-times scale of water needs coming over, potentially, the next two, three, four, five years.

CORRECTED

The Hon. SCOTT FARLOW: Apart from the stop and go signs, what's the impact on you, as local residents? Are you finding that your water's turned off more often, that you're not able to access water at home?

PAUL TRAINOR: There have been times when they have turned it off to work on it, to flush the system. A lot of it is more the traffic snarls that go around Lane Cove because, remember, the public school is there in the corner. Parents are picking up their kids et cetera. There are a number of other impacts, and our water pressure has dropped because AirTrunk has used, let's say, the community's pipeline over the last period of time.

The Hon. SCOTT FARLOW: They're currently putting in another pipeline, are they?

PETER EPHRAUMS: They're still working on it.

PAUL TRAINOR: That's our understanding, and it is not clear, because Interflow is their consultant and operator and they, up until very recently, have given us no information—just saying it's for the community—whereas we've subsequently found out it was for AirTrunk.

PETER EPHRAUMS: The other things we've had impact from were all the nature strip on Wood Street, and other streets leading down to the end of Mars Road being dug up to fit cable. I've ended up with two pits in the nature strip, and I've already had a truck turn up at three-thirty in the morning, start a winch and start pulling cable. I went out and said, "What are you doing?" "Oh, there's an outage. It has to be fixed. We've only got six hours to fix it." I said, "At three-thirty in the morning you're running a big winch and you've parked on my nature strip." Mission critical—bad luck for you, if you're a resident.

The Hon. SCOTT FARLOW: Who was that being conducted by?

PETER EPHRAUMS: That was being conducted by whoever does the cabling for AirTrunk. I didn't get the name of the company. It would be a subcontractor, but they have service obligations to AirTrunk, and they will meet them come hell or high water.

The CHAIR: I think the point that you raise in the submission, and that you've just mentioned now, around the approval conditions that are put on these sorts of approvals—we have this assumption, I guess, that residents raise concerns. The response from the planning authorities is to say, "Okay, fine, we'll put these particular restrictions on the development and on the company and its operation." But then when one of those conditions is breached and isn't complied with, what are the options for the community? What should be the options, and what's your experience?

PETER EPHRAUMS: In an ideal world, there'd be somebody responsible for meeting all of those objectives—somebody with a balance sheet, somebody who could be taken to court and held to account. I don't think that's the situation we're in at the moment. I am surprised that AirTrunk has not been made to remedy the noise problems that it's created. I'm very concerned that, again, a very hypothetical noise report like has been submitted for the Goodman Fielder site could end up with noise levels substantially higher than have been predicted. And what would we do? We have a \$900 million facility sitting there. Is anyone going to turn it off just because it doesn't meet its noise standards? No. The residents, the users of Blackman Park and the animals will just have to wear it. I think that's very risky, given how close it is to the park, how close it is to users and how close it is to the school.

The CHAIR: Given your interactions so far with these data centre operators, do you have any faith that they will be good corporate citizens in that regard?

PETER EPHRAUMS: I have only had very limited interaction and really just watching this Committee process and talking to people at the council as well. This is only in one example, so it may not be fair. I'm very surprised the industry has refused to front up here and actually represent itself. It sent somebody last week who I think you characterised as a lobbyist and clearly, under slight pressure of questioning, wasn't really able to answer anything. So I'm surprised that they haven't participated in that. I believe from discussions with local council members they didn't find the process particularly constructive when they were trying to get their points across either. I am very nervous about it, but it'd be unfair to, say, apply that to the whole industry, perhaps.

The CHAIR: It's very hard to build the so-called social licence when you're not engaging with community. I share the frustrations that we haven't had any of them turn up.

PAUL TRAINOR: As a follow-on with regard to Goodman and Goodman's approach—it comes down to the planning process and the planning authority. The department of planning is up-front with their SEARs, saying "This is how it should be done," talking about community consultation. They actually say, "You have to run a comprehensive, open and transparent consultation, and then you have to justify the approach that you're taking with your consultation method." They've employed a group called Urbis, who've written a whole EIS—not Goodman—and they've signed off on it as the REAP, or the registered environmental assessment practitioner.

CORRECTED

They've also done the visual impact survey. The starting point, I suppose, is a social impact assessment. This is where they underplay everything so they don't have to at the next step, in the engagement survey—and this is what upsets the community, because you're not engaging.

They've downplayed everything with very limited work. Even though that they had 86 per cent of respondents to "Would the proposed development impact the community?" saying, "Yes, it will impact it negatively," they still say this has a low to moderate impact. Therefore, with their engagement survey run by Urbis, they actually downplay everything. We as a community don't feel engaged in this process. Then we see their EIS—again put together by Urbis and signed off—is full of mistakes and errors. You just think, is the department of planning going to base their assessment on that document that is inherently not independent, terribly put together and fundamentally flawed? So us, as a community, think how can we—dare I say—trust Goodman Group if that has been their overarching approach all the way through this process?

PETER EPHRAUMS: I know we don't have time to deal with that proposal in detail; we don't have time much at all. In their social impact statement, the first thing off the bat, the first thing they talk about, is the impact on the environment. To build this project they're going to cut down 90 trees, endanger the surrounding riparian areas and dig out 69,000 cubic metres of sandstone, which is equivalent to a 41-by-41-by-41 metre block of sandstone, or imagine a 10½-storey-high cube of stone. How are they going to do that without damaging the environment? It's implausible; it's impossible. But their assessment is that it will have negligible-to-low positive impact on the environment. The moment you see that, you know you're in for a laugh if you weren't living there and a lot of distress if you are. They have downplayed or misrepresented just about every aspect of community—apparently putting in a 24/7 noise source, with 49 generators, noxious fumes, damage to the environment and 33 months of construction noise if it doesn't rain, is going to have a positive impact on our lifestyle in Lane Cove West.

The CHAIR: The sheer logistics of trucks having to get that amount of sandstone out, I don't understand how that was—

PETER EPHRAUMS: And you can never fix it up. It's not like you can knock the building down and get back to the bare earth. In the meantime, they claim that enhances the Aboriginal heritage of the area too, whatever that means. I could talk for hours about how outrageous this particular proposal is, but what's really drawn us in is that it has exposed us to the implications of more of them piling up. They're potentially very significantly negative for communities and for the environment.

The CHAIR: I think that's the key. When I was looking at your submission, even if you defeat this Goodman one, you've got another at least five or something in the pipeline.

PETER EPHRAUMS: Now that we know about clusters, we know we've got a real issue brewing in our suburb.

The CHAIR: We need some guardrails. That does unfortunately end our time. Thank you so much. It's really useful just to get that human impact. Understanding the interactions between yourselves and the proponents has been interesting to hear as well. To the extent there are supplementary questions, the Committee secretariat will be in touch. Thank you.

(The witnesses withdrew.)

CORRECTED

Mr MICHAEL McCABE, Director, Communities and Place, Wingecarribee Shire Council, sworn and examined

Ms JESSICA ANN LINTERN, Manager, Strategic Outcomes, Wingecarribee Shire Council, affirmed and examined

The CHAIR: I welcome our next witnesses. Would you like to start with an opening statement?

MICHAEL McCABE: I'd firstly like to thank the Committee for the invitation for Wingecarribee Shire Council to participate in today's hearing. My colleague Jessica Lintern and I will be representing Wingecarribee Shire Council. It's a regional shire council located on the Wingecarribee River, approximately 75 kilometres south-west of Sydney. It has a population of approximately 54,000. The shire comprises the three main towns of Bowral, Mittagong and Moss Vale and is heavily reliant on tourism and agricultural industries. Earlier this month Bowral was named the best country town and ranked fifth overall in the *Australian Traveller* top 100 best towns in Australia. The nearby town of Moss Vale has established an emerging development interest in data centres and power plants as a direct result of the data centre industry. At this time, we have a DA that is subject to a deemed refusal. We had a section 34 in the Land and Environment Court this morning, where I was in attendance. We had a State significant development application for the same site, which has SEARs issued by the planning secretary. It's a very dynamic issue in our shire.

The Hon. SCOTT FARLOW: Thank you very much for your attendance today. We've heard previously that there were other alternatives that were looked at on this site in the past in terms of power facilities or mining facilities. Is that correct?

MICHAEL McCABE: I'm not sure if you're thinking of the Hume Coal site, which is a former mining site, or whether it's the site of 30 Douglas Road at Moss Vale.

The Hon. SCOTT FARLOW: Was there any proposal on that?

MICHAEL McCABE: Not for anything data centre related.

The Hon. SCOTT FARLOW: There was some mention this morning; that was all. I was intrigued because it seemed to be a fair way away from Moss Vale. In terms of your experience with the deemed refusal, we've heard quite a lot that most of these are proceeding through State significant planning pathways. This was actually put to the council first off?

MICHAEL McCABE: This relates to a site at Moss Vale. An initial DA was lodged in 2021. This is now the fifth DA for the same site. You basically see a cumulative impact on the site. In parallel, we now have a State significant development application related to a proposed gas-fired power plant, not a data centre. The community has become more active in response to the DA simply because of the SSD application being afoot. The other thing I'll point out is around the SEARs for data centres in general. There are generic SEARs that the department of planning have, but this proposal is for a power plant, so those SEARs do not apply and separate SEARs were issued. In comparison, with the DA at the moment, there are cumulative effects. Data halls are added. There has been no master plan approval for this site—so each staged DA comes on, let's say, and this one proposes power generation of 16 megawatts, where the proposed SSD is 673 megawatts, which I believe will be one of the biggest, if not the biggest, power generators in New South Wales if approved and operating.

The CHAIR: With that, this is only to power the data centres—

MICHAEL McCABE: Exclusively.

The CHAIR: —and it's not going to go to the surrounding community at all?

MICHAEL McCABE: Not at all. I believe it'll be the biggest privately owned power generator in New South Wales. The other ones like Kurri Kurri, which is under construction at the moment, benefit the public.

The Hon. SCOTT FARLOW: Sixteen to 673 is quite a big disparity. On that basis, that'd be powering a lot of data centres, wouldn't it?

MICHAEL McCABE: As I said, there's no master plan approval for the site. The proponents of each DA add a new hall or a new number of halls with the relative energy needed, where the SSD is proposing power for a much bigger campus. There's no master plan approval for that much bigger campus, so the detail of that is only limited to a scoping report and the SEARs at the moment. We're awaiting an EIS being issued by the proponent in future months. When you talk about power, it's actually the cumulative effect. That 673 is related to the SSD only. You've already got established megawatts that have either been approved or are pending approval through the courts or a decision through the courts at this moment, so it could be over 700 megawatts.

CORRECTED

The Hon. SCOTT FARLOW: With respect to alternative uses that the council would see in the industrial park there, when council first approved a master plan for the area, what were the sort of uses that council envisaged would take place in that industrial park?

MICHAEL McCABE: The location is in a precinct called the Southern Highlands Innovation Park. We actually have an adopted master plan with council, and there's actually a planning proposal proposed to rezone part of that industrial park from E4 to E3 zoning. The rezoning comprises about 70 hectares, but the hectares of the innovation park itself, I think, are like a thousand hectares. We have it separated into three distinct precincts. There's an agricultural precinct and an innovation precinct. What was the other one?

JESSICA LINTERN: Education and research.

MICHAEL McCABE: The shire is open to stimulating the local, regional, State and national economy, but in a sustainable way. There is land and there is opportunity to plan things well. We're just not convinced at this stage that that is being done.

The CHAIR: When I think of the Southern Highlands, I think of weekends away by the fire with wine and having a lovely time in nature; I don't think of heavy industry. Was it ever envisaged?

The Hon. SCOTT FARLOW: You haven't been to New Berrima then.

The CHAIR: No, clearly not; I'm living a privileged life. But was it part of council's plans to have a massive gas plant and heavy industry on that site?

MICHAEL McCABE: No. The DA is related to that site—I'm giving that site just as an example. To date, the DAs have largely complied with controls, but this current DA with the deemed refusal has height breaches of several metres over a height limit of 20 metres—approximately 24-metre exhaust stacks. That is just for the 16 megawatt and that associated hall. If you imagine the 673 megawatt—of which we don't have detail yet—what infrastructure is needed there and what that impact has, both environmentally, whether it be air quality, carbon emissions and visual impacts, they're significant.

The CHAIR: Has the impact on tourism been modelled? Is there any sort of assessment of that?

JESSICA LINTERN: With this area—the Southern Highlands Innovation Park particularly—it is industrial land, so it's a regionally significant employment area. In terms of tourism, no, it's not a high-visitation area in the highlands. However, I think with data centres, one of the key things as well is the employment figures that actually happen post-construction. We know industrial land availability is decreasing and there are not a lot of areas left like the SHIP that actually provide employment lands. With data centres, they usually have a very small full-time employment capacity, so when you start to look at the cumulative impacts of having more data centres potentially in areas like the SHIP, they're actually not going to deliver more employment opportunities for our community.

The CHAIR: I must explain myself—I've been out to the site, so I've seen where it is. From my perspective, it's beautiful. As far as you can see, there are trees and lots of grassy areas. It doesn't look industrial, but I understand it has been zoned industrial. When the proponent first decided that they were going to build data centres there, what was the interaction like with council? Have there been discussions directly with the proponent?

MICHAEL McCABE: Jess and I have been with council under two years. The engagement with the proponent has been going on at least five—if not 10—years with council. In relation to when the proponent first introduced themselves to me, they were describing what they were planning to do in relation to data halls. That's the way it was described, not so much to the scale of energy consumption of what's being proposed now. I also believe, and it was represented to the proponent this way and I believe they're representing it this way to the community, that they entered this market with an envisaged demand from customers, let's say, but that has expanded. So, in response, they're wanting to expand their offering as well. That has been the conversation that they've had to me in only the last 12 or 18 months.

The CHAIR: We've heard about clusters being in places where—we've heard about the ones in Sydney being connected to the submarine network cables and in places where there's good fibre optics, those kinds of things. Why here? Why do you think that data centres want to set up in your shire?

JESSICA LINTERN: I've heard it's because of the high rainfall.

The CHAIR: Because of the cooling?

JESSICA LINTERN: Yes, and water storage.

The Hon. SCOTT FARLOW: So potatoes and data centres?

CORRECTED

JESSICA LINTERN: Yes.

The CHAIR: Interesting. And going from 16 megawatts to 600—

MICHAEL McCABE: It's 700.

The CHAIR: Seven hundred. How does that happen? Was this an honest underestimation by the proponent? Do you dare to—

MICHAEL McCABE: It wasn't proposed until the last six months, really. They lodged an application with the planning secretary for an SSD. Details—apart from their website, the SEARs and their scoping report, there's not much detail there. Obviously, they're going through a process of a DA, a process for a deemed refusal in court. We're dealing with that information. But that is a DA on its own, 16 megawatt. I suppose there's more information to come from the proponent when they do their EIS and if the application goes forward.

The CHAIR: The fact that they've already got a little bit of a footprint and they're running some diesel generators already, presumably, does any of that help their process now, when they're looking at this massive—in terms of the planning pathways? I'm just going to put it out there. If I was trying to be a bit tricky, I'd go in, perhaps, and start up all the little ones and get my little ones approved. Having got that footprint, I would perhaps then go for the larger one, knowing that then I'm going to get a more favourable pathway. Do you think that's—

MICHAEL McCABE: I think what you're describing is not unusual in the development industry, regardless of proposed uses.

The CHAIR: A very diplomatic answer. Thank you. What's the next step then for the community? I have seen there's quite a lot of backlash already. How, as council, do you try to facilitate better consultation? What do you think your options are at this point?

MICHAEL McCABE: At the moment we have a profile on our webpage to do with the site, and it simply directs readers to look at our DA tracker, which has all the details of the DAs, past and present, as well as the details of the SSD on the State Government's website. We had a notice of motion by one of the councillors, in recent months, that gained community interest as well. Council resolved to await the EIS to understand the project further, because there is a limited amount of information on the SSD at the moment. It's hard for us professionally to recommend a position of council when there is such lack of information.

The CHAIR: In your submission you recognise that data centres are critical infrastructure. We delved into that a little bit with previous witnesses, around how much of it is critical and how much of it is not actually critical and that, in order to get community buy-in, there needs to be some element of the community understanding what they're giving things up for. If you're going to have a massive gas-fired power station and a massive data centre in what I would view as a quite rural area, is it worth it? What is it being used for? Do you have any visibility over what this particular proponent is planning on using this particular data centre for?

MICHAEL McCABE: The proponent's representations to me are based on data of varying customers, particularly even government customers—Federal Government—but I'm not aware of the detail of that. They've made representations to me because of those arrangements. They're accelerating and expanding their offering. That's their desire. In relation to the impact, I was out at the section 34 LEC onsite conference this morning before it went back to council offices. There are hundreds of community members concerned. There were speakers addressing the commissioner. Those speakers were specifically objectors to the DA. They mentioned impacts to lifestyle, tourism and the speculative approach that business can have compared to what a resident or a local business can have in relation to entering or exiting or expanding offerings within the shire, let's say.

In relation to agriculture particularly, there was a nearby farmer talking about their farm and the numbers of cattle and the impact on our regional livestock exchange, which is growing in demand. The reason why that's growing is because other livestock exchanges have closed, so we have more head of cattle coming through that than ever before—record numbers. This farmer was describing what they speculate would be the impact of them actually even being able to sell their cattle or raise it, simply because of the environmental impact based on the air quality and gas emissions associated with the gas-fired power plant.

The CHAIR: One of the things we've not discussed in relation to this particular potential data centre cluster—whatever we want to call that; "the ship" is what you refer to it as—is the water usage. What's the potential water usage of this project? Have they said?

JESSICA LINTERN: We've been assessing it on a DA by DA stage, because the master plan for the site hasn't happened. It is really difficult to assess what the cumulative impacts of water will be for this proposal. They do provide large rainwater tanks, which is obviously very beneficial as well, but there is still a reliance on our potable water supply.

CORRECTED

The CHAIR: I understand that council has declared a climate emergency and adopted a community emissions action plan. Has there been an assessment yet as to how this particular project might impact on your ability to achieve that or how it might influence the plan?

MICHAEL McCABE: In relation to our action plan, the primary thing is to get to net zero and reduce carbon emissions. I would say, based on the proposition to date—of a reliance on fossil fuels—by the proponent, it's going in the opposite direction.

The CHAIR: A gigantic gas plant was not part of your climate action plan?

MICHAEL McCABE: No, not at all.

The CHAIR: I'm so surprised. Are there any wetlands around that area that you are concerned about? If they're collecting rainwater, how're they doing that—and whether that impacts then on the flows going elsewhere?

MICHAEL McCABE: There is the Wingecarribee River, which is the main natural water source for our shire. Our shire is a catchment for Sydney Water. We are our own water and sewer authority. We carry out testing et cetera. We're challenged by environmental issues like PFAS now, let alone air quality emissions that would be, in my opinion, probably low in the atmosphere because of the altitude we're at and because of the weather conditions we face with fog and cloud et cetera. I'm not a scientist by any means, but those types of things of how would the air quality sit in suspense and scatter and fall, they are unknown questions. The site that we're referring to is actually adjoining C3 environmental zoning as well. We do have two koala habitats in the shire and that represents 10 per cent of New South Wales koala population. They're endangered. At the same time, we just resolved in council this week to write a koala plan of management which can be legislated through what they call the koala SEPP. There are other animals, flora and fauna which feature prominently in our shire.

The CHAIR: So a gas plant in a place with heavy rainfall is like a recipe for acid rain. Yes, that's lovely. The other issue that we haven't heard much about is in relation to fire risk. Are you concerned at all in terms of the risk of fire at this site?

MICHAEL McCABE: In relation to the specifics on fire, obviously you have facilities that generate high heat load and across different seasons and—you described it yourself—in what would look like a pretty rural bushland setting. Obviously, access and egress in relation to fire situations, do you believe you have appropriate infrastructure? Appropriate measures on site? With other industry around, and what that impact is, and the ability to respond 24/7, definitely when you're talking about what could potentially be the State's largest power plant, yes, it escalates much higher, the risk.

The CHAIR: Wow. It's extraordinary. Any other questions?

The Hon. SCOTT FARLOW: Let's take the gas plant out of it, in a sense. Throughout the inquiry we have heard a lot of submissions saying that there are opportunities for regional communities in hosting data centres, and particularly tying to REZs and the like. If you were to take the gas plant out of the equation, would you see opportunities for data centres to be housed in the Southern Highlands? Taking up your point as well, in terms of the desirability with high rainfall, would there be an opportunity at all that you would see?

MICHAEL McCABE: Yes, I believe so. Our submission spoke about the principles of how data centres could contribute to society by way of doing that in a more sustainable and passive way—so the reliance on renewable energy as much as possible. We totally concede that data centres will need to mitigate risk in relation to the service they're offering, which is data security. The likes of generators and alternate energy sources would always be required and we fully understand that. The other thing is the scale of data centres varies immensely. You can have one in an urban area that's quite small, but what we're seeing are large portal-frame-type industrial buildings with large hardstand, reliant exclusively on fossil fuel. I suppose it depends on the offering each time but, yes, data centres—I mean, they're inevitable. You read the discussion papers both at Federal and State level, and there seems to be an interest to support economies based on data.

The Hon. MARK BUTTIGIEG: I'm sorry, I was a little bit preoccupied there. Has this been approved?

MICHAEL McCABE: For the site itself, there have been five DAs—four approved and one subject to a deemed refusal in the Land and Environment Court. There's a separate State significant development application pending. It's pending an EIS for the power plant.

The Hon. MARK BUTTIGIEG: So this prospective gas power plant is subject to the State significant process?

MICHAEL McCABE: That's correct.

CORRECTED

The Hon. MARK BUTTIGIEG: Where is that up to? What stage is that up to?

MICHAEL McCABE: A SEAR has been issued a couple of months ago, and we're awaiting the proponent to prepare their EIS. The SEAR has been issued in response to a proposed natural gas power plant. The secretary has standard SEARs for various uses, including data centres. The standard SEAR has not been issued for their proposal because it is a power plant, not a data centre, for that SSD.

The Hon. MARK BUTTIGIEG: I guess this will be a bit of a test of how well the planning system is dealing with these concerns you've raised.

MICHAEL McCABE: Definitely. I mean, I see data centres—the industry is a bit of a disruptor economy, like Uber was, let's say, and the world, communities and governments are catching up. Definitely.

The CHAIR: Thank you very much. I think that was a very interesting case study. Thank you for your submission as well. To the extent there are supplementary questions or questions taken on notice, the Committee secretariat will be in touch. That concludes our time with you.

(The witnesses withdrew.)

(Short adjournment)

CORRECTED

Mr TOM EDWARDS, Research and Policy Officer, Unions NSW, affirmed and examined

Mr CON TSIAKOULAS, Organiser, Plumbing and Pipe Trades Employees Union, sworn and examined

The CHAIR: I welcome our next witnesses. Would you like to commence with a short opening statement?

CON TSIAKOULAS: Firstly, thank you very much for the opportunity to contribute to the Public Accountability and Works Committee inquiry into data centres in New South Wales. It's an important, timely inquiry, given the scale and pace of data centre development in New South Wales. Our submission is from the Plumbing and Pipe Trades Employees Union, in partnership with the Plumbing Industry Climate Action Centre, and on behalf of the leading stakeholders in Australian plumbing, heating, ventilation, air conditioning and fire protection industries, as well as the Fire Industry Alliance. The growth of data centre infrastructure presents significant opportunities for skills development, careers pathways, apprenticeships and employment. Data centres have impacted the sector significantly already and, in many respects, have changed the position of mechanical services, fire protection and specialised piping work within the construction hierarchy.

Our main focus is about the training and development and the need for this training and development and skills to enable these data centres to be built and operated, and also to be at the forefront of any technological changes moving forward to make it more efficient. Because, at the moment, what we have is an industry that's rapidly on the rise with all the information that we've got there, but people are estimating whether that can be delivered. What I'm saying is, is there a skilled enough workforce into the future? Are there enough people in the industry to sustain all this? Our belief is that—unless you actually train the industry from now to be job ready to enable this work that's going to be going on for the next five to 10 years—if we miss the opportunity now to invest in the training infrastructure that's required for this specialist work, we're going to get to a point where the cost will exceed. There won't be enough people to be able to do the work and, at the same time, we'll be still doing old technologies.

As you've heard today, I can imagine there's been a wide variety. You're hearing different figures and numbers about data centres—megawatts, hyperscale. You've heard it all today. Yet, we can't test any of that theory. We don't have a centre of excellence where the industry can go and train and test these figures prior to being installed. The previous witness mentioned why they're building data centres down south. The skills can go there, but there's a different water supply down there—it doesn't come off the Sydney catchment. They're a lot of the factors—that water supply is what it needs—but the skills required in that space for these new technologies, and to make them more efficient, aren't being trained anywhere and aren't being tested anywhere. Unless we do that from today—start now—all these theories and guesses and scales and extensions and all these things are just going to fall into this backlog because there won't be enough people skilled enough to do it, especially in the construction phase. If you can't do it then—the demand might be there, but the supply won't be there.

We believe that for all these works, especially in the base build and in the infrastructure in the building—the fire protection, the air conditioning and that—we need to increase the training in there from now to be ready to enable moving forward efficiencies within the data centre. We can't just have a data centre. Data centres have been around a long time in buildings in the city, which you may or may not have realised. You've driven past. If you drive on the Western Distributor, there's one that's been there for 25 years—Global Switch. So what happens is, they're there already, but in the new wave that's coming in, there's obviously going to be newer technologies. With these newer technologies, we have to be up to speed to be able to install these newer technologies within the data centres as well. At the moment we don't have that, per se. It's the industry training their own in that space to deliver these projects. We don't have a mechanical services apprenticeship in New South Wales. It's been sitting dormant. For the past 25 years, the industry is the one that's kept the standard high and delivered all these projects on time and on budget, based on the fact that they've driven the training themselves and they've driven the advancements in installation technologies and offsite fabrication and prefabrication. They've actually done it. The industry has done that.

The training part of it now is the part that's required to go for the next 25 years. We believe the Government has to look at establishing a centre of excellence, not only to test the products. The idea would be that you have the centre of excellence. The skills required in this specific data centre space, which is going to be a big chunk of construction moving forward, to enable—we can test new racking systems, heating, ventilation and water mining. All these things can be tested and proven. And then we can also go to rollout, because a lot of these things are all theory based and not tested yet. Having a centre of excellence will enable that theory to be tested. And then we'll know how much water we need and how much energy we need, and all these other things.

TOM EDWARDS: Thanks for the invitation to appear. In our submission we focused on three key opportunities: firstly, ensuring data centre and renewables development supports local manufacturing; secondly, making it a platform to train apprentices for our future workforce; and thirdly, making sure the community benefits

CORRECTED

through good, safe jobs. For New South Wales to take up these opportunities, our submission recommended implementing rules for local content quotas and apprentice quotas, and training, work health and safety and industrial standards. Doing this is important for rebuilding our manufacturing industry, filling skills demands and supporting regional communities. Our recommendations for quotas and minimum standards are consistent with other areas of Government policy regarding energy and procurement. I'll keep my opening remarks brief and look forward to the Committee's questions.

The CHAIR: Thank you so much. For people who have no idea—I mean me—how these things are constructed, can you explain what the components of the workforce would be for an average data centre? How much of that is common with other industries and how much of it is data centre specific for that construction?

CON TSIAKOULAS: The best way to think of a base build is a coolroom, where they have to keep the temperature constant at a certain degree. Then the servers go in. That's the post-construction part which creates the heat. Going back to the base, you need the traditional construction methods of slabs on the ground, base build square buildings and basic power. Then you go to the next level: the air conditioning and the electrical supply. The air conditioning requires a lot of individual little air conditioning units called CRAC units—computer room air conditioning units—to be in that room to create that cool air going in there as well. That enables the servers—with the heat, the balance—to be kept cold in that room. The more heat, the more air conditioning units required. Now they're talking cooling using water to actually cool the server racks. These are the emerging technologies.

Basically, it's air conditioning, mechanical plumbing, but on a grander scale. For example, at Parliament House, you would have four chillers to cool the water that goes through the system through a series of pipes and goes into a duct. Then the air goes through the coil, heats off the pipe and goes into our room so it makes it cool. Four chillers are easy. A data centre could have 100 of those chillers. That's the scale. Basic same idea—water gets shored by a machine and then it goes through the system—but the scale is much greater.

The CHAIR: So there's the basic construction of building the concrete shell, I guess, and then there's a different level of workforce that goes in that's responsible for air conditioning and cooling and making sure—

CON TSIAKOULAS: Fire protection, electrical—yes. Service trades.

The CHAIR: If they weren't doing that project—where are we draining workforce from, if we don't have enough people? Would they otherwise be doing renewable energy projects? What other kinds of things would they be doing?

CON TSIAKOULAS: It's twofold. You've got to think of it—historically, industry trained their apprentices. When I was an apprentice I was taught on the job, and I worked on a project for two years. That cost was paid by that builder, in effect, in that development. I learnt my trade. At the same time, I learnt the skills of the work. Those skills are interchangeable. The difference is that everyone's going to want to work in this space because of the demand to finish these projects quickly. For example, there's a project out west that started six weeks ago, and there are already 50 plumbers working on the job. Again, if we don't have enough apprentices training and upskilling, that drain—that means they've come from somewhere else. There are still going to be enough plumbers everywhere else, but there are not going to be as many because they all want to go to these bigger jobs because of the overtime, the target dates and all of those things.

The CHAIR: What's the real world impact, then? Are they not building houses as quickly?

CON TSIAKOULAS: If we're not training and keeping up the supply of apprentices, potentially the cost—because these are all potentials. People are always looking to work in that space to make the most amount of money. They'll think, "I might work there instead of working there." Is there a drain? If we don't have apprentices. Because if we do have apprentices then, naturally, you find your own path.

The CHAIR: If we don't have enough apprentices and we're not training people up properly—and I think it's clear from what you're saying that's absolutely what we need to do. But at the moment, if we're not doing that, do we have slower houses being built or slower supermarkets being built? Where else would these—

CON TSIAKOULAS: It's twofold. It's a price thing.

The Hon. MARK BUTTIGIEG: What will happen is the price of skilled labour will get bid up. They'll be attracted to whichever sector is prepared to pay the most money, be it data centres or housing—probably data centres—and then you've just got a shortage of skilled labour.

CON TSIAKOULAS: Yes, if we don't train up—because what happens naturally is, especially in some of the service trades like plumbing and air conditioning, if you start on the bigger sites, you might not want to stay there. When you get your licence, you might then diverge and go to a different sector of the industry. The industry is very broad. You can work in different parts of the industry. As a plumber, for example, you can work for

CORRECTED

yourself and do maintenance. You can do drainage. You can do mechanical services. You can do all these different streams. The problem that we've got is that we know this is coming. We can predict it. We can see it. You have to be ahead of the game to enable the sustainability of the trade across the whole industry.

When we look at all these things and think, "How do we do it?", we develop skills and industries. The best part about the skills that you learn from the data centres is they are interchangeable with the rest of the industry. You can go back and work in construction. You can go back and still work maintenance. You can still work in all the other aspects. That's the big part that sort of answers your question—developing these skills, making sure we have enough people in the industry, and then it's interchangeable. That's a tricky question: Is there a skills shortage? I look around Sydney and think that there are not that many projects that aren't finished. We're not driving down the street saying, "Look at this empty building. We didn't have enough plumbers and we weren't able to finish it."

The CHAIR: But there are a lot of cost blowouts.

CON TSIAKOULAS: Again, who's at fault for these cost blowouts? I estimated the job and I wanted to pay someone \$5,000 to do the job, but the reality of that price is \$6,000. Is the problem the person that's doing the job, or is the problem the person that estimated that they only wanted to pay \$5,000 for the job? Is that a cost blowout? Is it?

The CHAIR: No.

CON TSIAKOULAS: That's me now telling you, the tradesman, "I want you to do the job for \$5,000." The reality is it costs \$6,000. It's very hard to give you a real answer because there are so many factors.

The CHAIR: Is one of the reasons why data centre operators want to have more data centres near the city because they have easier access to workers?

CON TSIAKOULAS: I think it's more the services. There's certain power that you need. Again, I'll take it on notice, but I'm pretty sure there are only certain areas of Sydney where you could build a data centre, like the west. The two previous witnesses said the Lane Cove area. There's Alexandria and there's the city. It all depends on what power goes in, but there are reports on where those areas are. Water's a big one. It's going to get even bigger. My daughter said to me, "Dad, what happens if they use all the water? Who gets the water first?" You have a centre of excellence that can do these things and answer these questions at the start.

The CHAIR: Mr Edwards, I know we've been talking a lot in the last few years about local content requirements, and I did ask Business NSW before if they would support local content requirements, and they very diplomatically said it would depend exactly what it looked like. What are your recommendations around that and how do we get past the fact that some things just aren't available locally?

TOM EDWARDS: There are already local content requirements for renewable energy zones that are set by the Renewable Energy Sector Board. They set two different sets of requirements; the first being the minimum requirements. They're based on what's already provided essentially in the supply chain, so when you build that type of renewable energy, where that already generally comes from. Then the next is the stretch goals, which increases that with the idea of investing over the next few years in building new capacity. Our recommendation is to require local content no less than that set by the Renewable Energy Sector Board, both for data centres and the associated renewable energy.

In terms of the way that renewable energy is funded through data centres, we need to have a think about that as well. I know that the New South Wales Government has proposed one model where essentially operators would sort of put money into the market and that would fund renewable energy, as opposed to directly building it and funding it that way. Which of those is going to be more reliable? I think we need to have a think about that. I think there's a risk if there's a market mechanism of it being opaque and us not being able to really track the local content that then goes into that renewable energy.

In terms of what we can do where there's just not capacity to build it, I think that was the purpose of the stretch goals under the Renewable Energy Sector Board. Another way of doing it potentially is you could use a weighting as well to incentivise the use, and that way if it's not available, then no tenders get the weighting simply, but if it is available, it incentivises local manufacturers to invest in that local content. That could potentially be another way to do it. As for where within the supply chain you could upscale that local manufacturing, there's certainly a lot where we're already producing that, but those are probably good questions for those industries.

The CHAIR: One of the things we haven't really touched on very much—I just wonder if it's something that you've thought about in Unions NSW at all—is the modern slavery risks, particularly with chips and the importing, and whether there is anything we should be doing around trying to incentivise a more ethical supply chain for data centres.

CORRECTED

TOM EDWARDS: In terms of modern slavery regarding chips, I'd be really happy to take that part on notice. We'd be really happy to look into that. In terms of ethical supply chains in general, we would support having a system where any supplier which has had breaches in the past, particularly work health and safety breaches but also of minimum conditions, would not be successful in the tendering process—or maybe tendering process is the wrong word, but would not be successful in its application to build a data centre.

The Hon. MARK BUTTIGIEG: I just have a few things relating to your evidence, Mr Tsiakoulas. The problem we've got here, which you identified, is how are we going to get all this skilled labour. Because it's the private sector building it, it's a bit hard to forecast the demand, even though we know it's going to be huge, right? Historically, what would have happened is you have a big employer—be it a builder or an electricity company or whatever—and they know the demand for a premises in advance, based on whatever market they're trying to service. How do we work that out?

CON TSIAKOULAS: I think there's a few ways you can do it. One of them is we simply sort of know what the data centre is going to be—the hyperscale, the megawatts, whatever. Everyone talks about training. The easiest way is to say, "This is the outcome we want." How is there an allowance for training? Each contractor that works in there, they get credits, or somehow it's involved where they pay for it. Because someone has to pay for this training. I mean, pay in the way where you can't expect the industry to train at the same time as trying to deliver these projects without having the support of a government or the support of the actual training centres to deliver this future unless there are centres of excellence, knowing that they've got to contribute to the centres of excellence based on the size of their project and the demand of their project. They're building a hyperscale data centre. There's a reliance for this much has got to go towards training to enable them, when they build the next data centre, so that we can still build the next data centre. Is it a fee? Is it a set-up? Something has to be in place where the data centre building, which is worth \$3 billion, a percentage of that money for the skilled trades or the contract trades—

The Hon. MARK BUTTIGIEG: Like a training levy?

CON TSIAKOULAS: Similar to that.

The Hon. MARK BUTTIGIEG: Let's go back to the problem for a minute. I don't know; we've probably had it in evidence and I probably haven't been paying attention. There's probably X amount of data centres in the planning and approval stage now, right? Are you saying that even if they were all approved—say they're all approved in the next couple of years—you would not have the skilled labour to be able to deliver them anyway, unless you stole them from the housing sector, for example, by bidding the wage up? Is that what you're saying?

CON TSIAKOULAS: You couldn't do that because the work would be more specialised. You've got to start from the training part.

The Hon. MARK BUTTIGIEG: Sorry to interrupt. That was going to part of my other question. It's not just generic plumbing X that can do data centres and can do whatever. It's got to be a specific skill set to do them.

CON TSIAKOULAS: Yes, there's more welding involved on the pipes because they're bigger pipes. They're stainless steel. There could be mould still. They're bigger pipes; they're not your traditional little half-inch pipes or three-quarter pipes that you would use in a house, for example, and your plastic PVC. It's a different base skill set which you need that base training and expertise to do. Not that you can't do it, but you can't just say, "Okay, we've got this data centre and we've got to deliver it in 12 months. We'll just grab 50 plumbers." Even though they're plumbers, there's a slightly different skill set. They can't do that work. They'll give it a red-hot go—

The Hon. MARK BUTTIGIEG: When you say they can't do it—

CON TSIAKOULAS: Because they haven't got the experience.

The Hon. MARK BUTTIGIEG: —do you mean in a regulatory, licensing sense or just skills?

CON TSIAKOULAS: No, skill set.

The Hon. MARK BUTTIGIEG: So you can do it, but it'd be dodgy.

CON TSIAKOULAS: Not so much dodgy. You'd have a go, but you wouldn't know how to TIG weld a stainless steel pipe, for example. You just don't have that skill. You can't just learn it on the spot. You've got to have a bit of experience, training and guidance from that company. You can do it, but it might take you three months or six months to get your ticket and then to be able to weld. You can do the other components of the job. It's slightly different. But, yes, that's what you'd do. At the same time, if you had that base training from the start for the new cohort of apprentices coming in, you would alleviate that. In the data centre space, they push bigger hours—

CORRECTED

The Hon. MARK BUTTIGIEG: So all these proposed data centres are going to bump up against skill shortages. You're not going to be able to do them in the time they want.

CON TSIAKOULAS: No, you will, because they will do night shift. They'll still deliver it on time.

The Hon. MARK BUTTIGIEG: But what about the skills you were just talking about?

CON TSIAKOULAS: These same guys will be doing 12-hour days—longer days—to deliver the same projects. That's what's happening at the moment. Because of the short time frames, the demand is a longer working day, as in a 10-hour or 11-hour day, or 12-hour days. It's a good thing, too, when there's overtime. People like that because of all the other things that are going on in the world that you live in. Coming from that space, I don't fear the work and the skill shortage that it won't get done. The skill set is there. The abilities are there within the people that are currently working in that space. It's just that we're going to need more. To get those more, you've still got to go to those existing people because they've already got those systems to train and develop.

The Hon. MARK BUTTIGIEG: Has our TAFE system got any capacity for this sort of stuff?

CON TSIAKOULAS: I think the TAFE system could have capacity, but they've already got enough. This can be targeted and specialist for data centres, right? TAFE already has enough in the plumbing space that they specialise in, and they deliver all the cert III components. There's enough there. This can be like a standalone, centre of excellence, super-scale testing facility.

The Hon. MARK BUTTIGIEG: Do you need a separate qual, like a new qual, or can it be under—

CON TSIAKOULAS: It would be under the mechanical services—that is currently not being delivered. It would sit under that qual.

The CHAIR: What does qual stand for?

CON TSIAKOULAS: Qualification. You can still do it under your current cert III as well, but a bit more targeted aspects of it too. Again, there are sites around Sydney that are ready to go for something of this scale, out in the western suburbs. Just need a bit of retrofitting and investment to make it like a true one-stop shop: testing for service, testing all this equipment. Have this one place where we can do it all.

The Hon. JACQUI MUNRO: May I ask a question on that apprenticeship pathway option? Are you engaging with any data centre operators at this point to speak to them about an apprenticeship workforce directly?

CON TSIAKOULAS: We engage with the contractors that are actually doing the work. In Sydney, for example, the five main air conditioning companies that are doing it—I won't name them. We work alongside them. Our members work in that space, too. We know exactly from that—and, respectfully, I'm from that space, and so is my boss. A lot of us, where we are, at the union—we're actually from that space as well.

The Hon. JACQUI MUNRO: I'm aware that apprenticeship dropout rates are quite high: around 50 per cent. You mentioned earlier that one of the reasons that this new data centre environment might be challenging for other projects is because data centre operators might be paying more for staff, so they're more attractive for workers. Is that a fair assessment?

CON TSIAKOULAS: No, I think that the fair assessment would be the time frame. Let's talk economics, basic 101: An apprentice working on that data centre, because of their working, is getting the overtime, he's getting the Saturdays—

The Hon. JACQUI MUNRO: Not necessarily getting paid more by the hour, but getting paid more as a function of working more, and the hours they work?

CON TSIAKOULAS: Because they want these data centres finished—that section of work done quickly—so they can fit it out after.

The Hon. JACQUI MUNRO: Do you think that there is going to be a lower dropout rate amongst those apprentices who are working on data centres because of that additional remuneration? Is that something that you track?

CON TSIAKOULAS: Yes, we do. We're in partnership with an RTO, and we have a very high—well over the 90 per cent.

The Hon. JACQUI MUNRO: That's quite different from other figures that I'm aware of. Is that right?

CON TSIAKOULAS: It's more targeted, those specific companies that work in that hiring space, like we said.

CORRECTED

The Hon. MARK BUTTIGIEG: That's a really good point, though. Is that a function of the amount of money they're earning, so they stay in the apprenticeship?

CON TSIAKOULAS: And the licence.

The Hon. JACQUI MUNRO: I mean, air conditioning in particular is an industry or trade that is specifically, consistently understaffed, or underemployed.

CON TSIAKOULAS: Yes. It's the nature of the work, because to do this data centre, you need a highly skilled and highly versatile workforce, to be able to work these big hours. But at the same time, air conditioning is very broad. A lot of people in air conditioning—we're plumbers, we're mechanical plumbers. The mechanical plumbers work in the air conditioning space, but they're plumbers. There's electricians that work in the air conditioning space, as well. We only talk about the plumbing side of things.

The Hon. JACQUI MUNRO: In terms of the broader trades or groups represented by unions, Mr Edwards, would you say that other unions are also experiencing a greater retention of apprenticeships in jobs that are—

CON TSIAKOULAS: Just on that—if you've got to finish your apprenticeship to get a trade licence, you would find the numbers would be a lot higher. Some apprentices might start off in the building trades and do their time, but not finish it, but also then divert to the building to try and get their builder's licence. A lot of carpenters go, "I want to become a builder." They might not totally finish their apprenticeship, but then they go to that building side of things, too.

The Hon. JACQUI MUNRO: Related to data centres and retention of apprentices through to the completion of their apprenticeship and getting into a licence—in addition to plumbers, are there other trades that would experience a higher retention rate because of work undertaken in data centres, specifically?

TOM EDWARDS: I'd have to take that one on notice.

The CHAIR: Do we know, at the moment, on a large data centre project, how many apprentices they're using?

CON TSIAKOULAS: Yes. I could make a phone call.

The CHAIR: Is there a percentage? The Unions NSW submission said something like 20 per cent would be a good target. How far off of that is the average data centre at the moment?

CON TSIAKOULAS: I can only speak for the plumbing and the fire protection side of things. I can take it on notice and I can give you an exact number.

The CHAIR: That would be good.

CON TSIAKOULAS: I can ring the contractors that are involved and give you real-time data.

The CHAIR: That would be really useful.

CON TSIAKOULAS: It's sort of the same four or five contractors that we've got working in that fire protection space, the same five or six that are in plumbing side of things. We can get some real, accurate data.

The CHAIR: That would be really useful.

The Hon. MARK BUTTIGIEG: On the local content, Mr. Edwards—this is obviously the private sector, right? We've got quite a few initiatives going on with procurement using government purchasing power. What's the proposition for government to ensure local content with this? Is it just mandating it in legislation?

TOM EDWARDS: I believe that there's an application process for data centre operators to go through—or I understand that's what the New South Wales Government is developing currently, through a consultation on guidelines. I believe that implementing those minimums—or, as we discussed earlier, in addition to those minimums, a weighting through that application process, so that if you've got a data centre operator, they make an application for a data centre and they've got their application there for the associated renewable energy as well. They have to demonstrate in that application that they're going to be meeting those local content quotas in order for that to be approved.

The Hon. MARK BUTTIGIEG: They're not in stone at the moment, are they? They're still developing them as guidelines, are they?

TOM EDWARDS: That would be my understanding, yes.

The Hon. MARK BUTTIGIEG: Is that being done out of Planning?

CORRECTED

TOM EDWARDS: I think Infrastructure NSW is leading it.

The CHAIR: Thank you so much; that's been really useful. For the questions taken on notice and for any supplementary questions, the secretariat will be in touch.

(The witnesses withdrew.)

CORRECTED

Mr DAVID EYRE, Chief Executive Officer, University of New South Wales Institute for Industrial Decarbonisation, affirmed and examined

Professor NICHOLAS EKINS-DAUKES, Head of School of Photovoltaic and Renewable Energy Engineering, University of New South Wales, affirmed and examined

The CHAIR: Did you want to start by making any opening statement?

DAVID EYRE: Chair and members of the Committee, thank you for the opportunity to appear today. In the interest of time, Professor Ned and I will each make brief opening statements. We propose to take the detailed recommendations and analysis in the UNSW submission as read and to explore those issues further through the Committee's questions. The institute coordinates UNSW cross-disciplinary research and industry engagement relating to industrial systems, critical infrastructure and climate resilience. Once data centre capability reaches industrial scale, governments need to think consciously about what kinds of systems they're enabling and why. It's clear that modern data centres operate less like conventional commercial buildings and more like strategic industrial systems, with significant interactions across electricity networks, water systems and land use planning.

In metropolitan growth corridors, data centres are competing for industrial land, altering urban form and creating cumulative amenity impacts associated with bulk, noise, heat rejection and concentrated utility demands. The sector's growth trajectory is both rapid and occurring at very large scale. Established operators are raising substantial capital for expansion, while new and less mature entrants are also entering the market and seeking electricity, water and land-use access in order to support project finance and development pathways. As a result, booked demand can materially exceed realised operational utilisation. This reflects the way that large-scale projects are financed and staged, but it also creates genuine challenges for sequencing, transmission planning and cost allocation across electricity, water, telecoms and related public systems. The key issue for government is, therefore, not whether these facilities have significant impacts—they clearly do—but whether planning and coordination frameworks are operating at an appropriate scale to manage cumulative effects and achieve net benefits under conditions of uncertainty.

It is important to recognise that not all data centres serve the same purpose. Multi-client metropolitan facilities are structurally tied to Sydney's enterprise and network ecosystem and require proximity to major population and business centres. Other classes of compute capability—including certain hyperscale and industrially integrated systems—may have greater locational flexibility and different operating requirements. There is also significant variation across proponents in technical capability and sustainability approach. Some operators are actively investing in firmed renewable energy, recycled water integration and advanced cooling architectures designed to reduce pressure on public systems. Other proposals remain more dependent on conventional grid supply and potable water sources. From a planning and public interest perspective, these differences matter.

A related issue concerns the relationship between sovereign digital capability and export-oriented digital services. New South Wales digital infrastructure should not be understood solely through the lens of hyperscale data centres. Sovereign computing capability does not necessarily require AI data centre deployment at multi-gigawatt scale. There is an important distinction between computing systems that are structurally integrated into the Greater Sydney economy—supporting scientific research, manufacturing, hospitals, transport systems, banking, government functions and so on—and facilities whose primary role is servicing globally distributed AI and commercial workloads as an export-oriented industry. Export growth potential of the data centre sector, however, is real and must be recognised and managed carefully. There is now a rapidly expanding global market for AI, cloud and digital services, and Australia is an attractive destination for some forms of that investment.

Different data centre functions may imply very different optimal infrastructure geographies. Some forms of export-oriented hyperscale compute may ultimately align more naturally with locations offering large-scale clean electricity. They also create opportunities to better align some forms of compute with renewable energy zones and emerging clean manufacturing ecosystems in regional New South Wales. One broader issue in this context is the interaction between digital infrastructure, the energy transition and regional industrial policy. For several decades, successive New South Wales governments have sought to deepen economic capability outside Greater Sydney through initiatives such as special activation precincts and regional industrial strategies.

One of the challenges facing current regional energy transition policy is that renewable generation infrastructure alone does not automatically create durable regional economic depth. In some regions, communities increasingly perceive themselves primarily as hosts for energy extraction and transmission systems supplying metropolitan demand, without sufficient long-term industrial participation locally. Alongside this, commercial data centre operators, understandably, often argue that regional deployment pathways remain difficult because

CORRECTED

telecoms capacity, electricity infrastructure, workforce capability and industrial ecosystems are currently concentrated around Greater Sydney.

These conditions are not necessarily fixed. Historically, industrial capability and critical infrastructure have evolved through long-term sequencing and strategic investment. In practice, the question for government is whether parts of regional New South Wales could progressively develop more integrated compute energy and industrial capability over time if supported through sufficiently coherent planning, infrastructure and industrial development frameworks. There's obviously a tension here between speed of development, market-led investment and longer term public interest strategy. Historically, Australia has generally preferred to allow industrial geography to emerge through market forces. However, once compute systems begin interacting directly with constrained infrastructure, governments inevitably become involved in questions of sequencing, spatial allocation and long-term infrastructure direction. Importantly, governments are not limited to reactive approval processes. They already possess a range of policy tools capable of shaping outcomes.

Singapore, in relation to your terms of reference, provides an important example of a government moving from reactive assessment towards a more strategic allocation approach, requiring proponents to demonstrate broader economic value, sustainability performance and systems integration as part of the approval process. Finally, UNSW also sees potential opportunities for New South Wales in the development of climate-adapted data centre technologies and related engineering systems. In summary, New South Wales already hosts advanced digital capability at significant scale. This is a foundational strength. The challenge is shaping development across the sector in a way that supports public amenity, systems integrity, climate resilience and long-term economic competitiveness. I'll now hand over to Professor Ekins-Daukes to briefly address the electricity system.

NICHOLAS EKINS-DAUKES: I'll make a few remarks just on the energy aspects, because that's the competency of my school, renewable energy. First, I'll note, quite evidently, that this data centre growth is occurring simultaneously with a renewable energy transition, so there's a lot of moving parts. As we know, it involves deploying infrastructure quickly and effectively to bring renewable energy onto the grid, and in this case, the data centres also deliver the product data. A few obvious remarks, but I think they may be important for framing the discussion that electrical power is extremely high density, and a cable to a data centre is effectively unchanged no matter what the source of that electrical power is. I think that helps us with our thinking and planning. The cooling technologies are more diffuse. The energy is diffuse and much more site dependent. As I'm sure you're aware, the cooling technologies can span from things that are hugely expensive in terms of water, or hugely expensive in terms of electricity, and anywhere in between. There's a lot of space for innovation there, but that is a piece of infrastructure that is harder to upgrade after the fact. Then of course, fibre communication is obviously a prerequisite for any site, as David has mentioned.

I want to note a report that I think is of particular interest to this inquiry, which was authored by members of the UNSW Collaboration for Energy and Environmental Markets. They specialise in understanding the impact of certain policy choices on the costs of electricity. Their report, titled *24/7 TRUZERO: Tracking Renewables Utilisation for Zero Emission Reporting and Operation*, explored the cost of time matching that intermittent renewable energy supply with demand. What they did was they modelled 51 different commercial and industrial loads under eight power purchase structures to figure out the effectiveness of matching the intermittent supply with the demand, and the financial consequences of that. One of the key findings is that achieving 90 to 95 per cent 24/7 carbon-free energy incurs a relatively modest cost premium. Pushing that to 100 per cent is significantly more expensive. So one of the key points is that if you have a data centre and you wish to supply it with renewable power, you can do so substantially today without a large financial cost. Pushing it to 100 per cent is, of course, the challenge that we wish to get to around about 2050. But I think in this conversation, with the rapid growth of data centres, this is a point that's worth noting.

I'll make one or two more comments about the infrastructure, which evidently has to be built out rapidly. We're very familiar with these challenges with solar, wind and renewable energy, and co-locating data centres with that represents an opportunity. I'll note that the technology is evolving rapidly in terms of its performance and cost, both on the renewable energy side and on the data centre side, not only around the compute efficiency but also the cooling technologies. The last point I'll make is that with the renewable energy transition, we're continuously reminded that the availability of a skilled workforce is a critical barrier to rapid deployment. I think the same is true for data centres. To make them sustainable requires trials, and if you're going to have an effective trial, that means you need even more highly skilled people to deliver that trial effectively. In summary, our ability to innovate towards environmentally benign and sustainable data centres in the highly competitive arena globally, in my opinion, hinges on the skill of our workforce as well as the stable regulatory framework in the sector.

The CHAIR: Thank you very much. There's a lot that's really interesting in your submission and in your opening statements. One of the things I wanted to touch on first was in relation to the ability for data centres to incorporate this behind-the-meter renewable generation—battery storage and even what I understand is called

CORRECTED

islandable microgrids. I guess the question is how realistic that is at the scale that's required. Do we think that that behind-the-meter solar and batteries can provide a 500-megawatt data centre or even gigawatt-scale data centre when it's operating at 80 per cent operating capacity? How realistic do you think that is?

NICHOLAS EKINS-DAUKES: With the renewable energy transition, that's where we need to go. I think today what we can say is we have the technologies. Solar photovoltaics is now so cheap. I think it's cheaper than anything we ever believed was possible. When I started in this, it was \$4 per watt; it's now 10¢ per watt. I started maybe 30 years ago, but that's an incredible reduction. It's true—we can deploy solar photovoltaics at scale. Battery technology is also now extremely inexpensive. There are a number of new technologies—for example, sodium ion storage, not lithium ion. Sodium ion is extremely well suited for stationary storage. It obviously doesn't use lithium, so it doesn't have any concerns about resource constraints. There are also vanadium redox flow batteries, which actually have a lot of innovation coming out of the University of New South Wales. They are capable of storing electricity for extended periods of time with very little discharge and can be cycled tens of thousands of times.

The technology is there. What I think is lacking is the opportunity to demonstrate things at scale, and it's through demonstrating things at scale that I think we realise, first of all, where the challenges are but also what we can actually achieve. Maybe I could do one more comparison. When the Hornsdale Power Reserve installed that large battery about eight years ago now, they built it in less than 90 days and it was the largest battery that had ever been installed. Obviously 500-megawatt or gigawatt-scale capacities are larger than anything we've built today, but there's no fundamental barrier from the technical standpoint to prevent us from doing this. What there is—and this is feedback I've received from the industry—obviously is a skills shortage. There are also supply chain constraints—that if you build something of that magnitude, that isn't something that manufacturers, and particularly the Australian installers, have the capacity to deploy instantly. It's something we'd build up to.

The CHAIR: I don't disagree with you that the speed at which technology is moving—we're getting better and better renewable technology all the time. I'm looking at the proposal in the Southern Highlands for New South Wales's biggest gas plant or whatever to back up their data centre, and all of the backup diesel generators everywhere. It seems that we are now rushing so fast to build data centres. It's all good and well to have the theory that we could have renewables backing up all of them, but what is the reality? We're talking about decarbonising the economy, so how do we then achieve that when we're adding so much more demand into the grid at the same time as what is a very competitive space for limited supply of renewables in workforce, as well as raw materials? How realistic is it that we can just solve this with renewables?

DAVID EYRE: Government definitely has a role in setting the right framework that allows data centres to differentiate themselves by moving more quickly in this direction. I think it's important to look at what some data centres are actually doing. One of the companies we work with has 900 megawatts of solar PV on a roof of one of its ASEAN data centres. It was in their interest to do that because the government involved had set that as a condition. This is a sector that is attracting massive global capital. There's a tension between the speed with which that capital can be allocated in jurisdictions like Australia and those companies investing in technologies that on their balance sheets are going to be a little bit more expensive, but they're not deal breakers.

New South Wales could definitely be aligning your policy around future data centre expansion with your policy for net zero, for regional development, and your broader infrastructure policy around industry transformation. It is already happening. Some of the data centres we work with are investing massively in long-duration storage and investing in power purchase agreements. The opportunity, as Nicholas said, is to demonstrate industrial-scale clean energy at scale in regional locations aligned with the needs of other regional industries that you are supporting, like critical mineral processing. There are opportunities in the Hunter and in other special activation precincts in New South Wales—in your renewable energy zones—to triangulate, if you like, certain kinds of data centre development with other forms of industrial innovation that you are seeking to incentivise.

The CHAIR: Again, if we accept that these data centres at the scale and pace that we're currently seeing them being developed is necessary in New South Wales, then you take this one that's got the—I forget what it is, 700 megawatts? Did we decide the—

The Hon. SCOTT FARLOW: I think that was about close to it—623 and up to 700, or something like that.

The CHAIR: The gas power plant being suggested—if they don't attach to the grid, and instead they have these behind-the-meter renewables, how much land are we talking about if they were to have solar panels? And even if they were to build massive solar farms to be able to support that data centre, that means that that solar farm is not being built in the renewable energy zone to quickly decarbonise the rest of the State. How do we solve that?

CORRECTED

DAVID EYRE: It's an integrated planning challenge. In our view, it involves sandboxing multifactorial scenario modelling to figure out where the optimal location is for these centres. There's a spatial and temporal allocation challenge here. The renewable energy zones are a fairly crude instrument. The art of doing this is aggregating demand across multiple industrial users in a way which aligns with existing transmission infrastructure, suitable land and the interests of local communities. Currently, the majority of large solar farms or wind farms are simply positioned where land is available and where there's very easy connection to existing high voltage transmission infrastructure, and most of that energy is going to Australia's capital cities.

To bring regional communities along, we need to point out that these future power stations are creating depth in those regional economies by enabling knowledge-based jobs, not just in data centres—there aren't too many jobs in data centres; we have to acknowledge that—but also in things like food processing, critical mineral processing and forms of manufacturing which actually are going to be competitive in a regional location. This is quite realistic. We have quite dynamic regional economies. This can turbocharge the kind of growth trajectories that the New South Wales Government has been seeking to achieve. But modelling is extremely helpful to de-risk this kind of activity because there are many moving parts.

NICHOLAS EKINS-DAUKES: I might just add, if we look beyond our own borders, China installed more renewable energy capacity last year than the rest of the world combined, so in terms of looking for ambition and technical feasibility, it's there. The biggest battery in the world is being built in Saudi Arabia, which is about three times bigger than the biggest one ever built in Australia. Australia actually is doing pretty darn well in terms of battery storage. At the moment we've got countries racing ahead and installing these systems. What's new is, of course, bringing them together in the way you suggest, and I think there's always a tendency to install what you installed last year or 10 years ago which you, in a sense, know works. There's always risk in doing something that's different. It's not just technological risk; it's also your workforce. These, I think, are all of the challenges but, in terms of where I sit at the university, looking at the technical feasibility, there are no barriers. We just need to organise everything else to make it possible.

The Hon. SCOTT FARLOW: In terms of the regional argument, have you done any work in looking at where the ideal regional locations in New South Wales are to host data centres?

DAVID EYRE: The short answer is that we haven't yet developed the integrated modelling that I'm talking about, but we do have highly competent modelling teams looking at energy markets, looking at digital grid infrastructure modelling, looking at the land use planning challenges and also looking at the business economics of it. We have a very large engineering faculty with mixed skill sets across these areas. We have a planning and built environment faculty that's quite substantial and is spatial modelling for the New South Wales department of planning. We have techno-economic modellers who can really look at the trade-offs across time and cost of infrastructure. What we'd very much like to do is put all of that together in an AI-supported model, and we could do that quite quickly. It's not unusual for us to do that kind of work in camera for government agencies at Commonwealth and State levels.

The Hon. JACQUI MUNRO: Are you doing that at a State level in New South Wales?

DAVID EYRE: At the New South Wales level, we've done energy market modelling and we've done water system modelling, including water modelling for data centres—you're probably aware of that. We've done techno-economic analysis relating to things like critical minerals—which minerals are going to be most economically valuable to incentivise processing in New South Wales. That really requires access to massive amounts of clean energy, by the way, so that connects up. If you'd like, we could—

The Hon. JACQUI MUNRO: Sorry, just to clarify, is that being provided to government specifically, or is this work that you're doing independently because you believe it's important?

DAVID EYRE: All those things I've mentioned have been provided to government—either to ARENA, the department of climate change or whatever. These are separate contracts to specific research teams. Of course, where the institute is coming from is connecting that all up to give a more integrated picture. I think Australia's at a point where we need to think across the different silos.

The Hon. JACQUI MUNRO: Hear, hear!

DAVID EYRE: And technology can really help us do that, by the way.

The Hon. JACQUI MUNRO: I perfectly agree.

The Hon. SCOTT FARLOW: That's why we need data centres, isn't it.

DAVID EYRE: It's quite funny; it's quite circular in that regard, isn't it.

CORRECTED

The Hon. SCOTT FARLOW: Yes, indeed. To that point, one of the debates within this Committee is whether Australia needs data centres itself, or whether New South Wales needs data centres; could they be domiciled elsewhere; what are the advantages and disadvantages? I'm just interested in your work and your perspective as to what the advantages are for us in having that capability onshore and here in New South Wales?

DAVID EYRE: It's a really important strategic economic question and critical infrastructure question. That's why in our submission we've really tried to unpack the differences between kinds of data centres, kind of computing capability that New South Wales needs and the export development opportunities, which are very substantial, by the way. New South Wales is already the State that has the most installed digital capability in terms of telecommunications, fibre, and actual data centres, and we've got a very sophisticated industry operating here. As I said, that is a huge strength for New South Wales. The question really is, if you like, the marginal value of increasing that capability when some of our core infrastructure is already stressed—our water, electricity and land use planning constraints in, for example, the Western Sydney corridor or other corridors.

We've got that inherent opportunity but, in my view, it does need to be shaped. The reason I reference Singapore is that they paused data centre development. I'm not saying that New South Wales should do that, but they took a step back and said how can we flip this around so that we're literally releasing a call for additional capacity for data centres according to X, Y, Z criteria? I know New South Wales criteria are going to be different to Singapore's, but I do think it would be worthwhile to look very closely at how Singapore has done that and what they got out of it, and I do talk to data centres that operate in that jurisdiction.

Quite surprisingly, they don't really mind because the upside for them is it provides clarity around what the rules are and it's also a differentiator. In a crowded market, it enables a company that's capable of getting the energy and the water right and dealing with community expectations around things like sustainability. It allows them to achieve a contract, as opposed to a company that may be simply a property sector player realising this is a billion-dollar industry and they want part of it. I think there's a really important avenue for the New South Wales Government to explore in that specific area.

The Hon. SCOTT FARLOW: Sure. We had the discussion before, or I think it was in your opening statement, about the choices that happen with data centres and whether they're very expensive in terms of water usage or very expensive in terms of energy usage. In the New South Wales context, is there a model we should be looking at for a definition of what sort of mix there is in terms of, effectively, water-cooling technologies that are employed by data centres?

DAVID EYRE: Yes. Look, there is a risk that data centres will be driven towards things like air cooling simply because communities are concerned about water consumption of, let's say, evaporative cooling. That's not necessarily the best outcome. I do think it would be really valuable to have some technical workshops and pilots and studies that are really looking at what is the best mix of technologies for the Australian operating environment, and our different climate zones for that matter, because most of the technologies that are employed within data centres—probably about 99 per cent of them—are imported from overseas and are provided by companies that are really specialising the technology for Northern Hemisphere environments. They're not really climate-adapted.

Of course, it takes time to reinvent that stuff and the data centres are working very fast. But there is an opportunity to drive innovation and to identify suites of data centre technologies which are going to be better than what is currently being deployed, and to do the techno-economic analysis hand in hand with the sustainability decarbonisation analysis and so on, so that it fits the regulatory framework. This is something we talk to entities like NABERS about—the people who are actually responsible for the metrics that apply to data centres. There's no easy answer to that. It requires research and development, frankly, and also a very pragmatic conversation with the data centre operators and the OEM technology providers, now that we're working with those providers as well. There's an analogy for the mining sector, by the way—the mining sector deploys technologies which are also imported from international OEMs. One of the areas we work in is inventing new technologies that are more suitable for Australia, and I think we need to do the same thing with data centres, which are emerging as a major industrial sector with significant technology demands.

The Hon. SCOTT FARLOW: I'm interested in your reflections on the clustering of data centres that we have. Of course, where there's availability in terms of water, electricity and fibre optic technology, data centres will want to go. There are arguments from communities, as well, about the community impact and the cumulative impact of data centres. I'm interested in those trade-offs and what you're seeing internationally to best manage that, and how cumulative impacts should be looked at in terms of the assessment process.

DAVID EYRE: In very pragmatic terms, I think the first step is making sure we've got our sovereign compute capabilities sorted. We need AI functionality to run hospitals, to do high-end scientific research and so on, and just to operate the economy—to automate manufacturing and so on. This is a mainstream domestic need, and that's going to require a certain amount of modern data centre capability. Alongside that, as I keep saying,

CORRECTED

there's the export opportunity, and there is a global market for cloud and AI data services, which is distributed globally and it is globalised, effectively. New South Wales is a very attractive destination for companies that are actually serving that market—not necessarily the needs of our companies, our hospitals and so on. I do think it's worth diving into that and trying to make that distinction.

But having that conversation constructively with the actual data centre operators—the big, sophisticated ones in areas like Western Sydney, or co-located with some of our hospital precincts, absolutely understand what the domestic needs are, and they're meeting the sustainability criteria. Alongside that, there's that question of, "Can we build a major export industry around providing trusted data services, potentially to our regional trade allies like Singapore, for example?" We could be building a sovereign data centre in New South Wales to meet the needs of the Singaporean Government as part of some kind of alliance. This is purely hypothetical, by the way, but there are an array of opportunities like that, which I think Premier and Cabinet could be exploring, and you could be exploring as a committee. Your role is super important in providing this kind of strategic advice.

The Hon. SCOTT FARLOW: On that point, in terms of the export benefits, part of the discussion is that you're talking about global tech companies, often with tax arrangements that are outside of our jurisdiction. What are those economic benefits to New South Wales in terms of that export capability?

DAVID EYRE: For our presentation, one of our economists in the business school did some modelling of things like the direct benefits and the multipliers. Obviously, there aren't too many direct benefits in terms of jobs from a data centre—in a construction phase, yes, but that's over very quickly. The real benefits are around deepening our economy, and the reality is that all future manufacturing will run on electricity, and it will run on data. Our economy needs data, our social services need data, government efficiency really depends on better integration of data services, electronic service delivery, and so on. Things innovate at different paces, and the data centre issue is really pushing Australia's digital transition. That creates considerable discomfort; however, it also presents an opportunity for government to step across some of the barriers to the integration of decision processes, policy, and government processes using digital technology to enable that in a cost-effective way. That point might be a bit too abstract.

The Hon. SCOTT FARLOW: No. There was a question I asked Business NSW earlier today effectively on that point, and the multiplier effect of data centres. Can I ask, potentially on notice, if you'd be able to share with us that report from that academic. That would be wonderful.

The CHAIR: One of the things we're getting competing information on—and maybe it comes down to the type of data centre or what the data centre is doing—is this idea about whether or not a data centre provides stability to the grid. When we're looking at large inverter-based loads, I know that in AEMO's 2025 transition plan, they identified data centres as an area of concern for the grid stability. Further on they say that data centres can ramp their demand bi-directionally at the rate of hundreds of megawatts per minute, which places a significant burden on frequency regulation, and, critically, that AI workloads make it worse. Then they go on to talk about international observations that AI processing can cause the load to rapidly oscillate—in one example, by 40 per cent—within seconds during different processing phases. That seems to be directly contrary to the industry claim that data centres are stable and predictable. So are we talking about something different between traditional data centres running standard enterprise workloads versus these AI-intensive ones, or is something else going on?

NICHOLAS EKINS-DAUKES: I'll offer a perspective. Data centres can be modulated by about 30 per cent of their output, and I think the UK national grid had some data on that. In that sense, having what we call a flexible load is actually extremely useful, because it means that if you end up with your grid, however you're managing it, approaching 100 per cent capacity, you can then back off certain absolutely non-essential loads like your data centre—by, according to the UK National Grid, 30 per cent—without there being too much trouble. The virtue of a data centre is that it's a load that sits there on the grid that can be modulated. There are, of course, lots of questions as to whether the data centre wants to do that. In terms of its operation, when they're training these models, that's hugely energy intensive but that's something you can do at times of, shall we say, cheap electricity supply; whereas operating your data centre, depending on the criticality of what you are serving—if you're just serving my Sunday afternoon request, if I wait another couple of seconds for that, that probably doesn't matter. But obviously there are some applications where speed is of the essence.

The information I have from the British national grid is that data centres can actually be a good citizen on the grid because they're flexible loads. I think anyone trying to manage a grid doesn't like a demand that fluctuates wildly, so an unscheduled shutdown of a very large load is going to cause problems. Once again, the British are somewhat sensitive to this because, historically, they've had this situation when one of their favourite soap operas, *Neighbours*, would—

The Hon. SCOTT FARLOW: Kettle.

CORRECTED

NICHOLAS EKINS-DAUKES: Not *Neighbours*—*EastEnders* would end, then everyone goes and turns on their kettle, and it's the same with the soccer matches, that would then put huge surges on the grid. They would turn on their pumped hydro, specifically to meet that demand of a lot of British people all deciding, "I'm going to make my cup of tea."

The Hon. JACQUI MUNRO: Wow—making a cup of tea!

NICHOLAS EKINS-DAUKES: Yes. I say this in all seriousness.

The Hon. SCOTT FARLOW: It is the UK after all.

NICHOLAS EKINS-DAUKES: Streaming has rather kind of destroyed this, but in the old days this was a major challenge for the British national grid. Uncontrolled demand is something that we've had to cope with on grids for a long time. Data centres obviously bring significant demand, but I think, if engineered properly, and particularly if they've got some battery buffer, they can be good citizens because you can modulate the demand at time scales that are useful. So they're a bit like a pumped storage but in reverse, because you can back off your demand when you get uncomfortably close to 100 per cent and bring it back on when you need to.

The CHAIR: But do you need to know what that centre does? As in, if some of it is critical and some of it is not as critical, doesn't that mean we need to know which data centres do what in order to work out whether or not we can play with their energy?

DAVID EYRE: I think it's worth recognising, as I said before, the difference between kinds of data centres. A multi-client data centre doesn't have a huge amount of control over what the clients within that data centre are doing with their compute. It's in their commercial interest to have a pretty open-ended allocation of electricity, or actually almost the opposite—a firm allocation of electricity so, no matter what, they can meet the load of whatever their customers are putting through the service at any given point in time. Those companies have a very different synchronisation and load management profile than a hyperscaler, who is one company operating, let's say, a data warehouse or an AI model of some kind. It's very important to recognise, too, that not a huge amount of training or even inference is likely to occur in Australia. That's really been concentrated in the United States, in China, in Europe. The kind of data centres we're seeing standing up are trying to be as flexible as possible with regard to whatever is implemented within the shell that they're constructing.

The CHAIR: With these large inverter-based loads, do they still increase the inertia requirements? Is that something that's different to this? Can you explain that to me?

NICHOLAS EKINS-DAUKES: Inertia just means the spinning things that spin, and keeping the frequency stable. As you take your grid to, shall we say—in terms of increasing the fraction of the power that's being powered by, let's say, inverters, the analogy I will put to this is an inverter is a bit like an electronic synthesiser. It produces the waveform artificially, whereas obviously something that spins has that natural repetition. The faster it spins, the higher the frequency. In principle, there is nothing stopping you from having 100 per cent inverter-based load on the grid. By 2050, that's where we need to be. You can, if you wish, install synchronous condensers, which are large spinning masses that just sit there stabilising the frequency so that effectively your synthesiser, which is the inverter-based load, has got a regular beat that doesn't wobble around too much.

The CHAIR: They're super expensive and quite hard to get, I understand.

NICHOLAS EKINS-DAUKES: Right. It leads to these supply chain problems because we don't need to build many of them and we haven't built many of them. Frankly, modern electronics at least gives me confidence that we can build a fully inverter-based power system. What you need is synchronisation, because you need a central clock that is keeping everything together from the most basic principle here—because otherwise, if you imagine you've got all these different synthesisers and all these different generators, and if they're not coordinated, you're going to potentially have trouble on the grid. There's a lot of quite detailed electrical engineering taking place right now in terms of doing this. Once again, China is doing a lot in this space.

The CHAIR: For that inertia infrastructure then—those condensers or whatever—if data centres are requiring us to invest more in that in order to stabilise the grid, then should data centre proponents be paying their fair share of that infrastructure?

NICHOLAS EKINS-DAUKES: I would position that around a positive—because you mentioned the battery storage. I would say there's a real opportunity for data centres to be this flexible load with battery storage, which means that they can back off and potentially even supply power, or at least back off even more, because they've got their own storage and obviously draw power when electricity is cheap, there is a surplus of solar and so on. At a high level, I would say data centres have the potential to assist. But, obviously, whether you realise that potential or not depends very much on how you've actually installed it and how you've gone about configuring

CORRECTED

the system. We can look at it from a risk lens. Sure, there are risks. There are things maybe we haven't done before, at least not in our region. But, in principle, I don't see enormous barriers to this happening.

The Hon. JACQUI MUNRO: My question was around how quickly data centre companies and electrical engineering technology is advancing to improve efficiency of resource use related to computer power, and even looking ahead at things like quantum computing and interaction with data centres. Is there good modelling around how that efficiency might improve over the next 10 to 50 years?

DAVID EYRE: We reference this issue in the submission. There is significant innovation in chip technology and how computing is conducted globally. Apple silicon is potentially an example you might want to look at. They, of course, have prioritised computing on a mobile device, so it has to be energy efficient. It's possible that this large aggregation of servers within data centres will always have a place, but it's also quite likely we're going to see more distributed AI at enterprise or even device level. The whole ecosystem of advanced computing may shift. We do want to avoid an overshoot in infrastructure in any particular fixed location. It always comes back down to understanding the specific purpose and function of a specific data centre, because they do differ.

Ned mentioned long-duration infrastructure—our water infrastructure and our electricity network. Our grid is very expensive and has very long capex cycles associated with it. Somehow we have to balance the innovation trajectory within the computing industry with how we invest in critical infrastructure in Australia. UNSW is a leader in this domain. It does actually create IP, manufacturing or innovation opportunities for our community where we can legitimately compete globally. We are already a leader, as you know, in photovoltaics. That came out of Ned's team, by the way. That solar cell is in pretty well 98 per cent of all solar panels.

We're also a leader in digital grid technologies. We have a supercomputer that models the grid at a sub-millisecond level. We can design future power stations—not like a contract engineer, but in the modelling sense—that are digitally controlled. As we move from alternating current distribution to direct current distribution, as we're seeing in China where we are starting to connect industrial technology directly to distribution systems, we can start rethinking the whole electricity model for Australia. It's such an exciting area for New South Wales if we frame it in the right way.

NICHOLAS EKINS-DAUKES: I think I'd just add that you can probably categorise this in a few different ways. There are certain things that are absolutely no-regrets actions—obviously the renewable energy zones and obviously building out grid infrastructure. Bringing fibre into remote regions, particularly bringing fibre into renewable energy zones, is probably, in this context, going to be a no-regrets decision. The technology is evolving rapidly. My role at the university is perhaps to provide some insight. A data centre is hugely power hungry because you are effectively organising information. That's what it is actually doing, and it produces a lot of heat as a waste product of that. Our brains also do that, but much more slowly.

Interestingly, our brains operate in a different way. Computers are binary whereas our brains have this complete—we can have any manner of things going on in our brains. Our brains are effectively the same thing, but they've evolved to be energy efficient. We've got the two ends of the spectrum. What I think we can expect is for us to figure out how to build silicon-based technologies on some timescale. You mentioned quantum computing, which I would say is right at the far end of that timescale, but we're going to see evolutions in the efficiency of these chips. I don't think what we see today is going to be representative of what we have in 10 years. Rather like with vehicle emissions and various other things, as we make engines more efficient, we'll just find ways of building bigger cars. We have to be a bit careful with that particular argument.

The Hon. JACQUI MUNRO: I'm looking forward to us all zooming around in F1 vehicles.

NICHOLAS EKINS-DAUKES: Electric vehicles, I hope.

The CHAIR: Thank you very much. It has been a really fascinating session. That concludes our time with you. To the extent that there were questions taken on notice or supplementary questions, the Committee secretariat will be in touch.

DAVID EYRE: If the Committee has time, we would love to invite you to visit UNSW and meet with our many experts across this field. We could only touch the surface today.

The CHAIR: Thank you very much.

(The witnesses withdrew.)

CORRECTED

Mr SIMON DRAPER, PSM, Secretary, Premier's Department, affirmed and examined

Ms REBECCA McPHEE, Deputy Secretary, Investment NSW, Premier's Department, affirmed and examined

Mr TOM GELLIBRAND, Chief Executive, Infrastructure NSW, affirmed and examined

The CHAIR: We now welcome our final witnesses for today. Did you want to make a short opening statement?

SIMON DRAPER: No. We're very happy to be here.

The CHAIR: I was really hoping you wouldn't say that, because we didn't get a submission. It makes it very hard to know what to ask. We will have questions.

SIMON DRAPER: You've had so many other witnesses. I must say I was enjoying listening to the previous witnesses. I wish I'd come earlier.

The CHAIR: It has been an interesting inquiry. Can I ask why no submission has been made by the department?

SIMON DRAPER: We thought turning up and giving evidence in person would probably be more effective for the Committee.

The CHAIR: It's not.

SIMON DRAPER: We have a number of public documents, particularly one public document—

The CHAIR: We do have one.

SIMON DRAPER: The guidelines.

The CHAIR: My understanding, from questions to Ministers during budget estimates, was that there was going to be a government submission, as we would have for every other inquiry. Do you know why that changed?

SIMON DRAPER: No. I'm not aware of that evidence that was given at the budget estimates hearings.

The CHAIR: Were you preparing a submission at one point?

SIMON DRAPER: No.

The Hon. JACQUI MUNRO: Who makes the decision to submit to upper House inquiries?

SIMON DRAPER: Usually, we would prepare one and ask the Government if we could make a submission. We didn't actually prepare one from our department. I'm not sure, Tom, if INSW did, but we didn't prepare one and ask for permission to do one.

The Hon. JACQUI MUNRO: Generally, it would be that the department thinks that it's useful to make a submission and then requests permission from the Minister. Is that accurate?

SIMON DRAPER: Usually, the Government has approved making submissions, yes.

The CHAIR: We did send an invite asking you to make a submission. Normally, that then results in a submission being made. I think this is the only inquiry I've ever chaired where we haven't had any submission from government. It's really unusual. Do you know why this has happened?

SIMON DRAPER: There was no deliberate decision. We just thought it would be more helpful to turn up and give evidence in person, which we are doing today.

The CHAIR: When you say "we" thought that—

SIMON DRAPER: The three of us.

The Hon. JACQUI MUNRO: So you made the decision together not to make a submission?

SIMON DRAPER: Yes. Well, it's sort of a bit of a negative—we didn't make a submission, so we didn't make a decision to make a submission.

The Hon. JACQUI MUNRO: So none of you spoke to each other and said, "Should we make a submission?"

SIMON DRAPER: I don't know if we discussed it, the three of us. But, no, I don't believe so.

CORRECTED

The Hon. JACQUI MUNRO: Did you have a conversation with anybody about whether you should make a submission to this inquiry?

SIMON DRAPER: The only person I had a conversation with was the Secretary of the Cabinet Office. I said I was happy to turn up and give evidence and so we didn't need to do a submission. That was the extent of the conversation.

The Hon. SCOTT FARLOW: Who within government is responsible for data centres or data centre strategy? Is that Infrastructure NSW, which led the consultation paper, or is—

SIMON DRAPER: We don't have a department of data centres, if I can put it that way—

The Hon. SCOTT FARLOW: I appreciate that.

SIMON DRAPER: —or a Minister for data centres. There are a lot of different government responsibilities that have an impact. It's one of those areas of policy that involves a number of parts of government. But you're quite right to call out Infrastructure NSW. As this issue started to become a more pressing one—and I guess that's the reason why the Committee has been set up and the inquiry is underway—over the course of the last year, we did decide that we needed a more developed government position on data centres, a more informed position, that cut across a whole range of policy areas. We asked Infrastructure NSW to pull together a draft position and a consultation paper, which they did. I believe it was released earlier this year.

The Hon. SCOTT FARLOW: To that point, what are the next steps from that consultation paper and will that form a New South Wales Government data centre strategy?

TOM GELLIBRAND: Thanks for the question. The consultation closed on 8 May. I think we got about 70-odd submissions, which is terrific, and you can imagine, they came from a wide variety of different stakeholders and organisations. We're going through all of those submissions at the moment, with a view to updating the guidelines to the extent that we think they need to be, and then we'll feed those in. We're working really closely with the Treasurer's office, as well as the planning Minister's office. They're two really important stakeholders in developing these guidelines. Then we'll update what we've prepared, which essentially are referred to as principles, and then it's really going to be up to government, the extent to which they confirm them as guidelines, confirm them as requirements, and publish them. But the next step in our work is just to update them in response to the consultation and put them before those two key Ministers. That's the next step.

The Hon. JACQUI MUNRO: Minister Chanthivong is not considered a key Minister in that operation?

TOM GELLIBRAND: No, he would definitely be so. But the ones that we've been working through most closely have been the Treasurer's office and the planning Minister's office.

The Hon. SCOTT FARLOW: What about Minister Dib, as well? Any involvement with Minister Dib?

TOM GELLIBRAND: Not directly from—certainly not from my point of view. There's been collaboration across a number of agencies—Planning; department of climate change, energy and water.

The Hon. JACQUI MUNRO: Is the Premier's Office involved?

TOM GELLIBRAND: Yes, and we're working with the—

The Hon. JACQUI MUNRO: Would he be considered a key Minister as well?

TOM GELLIBRAND: Absolutely, yes. In meetings that we've had with the Premier, it's come up as a matter and we've discussed where it's up to and the different views. We've run a very deliberate consultation process, which I think has been really effective. We've had a lot of interest. But it's also apparent that a lot of those organisations have made direct representations to different people in government.

The Hon. JACQUI MUNRO: Which Minister is responsible for data centres?

SIMON DRAPER: I think this just goes to the answer I gave a little bit earlier. There is no Minister for data centres. It cuts across a whole range of policy areas. Ministers have responsibilities that relate to particular pieces of legislation. They administer Acts, and many of those Acts relate to or impact upon data centres. Planning is clearly one where there's a significant role to play, so the planning Minister obviously has a significant role. There's an interest from an economic point of view and the role that data centre investment and the availability of that critical infrastructure has on the economy. The Treasurer and the Premier and others are, of course, interested in those areas, as is Minister Chanthivong in his portfolio—you mentioned Minister Dib. The truth is that we have actually had Infrastructure NSW working with the secretaries across the whole of government. The work that's been developed by Infrastructure NSW has been socialised amongst all the departments in government, before it went forward.

CORRECTED

The Hon. JACQUI MUNRO: Can I ask what Ms McPhee's role is? What is Investment NSW's role here?

REBECCA MCPHEE: The Investment Delivery Authority is made up of four secretaries. There is a team that supports the Investment Delivery Authority in terms of assessing applications and progressing facilitation and support for those endorsed projects, and the team that does that sits within my team in the Premier's Department.

SIMON DRAPER: Perhaps I would add, in terms of Ministers, the other ones that are really important here are the Minister for Energy and the Minister for Water, of course. They've got a very deep interest in this.

The CHAIR: How many data centres do you have currently in the IDA pipeline?

SIMON DRAPER: I think it's 15. Is that correct?

REBECCA MCPHEE: Yes.

The CHAIR: The IDA was set up last August, was it?

SIMON DRAPER: That sounds about right. It was certainly about that time last year.

REBECCA MCPHEE: The establishment was announced in June, but the first EOI opened on 1 October.

The Hon. MARK BUTTIGIEG: Would all data centres have to go through that IDA process or not necessarily?

SIMON DRAPER: Not necessarily. I mean, it's a very separate process. If they want to get a planning consent, they're more than welcome to go through the planning process without coming to the IDA. In fact, many that are going for planning consents are not part of the IDA process. They're not directly correlated.

The Hon. MARK BUTTIGIEG: Do we have a sense of the global amount, inside and out?

TOM GELLIBRAND: I've got some numbers. I have to admit I'm not from DPPI, so some of these numbers might be a little bit out. You'd best check with them or the Planning Portal. In terms of the actual expressions of interest that were received by the IDA, I think at least one of those actually had planning approval, which I think is actually quite an interesting point, which I can get back to if you like. We also understand that there are at least nine data centres that have been going through the State significant development application process which haven't come through the IDA. As probably everyone can appreciate, data centres exist today—some of them were built some time ago—and they do actually operate within the normal economy. There are a number that are going through the normal planning process.

The reason why I mentioned that one of the expressions of interest received was for a data centre that already has been approved is that one of the key roles the IDA plays is to engage—in this case with data centres—to find out from them if there are any impediments for progressing their development, "You've got an approval. Is there something holding you up to make that investment in New South Wales?" In some cases there are, and the IDA says, "What is it?" It could be connecting to a utility—water or power or something. We then ask those questions, "What are the particular issues? Is there a role for government to play in supporting you in that development?" One of the reasons why Rebecca is so closely involved is that Investment NSW, by virtue of its name, is there to support good economic development in New South Wales, and the IDA will assess those expressions of interest to determine if there's a good role for government to assist or not.

The Hon. JACQUI MUNRO: When the Data Centres Australia representative came in, I asked specifically if they had been asked to contribute or engage in any process related to a data centre strategy, and the response was this:

No. We hear a lot of comments about a data centre strategy, whether it's Sydney Water or whether it's someone else, but they have not come to us, no.

How does that engagement with the data centre industry square up with your experience?

SIMON DRAPER: Obviously I didn't hear her evidence and I haven't read it, but I don't know what she meant by a strategy. We definitely—I know because I was there—met with Data Centres Australia and talked to them about the consultation paper that has been released and had their input on that. I know that for sure. I'm not sure what she meant by a strategy, but what we mean by a strategy is setting some parameters under which data centres can proceed on the terms that meet all the policy trade-offs that we've got on the table here. That's what we would call a strategy for data centres.

The Hon. SCOTT FARLOW: Is that what Infrastructure NSW has gone out on consultation with?

SIMON DRAPER: There was a consultation paper. That's not the final document.

CORRECTED

The Hon. SCOTT FARLOW: To your knowledge of what's being done in government with data centres and any parameters or strategy, for want of a better word, that's the work that Infrastructure NSW has gone out on consultations?

SIMON DRAPER: To try to be comprehensive for you, I think there's the work that Tom and INSW have done, which is to try to pull together a whole range of issues, and they've really done a great job of that. There's the work in the Investment Delivery Authority. That's dealing with a group of data centres who've proposed that they might need some additional focus from government to progress their proposals. We obviously pay a lot of attention to what's happening with data centres with energy and water demand. We have work going on in those areas. There's a whole range of things that we are doing, and obviously the planning department has a number of proposals. There's a fairly wide range of scope of activities, but we do try to stay very coordinated through the secretaries of all those various departments. I would say probably the binding process has been the Infrastructure NSW consultation paper, which everyone has been involved in.

The Hon. JACQUI MUNRO: To clarify, you're saying that there's not necessarily going to be a data centre strategy. Is that correct?

SIMON DRAPER: I don't know what you mean by strategy—and I'm not being clever about it. Do you mean are we going to—

The Hon. JACQUI MUNRO: It does sound like you're being a bit cute about it—a data centre strategy.

SIMON DRAPER: Maybe you could unpack what you mean by a strategy.

The Hon. JACQUI MUNRO: Will there be a data centre strategy that comes out of the consultation paper?

SIMON DRAPER: It's not a strategy. It won't be a strategy; it'll be a set of parameters that government will be seeking from data centres when they proceed.

The Hon. SCOTT FARLOW: Those parameters will be implemented across different agencies, will they—or will they be implemented across Planning in terms of the consent process?

SIMON DRAPER: It's really what data centres will need to satisfy. It's a requirement for data centres, not for agencies.

The Hon. SCOTT FARLOW: In terms of this strategy, data centres will have to tick all these boxes, but I imagine that they will be administered by a certain agency in terms of actually checking that and that they do comply. Will that be done by Planning, or will that be done by a multitude of agencies? For instance, you were saying about the interests that the Minister for Climate Change, and Minister for Energy, might have, or the interests that the Minister for Water might have. I take it that, of course, Sydney Water has certain requirements when it comes to connections. In terms of that, who will actually be doing the assessment that the data centres are complying with those requirements that will come out of that strategy?

TOM GELLIBRAND: The process that is underway at the moment is coming up with a set of principles by engaging with industry and trying to get feedback on it, with a view to firming up a set of—they may be called guidelines or they may be called requirements, but that's not necessarily settled at the moment. The purpose of that is to make it clear to the outside world that if you want to undertake this sort of development in New South Wales, it's not at any cost. Arrangements need to be made with respect to the energy you use, the water you consume and potentially the local impacts that you may have in terms of noise or air quality. Those things are really helpful because the data centres know that it's not just a blank sheet of paper. We are keen on having that development, but not at any cost. That also then enables the regulators, like the planning department that you mentioned before, to start developing specific secretary's environmental assessment requirements.

When the applications come in—I don't want to make light of it—they're dealt with the same as every other application that comes in. The secretary has to issue specific requirements. They will need to obtain, in most cases, a certificate from Sydney Water saying arrangements have been made in terms of the use and availability of water and wastewater. They also will need to get approval from Ausgrid to connect into transmission lines, or a retailer if they're going into a more local network. You can apply your own words, but that regulatory environment is effectively a safeguard. There needs to be consideration of water demand and energy demand as part of the assessment. If the data centres are unable to satisfy the requirements of those utilities, the approvals won't flow. The value of having a set of guidelines or a set of requirements out there in the market is it tells people that if you want to engage and do business in New South Wales, these are the sorts of things that'll be important as we undertake those assessments. That's how they operate.

CORRECTED

The Hon. JACQUI MUNRO: When did the conversation about data centres needing additional guidelines or requirements begin in government?

SIMON DRAPER: I would say probably about a year ago—a bit more than a year ago, I think—was when we started to see the issues coming from different policy areas, as I mentioned before, and also a significant ramp-up if not in planning proposals submitted to planning then at least in advice from industry players that they were planning to invest quite a lot in New South Wales. We took a view that that needed to be coordinated across government in a whole range of areas, particularly, I think—as Tom pointed out—for the department of planning, who are in some ways at the front end of this. Where they are receiving State significant development applications and they have to assess them, they need something to assess it against. They need to understand how that fits into the overall picture of energy requirements, energy demand, the capacity in that sector—similarly with water—the cumulative impacts of the number of data centres because the proposals do tend to be in particular areas and particular parts of Sydney. They needed some guidance around that from a planning point of view.

The Hon. JACQUI MUNRO: So a year ago?

SIMON DRAPER: About a year ago, yes. Sorry, I should have given that shorter answer.

The CHAIR: Does that mean that the IDA was set up for data centres specifically?

SIMON DRAPER: No.

The CHAIR: Was the IDA going to be set up anyway? Was that the idea?

SIMON DRAPER: I think the IDA really came from the early months of us operating the HDA—the Housing Delivery Authority—and the enthusiasm we got from proponents in the housing sector to use that channel caused the Government to think that there may be benefit in looking at other areas of investment that may benefit. The areas where the biggest volume of investment proposals appeared to be emerging from were those involved in the energy transition, and data centres or technology generally was another one.

The Hon. JACQUI MUNRO: Can I just ask about—is it the trade and investment or the trade and industry strategy statement? I think it is more of a statement than a strategy. The words "data centre" are only found once in that statement, and I just wonder if you think that is a failure of imagination, when we're sitting in this inquiry a year later and our State's Trade and Investment Strategy couldn't anticipate something that was clearly happening around the world since at least 2023.

SIMON DRAPER: Rebecca can answer for that, but I would say that strategy—or statement, as you called it—is designed to call out areas that need government focus to promote more investment in those areas. One of the beauties of data centres is that they are privately initiated and privately financed. There's no dearth of investment proposals in those areas, so it doesn't require a lot of government intervention to make that investment happen.

The Hon. JACQUI MUNRO: But it does. That's why we've got this whole inquiry.

SIMON DRAPER: Whereas some of the other areas covered in that strategy are designed to try to nurse along parts of the economy that need more government assistance.

REBECCA McPHEE: If I may add, whilst data centres on their own are not called out consistently through that document and through the other economic development documents that we've published, we call out a strategic focus on digital transformation, innovation and productivity. Clearly, data centres are one element that supports digital transformation but, as you know, to drive jobs, economic growth and productivity from digital transformation, data centres are only one part of the equation. So we're looking much more holistically at that sector.

The Hon. JACQUI MUNRO: But the Treasurer has just come out to say that our entire State's economy relies on data centres and renewable energy to keep growing, so it's difficult to take that seriously—that data centres are just a tiny part of digital transformation and therefore—

SIMON DRAPER: They're an enabling part of it, aren't they?

The Hon. JACQUI MUNRO: Of course they are.

SIMON DRAPER: It's like saying that—

The Hon. JACQUI MUNRO: That's why it's baffling that they're not in the Trade and Investment Strategy.

CORRECTED

SIMON DRAPER: Data centres are just infrastructure. Right now we're very excited about them, but in years to come they'll be the most boring part of the sector. It will be like talking about a water treatment plant. These are pieces of infrastructure that—

The CHAIR: Unless you live next to one.

SIMON DRAPER: The types of technology industries that we're more focused on are those that utilise the enabling capacity of data centres.

REBECCA MCPHEE: In terms of the Trade and Investment Strategy, if we talk about foreign direct investment into New South Wales in those sectors, some figures from the past three years: We've seen about \$32.6 billion of investment in renewable energy—that's about 57 per cent of the total—and about \$4.7 billion, so about 8 per cent of the total, in data centres.

The Hon. SCOTT FARLOW: What's in the pipeline for data centre investment, or even what you've gone through the Investment Delivery Authority? What sort of pipeline have you got?

REBECCA MCPHEE: The 15 projects that are in the Investment Delivery Authority—and I want to get this number right—is about \$50 billion.

The Hon. SCOTT FARLOW: Which has landed in your lap without any thought?

REBECCA MCPHEE: It is \$59.18 billion.¹

The CHAIR: If there were 15 as of October, how quickly are they moving through and where are we at now with those?

SIMON DRAPER: They are 15 proposals that the IDA has decided to engage with and work with to progress their investments. There's a whole range of other ones out there that are not in that process.

The CHAIR: But how many of those 15 have progressed? What is it that the IDA is doing to make it go quicker?

SIMON DRAPER: A lot of them are at quite early stages. They have different needs. Tom mentioned that one of them has a planning consent. Most of them have not yet got to the planning consent phase. They haven't got a consent, so that's the next stage they need to go through. They need to get power connection agreements. They need to get water servicing agreements. There's a whole range of things.

The Hon. SCOTT FARLOW: What's the advantage of going through the IDA rather than going through just a State significant development when it comes to a data centre?

SIMON DRAPER: I think it's primarily that all of the legislation that governs planning consents and everything else is exactly the same. That doesn't change. What changes is an accurate capacity to engage more directly with government to identify things that are moving too slowly, identify things that could cause risks to the projects and see if we can get attention to addressing those risks more quickly. There are a lot of different parts of government involved in this. Rebecca has a team of people from different parts of government. Sometimes something could be held up by a piece of enabling infrastructure that government is going to deliver in 12 months time. If we brought it forward, it would precipitate a much bigger private sector investment. It may be land, access to land. It may be environmental licensing arrangements and making sure that those get progressed. There's a whole range of things. The one thing that we don't contemplate is government participation financially. That's why I mentioned earlier these are all privately funded investments, which is unusual for infrastructure. Most of the infrastructure in New South Wales over the past many decades, but particularly over the last decade, has been paid for by government. This is an opportunity to get very large-scale investment funded by private investors.

The CHAIR: It's access to a better quality service, then. Instead of being stuck in the planning pipeline and having to speak to people and do all of the stuff yourself, if you're the head of whoever it is that is the data centre proponent, you can just call someone quite senior in government and go, "There's this problem that I've got with this thing."

SIMON DRAPER: The process is—and Rebecca might expand on this—that we sit down with the proponents. We identify very specific potential risks to their project's proceeding, develop a work plan with them and, yes, they have a contact in Rebecca's team who liaises with them, and they can pick up the phone and speak to that person at any time. Those issues can be escalated within government, which is why we have a number of

¹ In correspondence to the committee dated 22 June 2026, Ms Rebecca McPhee, Deputy Secretary, Investment NSW, Premier's Department, [clarified](#) their evidence

CORRECTED

different secretaries and CEOs involved in the IDA process. It often requires coordination within government and communication within government to resolve those things.

The CHAIR: Like VIPs.

REBECCA McPHEE: Yes, absolutely. I think probably the second thing, and quite important to add to that, is we've got that remit to facilitate for these large-scale projects that are supported by the IDA. But we also have a remit to identify systemic reform out of that engagement so that we can identify where we're seeing common points of friction within government processes and regulations in order to identify systemic reform that will help all private investment flow more freely.

The CHAIR: Excuse me for just not knowing—it's taken ages for me to understand what the IDA actually does—but to distinguish that from a system where someone who's very influential in business can just pick up the phone to a Minister and say, "I need you to fast-track my project," and it's all a bit dodgy, clearly that's not what the IDA is, but in order for that corruption risk to be managed, is there greater transparency? What kind of oversight is there then of that process to make sure it's not being misused?

SIMON DRAPER: I think it's a very structured process for something that does happen more informally in governments, not just in New South Wales but in other parts of the world. It's clear there's a governance structure. We have a group of Ministers who oversee the process. We have a panel of secretaries and CEOs. We have a secretariat. There's an EOI process with evaluation criteria, which are well documented. There are all the governance protocols that we'd normally have for these sort of things, with conflict-of-interest statements and all of those sorts of things. It's a pretty well-structured process, but also designed to be dynamic and help us work with those industry players.

The CHAIR: How is it different, then, to the Development Coordination Authority?

SIMON DRAPER: The Development Coordination Authority has got a very specific role in the consents process. Within government, after a planning consent has been given, there's usually a whole range of post-consent conditions that you have to work through. They help coordinate that, but they also help coordinate the input of different parts of government into the planning assessment process. Usually, the planning department will ask many parts of government for advice and they'll make referrals. The Development Coordination Authority will now play that role in making sure that works effectively. That's often not going to major projects like we're talking about here today, but often even housing projects and those sorts of things.

The Hon. SCOTT FARLOW: The land use conflict is one of the issues that we're confronting throughout this inquiry. Many have raised it—even Business NSW—in terms of suboptimal locations for data centres. Is there any work being done in government considering where the best places for data centres are and how there can be signals given to the market about where data centres should be located?

TOM GELLIBRAND: I'm aware that the Department of Planning, Housing and Infrastructure are looking at all of the industrial lands in and around metropolitan Sydney, because I think, arguably, they're really important in terms of different activities, employment—all sorts of things—and making sure that Sydney operates well. In doing that work, they're cognisant of the risks associated with data centres occupying existing industrial areas, because they could displace other activities. That's a piece of work that they're doing. I don't believe it's complete. But apart from that, I don't think there's any express planning work going, "Is there a particular zone of Sydney where data centres might go, versus other areas?" That could be a little bit problematic, because as data centres will use either a lot of water or a lot of energy, or a lot of both—because different data centres actually place different demands, depending on the way they're designed—an overt clustering of them saying, "We want data centres just here" could actually lead to some poor outcomes in terms of the way you have to reconfigure your water, wastewater and electrical networks.

The Hon. SCOTT FARLOW: But some of those challenges are that the clustering is already occurring. We're hearing evidence throughout the inquiry—Lane Cove, at the moment, and the Lane Cove West Business Park. I think there are five—there's one there, and there are four other proposals. We've seen, of course, with Macquarie Park, that there are several data centres there, where you've got three metro stations in that precinct, and planning aspirations from the Government for significant residential development throughout that precinct, as well. Without those signals and data centres being permissible uses under both commercial and industrial zonings, there seems to be a case where data centres can crowd out other forms of either industrial or commercial development, which may have been what the Government planned and intended. Are there no barriers that government is looking at in terms of those controls, at all?

SIMON DRAPER: I think it sort of goes to the point that Ms Munro was asking about earlier, about whether we've got a strategy. It's not so much about a strategy; it's more about saying, "These are the conditions under which investment in data centres will meet a whole range of other policy objectives", including the one that

CORRECTED

you just referred to. The data centre proponents themselves don't have a clear vision of what other proponents are coming up with, whereas we do, because of the proposals to the planning system. I do think that if you were looking for a place to develop a data centre, you'd be looking for places that have good access to power, good access to water if they need water—as Tom mentioned, not all of them need a lot of water—are away from sensitive receivers who may be affected by noise, and are not competing directly with other potentially high-value industrial land uses. A lot of areas that we reserve for commercial uses, the feedback from those who invest in those sectors is that there's not much demand, particularly for commercial building space, at the moment. We can't fight against the tides of the economics in that case. Whereas data centres are keen to invest, some others in those areas are not. It may be that there's a geographically sensible solution, but I don't think we've considered going as far as—as I think you're suggesting—more like a zoning approach to data centres.

The Hon. SCOTT FARLOW: There are examples of industrial land next to an intermodal—and we know of the scarcity of industrial land in Sydney—and a data centre proponent looking at an application there. You would think that next to an intermodal, a data centre wouldn't necessarily be something that you were intending for that zoning. But because they can pay overs, effectively, as we've heard throughout this inquiry, that's likely to be what's developed on that site. There's no consideration from government about limiting uses like that or being able to identify certain barriers so that the land could be preserved for what planners would have intended it to be used for?

SIMON DRAPER: Maybe Tom or Rebecca can answer this more expansively, but I will say it is considered. In my experience working with the department of planning, they very much take that into account as one of their deep considerations they provide feedback on, including when items were coming through the IDA process—about whether there are other high-value uses of that land where the proposal was going to be. That was one of the considerations that they would raise. So I think that's definitely something that we're considering in the IDA but, probably more importantly, in the planning system itself.

The Hon. JACQUI MUNRO: Is there a timeline? When do you expect to have those guidelines published?

TOM GELLIBRAND: There are always two elements to getting something completed. One thing is doing the work and the other thing is getting the tick to publish. We're going through the submissions now. We're at an advanced stage where we think they're in good shape. That's when we have to engage further with the different Ministers. Once government is comfortable with the guidelines or the requirements, they get published. But we're at a fairly advanced stage. One of the things—

The Hon. JACQUI MUNRO: In the first half of the year?

TOM GELLIBRAND: I'd love to be able to answer the question, and I just can't. Ultimately, our job is to produce the draft that we think, from our consultation, is in good shape and is best for New South Wales. Then it's a decision for government about whether or not they agree and when they'd like to publish it.

SIMON DRAPER: The first half of the year is only another five weeks. Given the budget process going through, it may not be in that period. But, yes, the work is advanced. The Government has taken a lot of interest in it. They were the ones who asked us to do it, or Tom to do it. I think you could expect to see it sometime this year.

The CHAIR: Coming back to the data centres that are currently going through the IDA, you mentioned before that there was some visibility of the cumulative impacts. Is that something that you are considering? If you've got three that are all in one particular part of Ryde or something, are you considering the cumulative impacts of all of those and feeding that back into what the anticipated energy demand or the pollution would be, or whatever? Is that fair to say?

SIMON DRAPER: Yes, that's part of the process. That would be considered in the planning process as well. Again, the challenge for any department of planning is that they're dealing with a submission or an application on a case-by-case basis. The planning department does see it; once an application is made, they can see it. The IDA process also has drawn out—like, we haven't quite hit that phase, so we're building a bigger picture of where there may be data centre proposals.

The CHAIR: My understanding under the planning framework as it currently stands is they can't look at it cumulatively; they can only assess each proposal on its merits.

SIMON DRAPER: They would assess each proposal on its merits, but they would look at the impacts more broadly. They would look at what the existing conditions are—for example, if there was a group of homes, they would take into account the number of diesel generators that were wanted to be installed.

The CHAIR: So including those that are approved, not just those that are existing.

CORRECTED

SIMON DRAPER: Yes.

TOM GELLIBRAND: I think a good example of that—I definitely don't want to speak on behalf of Planning, but if you think of a series of traffic-generating uses, they get progressively approved so then you've got a number of vehicles using the roadway. Then someone else comes in. They get assessed relative to what's going on in the background. That cumulative impact—and they might only generate a small number of vehicles relative to everything else. But it gets to a point where the assessor would go, "Wow, these additional cars are now going to cause a problem." Cumulative impacts do come into consideration when you're assessing proposals.

The CHAIR: Does that mean that, for example, if you did have three in one spot and you looked and went, "Well, no, the cumulative impact of all three of these in the IDA is going to be unacceptable", then presumably some of those projects could not proceed? It's not an implicit endorsement just because they're in the IDA that they're going to definitely get through the other end.

SIMON DRAPER: Yes. I think one of the benefits with the IDA process would be that we would be able to see all of those together and we'd be able to speak to those proponents and say—they may not see the risks. We would see the risks because we can see all the proposals. We may call out to them, "There's a risk to your project. If you go down that path or use that particular technology, that particular cooling technology"—

The CHAIR: You might want to change your approach or do this so that then you don't hit that roadblock later on.

SIMON DRAPER: Correct.

TOM GELLIBRAND: That's sort of one dimension. But the other dimension is that we talk to the utilities and we say, "Our intelligence is telling us that there's significant demand for these activities in this particular location. How are your assets and what's your capacity?" On many occasions, they're saying it's constrained. Then we say, "What would you need? What sort of augmentation is required if these things were to proceed?" Often they say, "We need a substation", or, "You'd need to intercept the sewer and have a recycled water arrangement", what have you.

That's where—I'm a bit nerdy—the exciting dimension of the IDA comes into it, because you've got three people in the room that have got the deepest pockets out of anyone on the globe, like Amazon, Microsoft, those sorts of people. They want to locate in a particular area and we've all identified that the provision of potable water might be limited. There are ways to fix that. You can use alternative technology which doesn't require water to cool. Or you could intercept sewage and then get that converted into a water resource that you can use to cool. The conversations we had with consultation is that many of those users of the technology—like Microsoft, Amazon et cetera—are up for that. Around the world, they do fund provision of alternative energy. They are moving towards not using water or creating—or identifying their own supply. You can't create water; it's finite.

That's where the IDA is really quite useful, because we have had interest from some proponents to invest in wind farms. We've given them details about wind farms that are in the system but haven't started. They're looking to sell their energy effectively. We've also spoken to Transgrid about potential upgrades to the broader transmission network that are currently unfunded and a little further in the future. But if data centres emerge, the demand for that will be brought forward. Is there an opportunity for the data centres to contribute that in such a way that doesn't change the cost exposure to consumers? There's a virtuous alignment between the demand generated by data centres and the longer term needs for augmenting water and energy networks. It can be quite virtuous.

The CHAIR: The obvious question then that I'm going to ask is: What is the cumulative energy requirement of those 15 projects?

SIMON DRAPER: We'd have to get it out of the gigawatts.

TOM GELLIBRAND: Actually, on day one it's probably really small. I'm definitely not an expert in data centres, but the way I understand it is—step one, try and find a client or someone and then the developer buys a big block of land. They get confidence that the market is going to buy or locate there or buy that processing capacity. Then they go about building the building, getting the services connected. On day one, what we've been told directly from Data Centres Australia, as well as most of the proponents, is they take up a fraction of the capacity. There'll be a bunch of servers churning away and then it gets built out over time. We had an idea where we would require them to sign up to a whole range of requirements in terms of water efficiency, energy efficiency, as well as alternative energy arrangements on day one, and they're going, "We want to do the right thing, but on day one our power usage requirements are really quite low. But they get built out over a number of years." Whilst ultimately they will have a maximum demand for water and energy, on day one it's a fraction of that.

CORRECTED

The CHAIR: It would still be nice if they just put more in than they took. I appreciate if you need to take it on notice, but what that aggregate or that combined—

SIMON DRAPER: We probably could take that on because I think we do have—as you know well, they express the capacity in megawatts or gigawatts. So we can figure that out and provide you with that on notice.

The CHAIR: I don't personally buy the data centre line on that, but that's all right. That does bring us to an end. Thank you. That has been very useful. To the extent there are questions taken on notice or supplementary questions coming from the Committee, the Committee secretariat will be in touch. That concludes our time with you and it concludes our hearing for today.

(The witnesses withdrew.)

The Committee adjourned at 17:05.